$$h = u_2 t_2 + \frac{1}{2} g t_2^{\,2} \Rightarrow 44.1 = 2u + \frac{1}{2}(9.8)(2)^2$$

$u = 12.2\ 5$ m/s

Answer: A

220. A driver of a train travelling at 115 km/h sees on the same track, 100m in from of him, a slow train is travelling at 25 km/h in the same direction. The least retardation that must be applied to faster train to avoid collision is?

A. $25\ m/s^2$ B. $50\ m/s^2$
C. $75\ m/s^2$ D. $3.125\ m/s^2$

Solution: Relative velocity = 90 km/h = 25 m/s

$v^2 - u^2 = 2as$

$0 - (25)^2 = 2(a)\ (100) \Rightarrow a = -3.125\ m/s^2$

Answer: D

221. A stone falls freely from rest and the total distance covered by it in the last second of its motion equals the distance covered by it the first three second of its motion. The stone remains in air for

A. 6 sec B. 5 sec C. 7 sec D. 4 sec

Solution: $\frac{1}{2} g t^2 = \frac{g}{2}(2t - 1)$

$$\Rightarrow \frac{1}{2} g (3)^2 = \frac{g}{2}(2t - 1) \Rightarrow t = 5\,\text{sec}$$

Answer: B

222. On applying brakes, an automobile stops after 5 second covering 25m with uniform retarded motion. The initial speed of the automobile is

A. 5m/s B. 10 m/s C. 15 m/s D. 20 m/s

Solution: $\left(\frac{v+u}{2}\right) t = s \Rightarrow \left(\frac{o+u}{2}\right) = \frac{25}{5} \Rightarrow u = 10 m/s$

Answer: B

223. The velocity of a stone thrown upward is halved in 1.5 sec. The maximum height attained by the stone is?

A. 20m B. 25m C. 30m D. 45m

Solution: $v = u - gt \Rightarrow \frac{u}{2} = u - (10)(1.5) \Rightarrow u = 30 m/s$

$$h = \frac{u^2}{2g} = \frac{(30)^2}{2(10)} = 45m$$

Answer: D

224. A car accelerates from rest at constant rate of $2m/s^2$ for some time. Then, it retards at a constant rate of 4 m/s^2 and comes to rest. If the total time for which it remains in motion is 3 sec. what is the total distance travelled?

A. 2m B. 3m C. 4m D. 6m

Solution: using $v = u + at$ or $v - u = at$ if a is doubled t is halved.

$$t + \frac{t}{2} = 3 \text{ or } t = 2\text{sec}$$

$$S = s_1 + s_2 = \frac{1}{2}(a_1)t_1^2 + \frac{1}{2}(a_2)t_2^2 = \frac{1}{2}(2)(2)^2 + \frac{1}{2}(4)(1)^2$$

$S = 6m$ **Answer: D**

225. A ball is dropped at t = 0 from a certain height. At t = 1 sec, another ball is dropped from same height. The distance between the two balls at t = 3 sec is

A. 5m B. 20 m C. 25m D. 125 m

Solution: $Y_1 = \frac{1}{2}gt^2$ and $y_2 = \frac{1}{2}g(t-1)^2$

$$\Delta y = y_1 - y_2 = 25 \text{ m}$$

Answer: C

226. A particle moving with a uniform acceleration travelled 24m and 64m in the first two consecutive intervals of 4 sec each. Its initial velocity is

A. 10 m/s B. 5 m/s C. 2 m/s D. 1 m/s

Solution: $S = ut + \frac{1}{2}at^2$

$$24 = 4u + \frac{1}{2}a(4)^2 \text{ and } 88 = 8u + \frac{1}{2}a(8)^2 \Rightarrow u = 1 \text{ m/s}$$

Answer: D

227. The velocity of a body at the end of 5 sec is 30 m/s at the end of 12 sec is 58m/s and at the end of 22 sec, it is 98m/s. the body is moving with

A. uniform velocity
B. uniform acceleration
C. uniform retardation
D. uniform displacement

Solution: $v = u + at$

$$\left.\begin{aligned} \Rightarrow 30 &= u + 5a \\ 58 &= u + 12a \end{aligned}\right\} a = 4\text{m/s}^2$$

Answer: B

228. The distance covered by a body is given by $S = at + bt^2$. The acceleration of the body is

A. a/b B. 2b C. a + b D. 3b

Answer: B

229. A particle is moving with constant initial velocity 4m/s till t = 1.5 sec. Then it accelerates at 10m/s^2 till t = 3 sec. the distance covered is

A. 17.25m B. 36.25 m C. 40 m D. 23.25 m

Solution: $S = s_1 + s_2 = vt_1 + ut_2 + \frac{1}{2}at^2_2$

$$= (4)(1.5) + (4)(1.5) + \frac{1}{2}(10)(1.5)^2 = 23.25 \text{ m}$$

Answer: D

230. A monkey in a tree 20m high drops a coconut directly above your head as you run with a speed of 1.5m/s beneath the tree. How far behind you does the coconut hit the ground?

A. 2m B. 3m C. 4m D. 5m

Solution: $y = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2y}{g}} = 2\text{sec}$

$$x = ut = (1.5)\ (2) = 3\text{m}$$

Answer: B

231. A body starts from rest and moves with a uniform acceleration of 10m/s^2 in the first 10 sec. During the next 10 sec. it moves with uniform velocity. The total distance covered by the body is

A. 1500m B. 2000m C. 2500m D. 3000m

Solution: $S = s_1 + s_2 = ut_1 + \frac{1}{2}at_1^2 + vt_2$, $v = at_1 = (10)(10)$

$$S = \frac{1}{2}(10)(10)^2 + (100)(10) = 1500\text{m}$$

Answer: A

232. A body projected upward with a velocity u returns to the starting point in 4sec. The value of u is

A. 5m/s B. 4 m/s C. 20 m/s D. 10 m/s

Solution: $t = \frac{2u}{g} \Rightarrow u = 20\text{m/s}$

Answer: C

233. A body thrown up reaches a maximum height of 100m. Another body with double the mass thrown up with double the initial velocity reaches a maximum height of

A. 25 m B. 100 m C. 200 m D. 400 m

Solution: $h_2 = \frac{u_2^2}{2g} = \frac{(2u)^2}{2g} = 4\left(\frac{u^2}{2g}\right) = 400m$ **Answer: D**

234. Galileo's experiment showed that if two bodies of unequal masses are dropped from the same height, they reach the ground at the same time. But if they are thrown up with equal velocity, the ratio of time they take to reach the ground is equal to?

A. one
B. mass one time mass two
C. the ratio of their masses
D. the inverse ratio of their mass **Answer: A**

235. A stone falls from a balloon that is descending at a uniform rate of 12 m/s. The displacement of the stone from the point of release after 10 sec is

A. 725 m B. 610 m C. 510 m D. 490 m

Solution: $y = ut + \frac{1}{2}gt^2 = 610m$ **Answer: B**

236. If a body having initial velocity zero is moving with uniform acceleration of $8m/s^2$. The distance covered by it in the fifth second is

A. 36m B. 40 m C. 100 m D. zero

Solution: $S = u + \frac{a}{2}(2t-1) = 0 + \frac{8}{2}(2(5)-1) = 36m$ **Answer: A**

237. The velocity of a car is given by $v = 6 + 8t - t^2$. The instantaneous acceleration at $t = 5$ sec is

A. $-1m/s^2$ B. $-2m/s^2$ C. $1\ m/s^2$ D. $2\ m/s^2$

Solution: $a = \frac{dv}{dt} = 8 - 2t = 8 - (2)(5) = -2m/s^2$ **Answer: B**

238. Two buses are moving in the same direction with a velocity of 40 km/h. They are separated by a distance of 10km. A truck moving in the opposite direction meets these two buses in an interval of 10 minutes. The speed of the truck is

A. 10 km/h B. 20 km/h C. 40 km/h D. 60 km/h

Solution: Let V be truck velocity. Then velocity of truck relative to each bus = v + 40

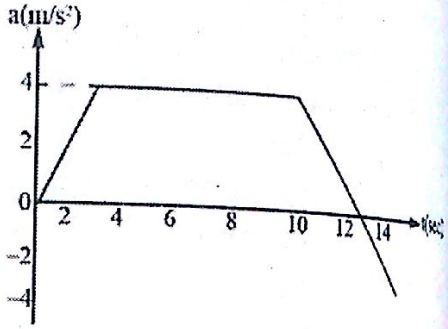
$$\text{relative velocity} = \frac{\text{distance}}{\text{time}} = \frac{10\text{km}}{\frac{1}{6}\text{h}} = 60\text{km/h}$$

$$\Rightarrow v = 20 \text{ km/h}$$

Answer: B

Graphical Description of Motion

239. From the acceleration – time graph, the speed of a particle at $t = 14$ sec is



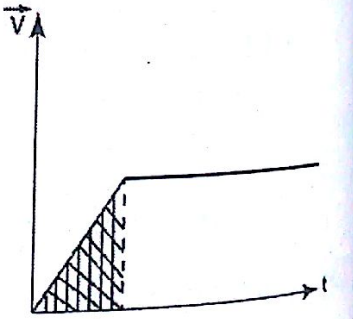
A. 2 m/s
B. 34 m/s
C. 20 m/s
D. 42 m/s

Solution: V = Area

$$V = \frac{1}{2}(4)(4) + (6)(4) + \frac{1}{2}(2)(4) - \frac{1}{2}(2)(2) = 34\text{m/s}$$

Answer: B

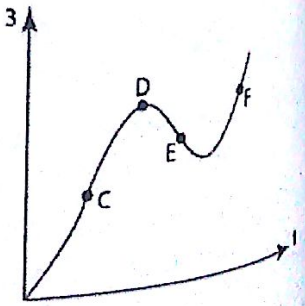
240. For the velocity time graph shown, the shaded portion of the graph has an area equal to



A. the average acceleration
B. distance moved during uniform motion
C. distance moved during accelerated motion
D. the average velocity

Answer: C

241. The displacement – time graph of a particle is shown. At what point is the instantaneous velocity of the particle negative?

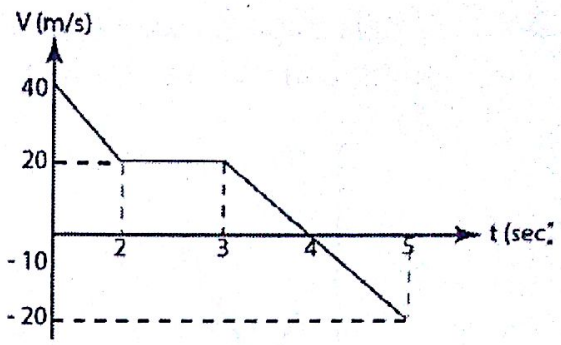


A. C
B. D
C. E
D. F

Hint: negative slope

Answer: C

242. In v – t graph shown, the distance travelled by the body in 5 sec will be



A. 50 m
B. 40 m
C. 80 m
D. 100 m

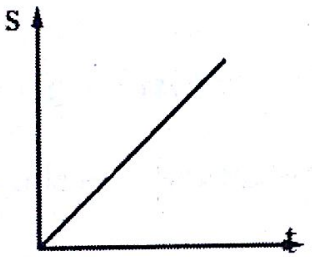
Solution: S = Area

$$=\frac{1}{2}(20)(20)+(2)(20)+(1)(20)+\frac{1}{2}(1)(20)+\frac{1}{2}(1)(20)=100\text{m}$$

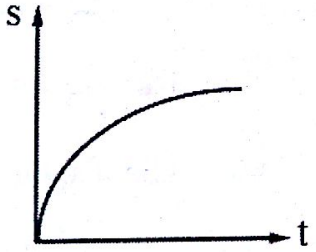
Answer: D

243. Which of the following graph represent the distance time varation of a body release from the top of a building?

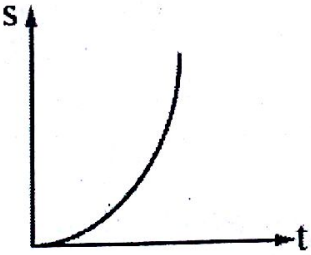
A)



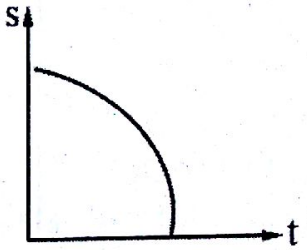
B)



C)

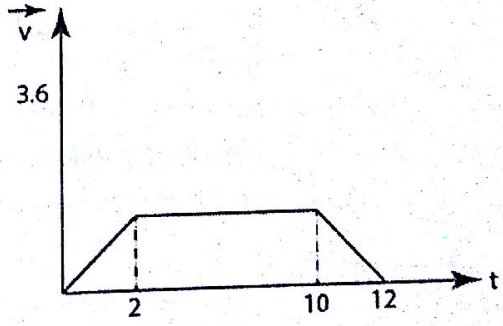


D)



Answer: B

244. An elevator is going up. The varation in the velocity of the elevator is as given in the graph. What is the height to which the elevator takes the passenger?



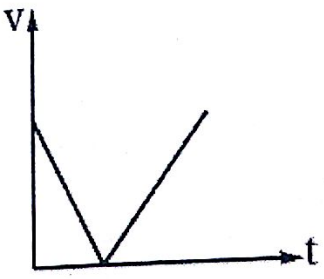
A. 3.6 m
B. 28.8 m
C. 36 m
D. 72 m

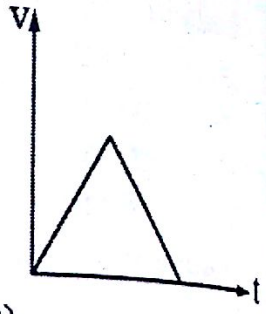
Solution:

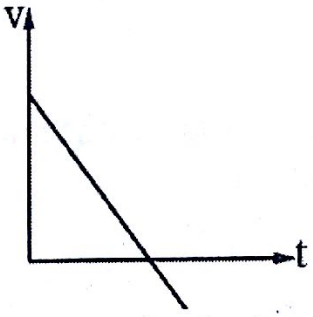
$$\text{h} = \text{area} = \frac{1}{2}(3.6)(2) + 8(3.6)+\frac{1}{2}(3.6)(2) = 36\text{m}$$

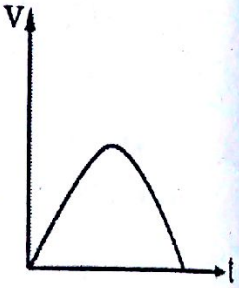
Answer: C

245. A ball is projected upward. Which graph represents the velocity of the ball during its flight.

A) 

B) 

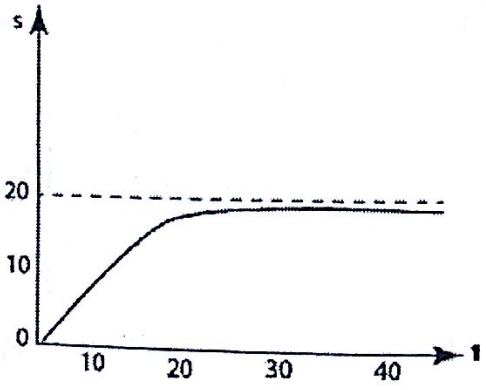
C) 

D) 

Hint: upward velocity positive and downward negative.

Answer: C

246. The displacement – time graph of a particle is as shown



The graph indicate

A. the velocity of the particle is constant throughout

B. the particle starts with certain velocity, but retarded and finally stops

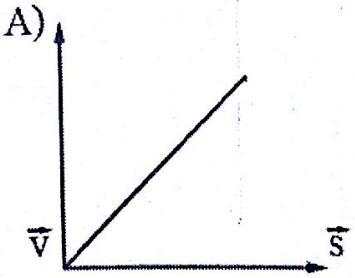
C. the particle starts with constant velocity and accelerated and then moves with constant velocity

D. the acceleration of the particle is constant throughout

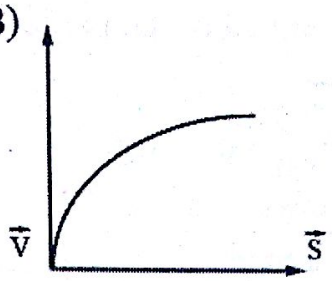
Answer: B

247. A particle starts from rest and moves with constant acceleration. Its velocity – displacement graph looks like.

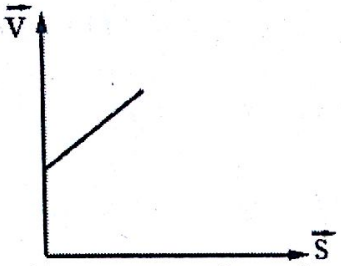
A)



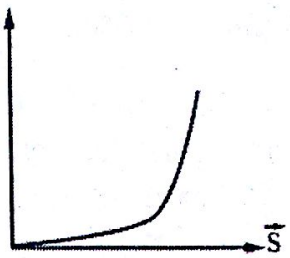
B)



C)

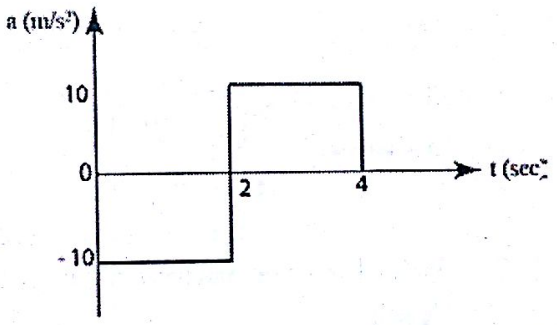


D)



Hint: $v^2 \sim s$ from $v^2 = u^2 + 2$ as **Answer: B**

248. A particle starts from rest and mover with acceleration. The speed of the particle with be maximum after

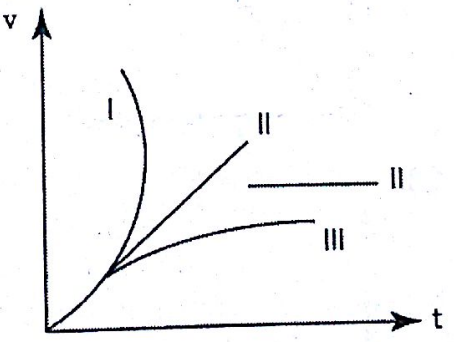


A. 1 sec
B. 2 sec
C. 3 sec
D. 4 sec

Answer: B

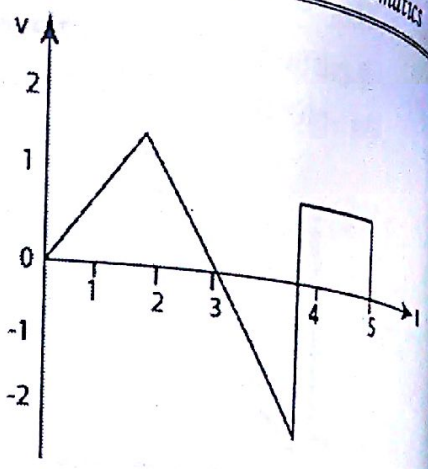
249. For the given velocity – time graph, in which case is the acceleration uniform and positive?

A. I
B. II
C. III
D. IV



Answer: B

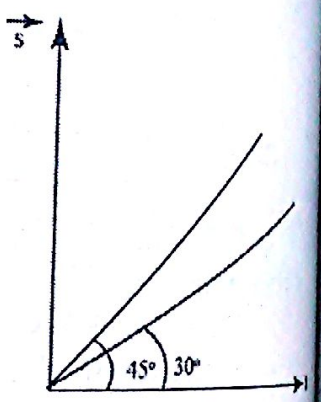
250. The velocity – time graph of a particle moving along a straight line is shown. What is the displacement of the particle in 5 seconds



A. 0.5 m
B. 1 m
C. 2 m
D. 3 m

Solution: $\vec{S} = \text{Area} = \frac{1}{2}(2)(3) - \frac{1}{2}(1)(2) + (1)(1) = 3\text{m}$ **Answer: D**

251. The displacement – time graph of two moving particles make angles of 30° and 45° with the x – axis. The ratio of the two velocities is

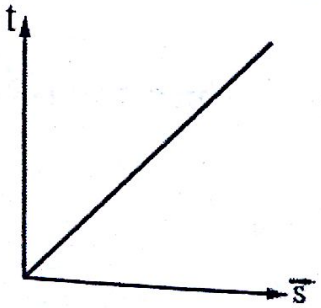


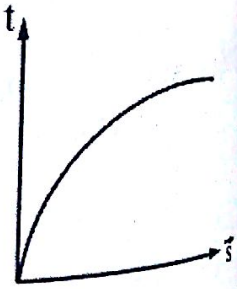
A. $\sqrt{3}:1$ C. $1:2$
B. $1:1$ D. $1:\sqrt{3}$

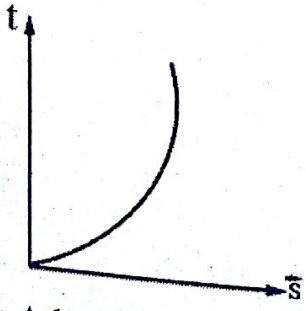
Solution: $\frac{v_1}{v_2} = \frac{\tan 30°}{\tan 45°} = \frac{\frac{1}{\sqrt{3}}}{1} = 1:\sqrt{3}$

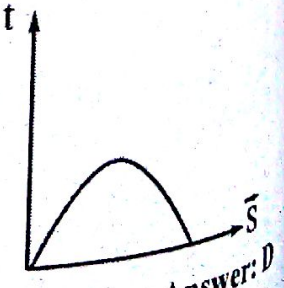
Answer: D

252. Which of the following represents an impossible situation?

A)


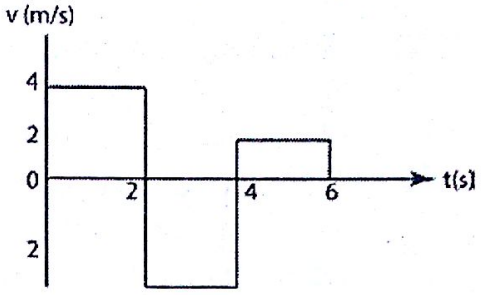
B)


C)


D)


Hint: A body cannot be at different places at one time. **Answer: D**

253. The velocity – time graph of a body moving in a straight line is shown. The displacement and distance travelled by the body in 6 sec are respectively.



A. 8m, 16 m
B. 16, 8 m
C. 16 m, 16 m
D. 8 m, 8 m

Solution:

$\vec{S} = (2)(4) - (2)(2) + (2)(2) = 8m$

S = (2) (4) + (2) (2) + (2) (2) = 16 m **Answer: A**

254. The magnitude of the acceleration of a moving object is equal to the

A. gradient of a displacement – time graph
B. gradient of a velocity – time graph
C. area below a displacement – time graph
D. area below a velocity – time graph **Answer: B**

Projectile Motion

255. If the initial velocity of a projectile be doubled, keeping the angle of projection same, the maximum height reached by it will

A. be halved C. be quad rupled
B. be doubled D. remain the same

Explanation: $h_{max} = \frac{u^2 \sin^2\theta}{2g} \Rightarrow h_{max} \sim u^2$ **Answer: C**

256. A particle is projected such that the horizontal range and vertical height are the same. Then the angle of projection is

A. $\tan^{-1}(4)$ B. $\tan^{-1}\left(\frac{1}{4}\right)$ C. $\frac{\pi}{4}$ D. $\frac{\pi}{3}$

Solution: $R = \frac{u^2\ 2\sin\theta\cos\theta}{g}$ and $h_{max} = \frac{u^2 \sin^2\theta}{2g}$

for $R = h_{max}$, $\tan\theta = 4$ or $\theta = \tan^{-1}(4)$ **Answer: A**

257. A projectile fired with initial velocity u at some angle θ has a range R. If the initial velocity be doubled at the same angle of projection, then the range will be?

A. R/2 B. R C. 2 R D. 4R

Solution: $R = \frac{u^2 \sin(2\theta)}{g} \Rightarrow R \sim u^2$

Answer: D

258. At the highest point on the trajectory of a projectile, its
A. potential energy is minimum
B. kinetic energy is maximum
C. total energy is maximum
D. kinetic energy is minimum

Hint: velocity is minimum at high point.

Answer: D

259. One stone is projected horizontally from a 20m high cliff with an initial speed of 10m/s. A second stone is simultaneously dropped from that cliff. Which of the following is true?
A. Both strike the ground with the same velocity
B. The ball with initial speed 10m/s reaches the ground first
C. Both hit the ground at same time
D. The second stone will reach first.

Answer: C

260. Two bullets are fired horizontally, simultaneously and with different velocities from the same place. Which bullet will hit the ground earlier?
A. It would depend upon the weight of the bullets
B. The slower one
C. The faster one
D. Both will reach simultaneously

Hint: $t = \sqrt{\frac{2y}{g}}$

Answer: D

261. In the case of a projectile fired at an angle equally inclined to the horizontal and vertical with velocity u, the range is

A. $\frac{u^2}{g}$ B. $\frac{u^2}{2g}$ C. $\frac{u^2}{3g}$ D. $\frac{u^2}{4g}$

Explanation: Here $\theta = 45^\circ$

$$R = \frac{u^2 \sin(2\theta)}{g} = \frac{u^2}{g}$$

Answer: A

262. A bomb is released by a horizontally flying aero plane. The trajectory of bomb is
A. a parabola
B. a straight line
C. a circle
D. a hyperbola

Answer: A

263. A stone is thrown at an angel and returns to the ground describing a parabolic path, which of the following remains constant?
A. speed of the stone
B. kinetic energy of the stone
C. vertical component of velocity
D. horizontal component of velocity **Answer: D**

264. An object is thrown along a direction inclined at an angle of 45° with the horizontal direction. The horizontal range of the particle is
A. four times the height C. twice the height
B. thrice the height D. equal to the height

Solution: $R=\frac{u^2\sin(2\theta)}{g}=\frac{u^2}{g}$ and $h=\frac{u^2\sin^2\theta}{2g}, \sin 45°=\frac{1}{\sqrt{2}}$

$\Rightarrow R = 4\, h_{max}$ **Answer: A**

265. A projectile thrown with a velocity u at an angle θ has a range R on earth. For same u and θ, its range on the moon is
A. 36R B. R/36 C. R/6 D. 6R

Explanation: on the moon, $g_m = \frac{1}{6}g$

$$R=\frac{u^2\sin(2\theta)}{g} \Rightarrow R \sim \frac{1}{g}$$ **Answer: D**

266. A projectile rises to a height of 10m above and then falls at a distance 30m away from the point of projection. Its vertical displacement is
A. 10m B. 20m C. 30m D. zero
Answer: D

267. If air resistance is ignored, then the horizontal motion of the inclined projectile takes place at
A. uniform acceleration C. uniform retardation
B. variable acceleration D. uniform velocity
Answer: D

268. A bag is dropped from an aero plane flying horizontally at a constant speed. If air resistance is ignored, where will the aeroplane be when the bag hits the ground?
A. ahead of the bag C. directly above the bag
B. far behind the bag D. data is not sufficient
Answer: C

269. A hunter takes an aim at a monkey sitting on a tree and fires a bullet. Just when the bullet leaves the barrel of the gun, it so happens that the monkey begins to fall freely. The bullet will
A. go above the monkey C. hit the monkey
B. go below the monkey D. information is incomplete

Hint: from $y = x\tan\theta - \frac{1}{2}gt^2 = x\tan\theta\frac{-gx^2}{2u^2\cos^2\theta} \Rightarrow \frac{x}{t} = u\cos\theta$

Answer: C

270. Two projectile P and Q are thrown with the same speed but at angles of 40° and 50° with the horizontal. Range of P will be
A. equal to that of Q C. greater than that of Q
B. less than that of Q D. ¾ time that of Q

Explanation: For Q and $90 - \theta$, range is the same in our case, $40^\circ + 500 = 90^\circ$. Range the same

Answer: A

271. If air resistance is ignored, then which of the following factors determine the time of flight of the projectile?
A. initial velocity
B. horizontal component of initial velocity
C. vertical component of initial velocity
D. horizontal component of initial velocity

Explanation: $t = \frac{2\,u\sin\theta}{g}$

Answer: A

272. The maximum height attained by an inclined projectile does not depend upon
A. velocity of projection C. acceleration due to gravity
B. angle of projection D. mass of projectile

Hint: $\text{hma}\,x = \frac{u^2\sin^2\theta}{2g}$

Answer: D

273. A body is thrown with a velocity of 10m/s at an angle of 60° with the horizontal. Its velocity at the highest point is
A. 7 m/s B. 9m/s C. 18.7m/s D. 5 m/s

Answer: D

274. A ball is projected upward. Its acceleration at the highest point is
A. infinite C. directed upwards
B. zero D. directed down

Answer: D

275. The greatest height to which a man can throw a ball is h. what is the greatest distance to which he can throw the ball?

A. h/4 B. h/2 C. h D. 2h

Explanation: $u = \sqrt{2gh}$, greatest distance $= \frac{u^2}{g} = \frac{2gh}{g} = 2h$

Answer: D

276. An inclined projectile has the smallest speed on its trajectory when it is

A. at the highest point
B. at the point of projection
C. hitting the ground
D. having equal range and height

Answer: A

277. What is the acceleration of a projectile at its highest point?

A. maximum B. minimum C. zero D. g

Answer: D

278. A body is projected at an angle of 30° with the horizontal with momentum P. At its highest point, the momentum is

A. P B. P/2 C. $\frac{\sqrt{3}}{2}$ D. $\frac{2}{\sqrt{3}}P$

Solution: $P = mv$ and $P = mu\cos 30° = \frac{\sqrt{3}}{2}mv = \frac{\sqrt{3}}{2}P$

Answer: C

279. In the case of an inclined projectile, the velocity is perpendicular to acceleration

A. once only B. twice
C. thrice D. four times

Explanation: At the highest point the velocity is perpendicular to the acceleration

Answer: A

280. A ball rolls off the top of a stairway horizontally with a velocity of 4.5 m/s. each steps is 0.2m high and 0.3 m wide. Which step will the ball strike?

A. 9^{th} B. 10^{th} C. 11^{th} D. 12^{th}

Solution: $y = \frac{1}{2}gt^2 \Rightarrow 0.2n = 5t$ ..... (*)

$x = ut \Rightarrow 0.3n = 4.5t^2$ ..... (**)

From the two equations, $t = 0.6$ and $n = 9$ **Answer: A**

281. A body of mass 1 kg is thrown with a velocity of 10m/s at an angle of 60° with the horizontal. Its momentum at the highest point is?

A. 2 kgm/s C. 4 kgm/s
B. 3 kgm/s D. 5 kgm/s

Solution: $P = mu\cos\theta = (1)(10\cos 60°) = 5\text{kgm/s}$

Answer: D

282. A body is projected at an angle θ to the horizontal with kinetic energy E. The kinetic energy of the body at the topmost point of the path will be

A. zero B. $E\cos\theta$ C. $E\cos^2\theta$ D. E

Solution: $K_E = \frac{1}{2}mv_x^2 = \frac{1}{2}mv^2\cos^2\theta = E\cos^2\theta$ **Answer: C**

283. A helicopter is flying horizontally at an altitude of 2km with a speed of 100m/s. A packet is dropped from it. The horizontal distance between the point where the packet is dropped and the point where it hits the ground is

A. 2 km B. 0.2 km C. 20 km D. 4 km

Solution: $y = \frac{1}{2}gt^2 \Rightarrow t = 20\text{sec}$

$x = ut = 2000\text{ m} = 2\text{km}$ **Answer: A**

284. A projectile is fired with kinetic energy of 800 J. If it has the maximum possible range, then its kinetic energy at the highest point will be

A. 800 J B. 400 J C. 200 J D. 100 J

Solution: R is maximum at 45°

$$K_E = \frac{1}{2}m(v\cos 45°)^2 = \frac{1}{2}\left(\frac{1}{2}mv^2\right) = \frac{1}{2}(800) = 400$$

Answer: B

285. A body is projected at an angle of 30° with the horizontal with kinetic energy E. The kinetic energy at the top most point is

A. E B. E/2 C. E/4 D. $\frac{3}{4}E$

Solution: $K_E = \frac{1}{2}m(v\cos 30°)^2 = \frac{3}{4}\left(\frac{1}{2}mv^2\right) = \frac{3}{4}E$ **Answer: D**

286. Two projectiles, one fired from earth with 5 m/s and the other from a planet with 3 m/s trace identical trajectories. If $g = 9.8 m/s^2$ on earth then the value of g on the planet is?

A. $1.5 m/s^2$ B. $3.5 m/s^2$ C. $7.5 m/s^2$ D. $9.5 m/s^2$

Solution: From, $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

$$\frac{g}{u^2} = \frac{g_p}{u_p^2} \Rightarrow \frac{g_p}{3^2} = \frac{9.8}{52} \Rightarrow gp = 3.5 m/s^2$$ **Answer: B**

287. A plane flying horizontally at 98 m/s releases an object which reaches the ground in 10 second. The angle made by the velocity of the object with the horizontal at the time of impact is

A. 30° B. 45° C. 60° D. 75°

Solution: $\tan\theta = \frac{v_y}{v_x} = \frac{gt}{u} = \frac{(9.8)(10)}{98} = 1 \Rightarrow \theta = 45^\circ$ **Answer: B**

288. An arrow is shot into air. If its range is 200m and its time of flight is 5 sec. the horizontal component of the arrow velocity is?

A. 12.5 m/s B. 25 m/s C. 31.25 m/s D. 40 m/s

Solution: $R = (u\cos\theta)\frac{(2\,u\sin\theta)}{g} \Rightarrow 200 = (u_x)(5) \Rightarrow u_x = 40 m/s$

Answer: D

289. If the height attained by a projectile is the greatest, then which of the following is the maximum?

A. range C. force of gravity

B. angle of projection D. time of flight

Answer: D

290. The velocity of an inclined projectile is $(6i + 8j)$ m/s. The horizontal range of the projectile is

A. 4.9m B. 9.6m C. 19.6m D. 14m

Hint: $\sin\theta = \frac{8}{10}, \cos\theta = \frac{6}{10}$ and $R = \frac{2u^2\sin\theta\cos\theta}{g}$ **Answer: B**

291. A marble is to be thrown horizontally from a height of 19.6 cm above the ground so that it hits another marble on the ground 2m away. The velocity of throw should be

A. 5 m/s B. 10 m/s C. 15 m/s D. 20 m/s

Solution:

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$y = \frac{1}{2}gt^2 \Rightarrow t = 0.2$ sec and $x = ut \Rightarrow u = \frac{2}{0.2} = 10$m/s **Answer: B**

292. The range of a projectile fired at 15° is 40m. If it is fired with same speed at an angle of 45°, its range will be?
A. 60 m B. 70 m C. 80m D. 100m

Explanation: $R = \frac{u^2\sin^2\theta}{g} \Rightarrow \frac{R'}{R} = \frac{\sin 90°}{\sin 30°} \Rightarrow R^1 = 2R = 80m$

Answer: C

293. A man can throw a stone 100m horizontally. The maximum height to which he can throw vertically is
A. 200 m B. 100m C. 50m D. 400m

Solution: $\frac{v^2}{g} = 100$, $h_{max} = \frac{u^2}{2g} = 50m$ **Answer: C**

294. A ball is projected horizontally at 20 m/s. the speed of the ball after 5 sec is nearly?
A. 24 m/s B. 34 m/s C. 44 m/s D. 54 m/s

Hint: $v = \sqrt{v_x^2 + v_y^2}$, $v_x = u$, $v_y = gt$ **Answer: D**

295. The velocity of projection of an inclined projective is $\vec{v} = (3i + 2j)$ m/s. The speed of the projectile at the highest point of the trajectory is?
A. 3m/s B. 2 m/s C. 1 m/s D. zero

Answer: A

296. Five balls A, B, C, D and E are projected with the same speed making angles of 10°, 30°, 45°, 60° and 80° respectively, with the horizontal. Which ball will strike the ground at the farthest point?
A. B B. A C. E D. D E. D

Answer: D

297. Three particles of masses 1g, 2g and 3g are dropped from different heights 1m, 2m and 3m respectively. Now
A. particle of mass 1g has maximum acceleration
B. particle of mass 2g has maximum acceleration
C. particle of mass 3g has maximum acceleration
D. all the particles have equal acceleration

Answer: D

298. A train covers the first half of the distance between two stations at a speed of 40km/h and the other half at 60km/h. The average speed of the train is

A. 48 km/h B. 50 km/h C. 52 km/h D. 54 km/h

Solution: $V_{av} = \frac{2v_1v_2}{v_1 + v_2} = 48\text{km/h}$ **Answer: A**

299. A food packet is released from a helicopter rising steadily at 2 m/s. What is the velocity of the packet after 2 sec?

A. 18 m/s B. 20 m/sC. 22 m/s D. 98 m/s

Solution: $v = -2 + (10)(2) = 18$ m/s **Answer: A**

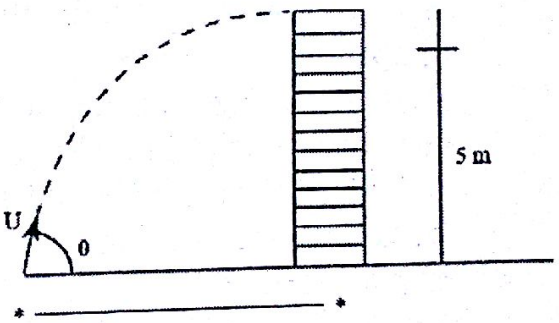
300. A ball thrown vertically upward has a speed of 10 m/s when it has reached one half of its maximum height, how high does the ball rise

A. 20 m B. 15 m C. 5 m D. 10 m

Solution: $v^2 - u^2 = 2gh = 0^2 - 10^2 = -2(10)\,h/2 \Rightarrow h = 10\text{m}$

Answer: D

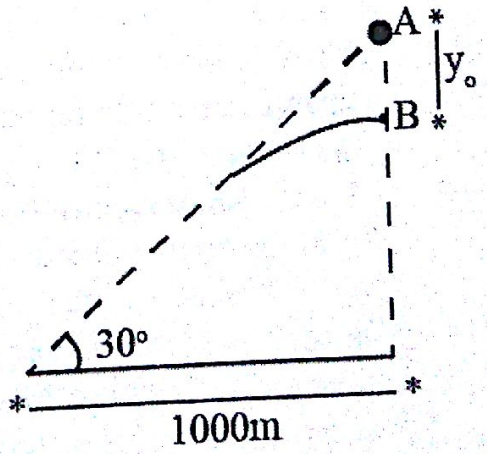
301. A ball is thrown upward at angle 30° to the horizontal and lands on the top edge of a building that is 20m away and 5m high. How fast was the ball thrown?



A. 10 m/s
B. 20 m/s
C. 40 m/s
D. 80 m/s

Hint: $y = x\tan\theta\frac{-gx^2}{2u^2\cos^2\theta} \Rightarrow u = 20$ m/s **Answer: B**

302. A shooter aims his rifle at an angle of 30° with the horizontal to hit an object at A but the bullet hits at point B, y_o below A. If the initial velocity of the bullet is 600 m/s, the valve of y_o is



A. 1.85 m
B. 18. 5m
C. 185 m
D. 1850 m

Solution: time taken to reach point A

$$t = \frac{x}{u\cos\theta} = \frac{1000}{600\left(\frac{\sqrt{3}}{2}\right)} = \frac{10}{3\sqrt{3}}\sec$$

Vertical deflection of the bullet in this time

$$y_o = \frac{1}{2}gt^2 = \frac{1}{2}(10)\left(\frac{10}{3\sqrt{3}}\right)^2 = 18.5\text{m}$$

Answer: B

Unit 3: Conceptual Questions

1. Can the instantaneous velocity of a body be ever be greater or less than the average velocity of the body?
2. Can the velocity of a body reverse direction when the body still have constant acceleration?
3. If the average velocity equals zero for some time interval can we say that the instantaneous velocity is also zero at some time in this interval? Explain
4. Is there a possibility for the velocity and acceleration to have opposite signs? Explain.
5. What is wrong with the statement "A car rounds the curve at a constant velocity of 60 km/hr"
6. A body throws a ball into the air and another boy drops a ball from the same height. Which ball has greater acceleration? Explain
7. You throw a ball up from a cliff and another time you throw the ball down. In which case will the ball landed with greater speed? Explain
8. Suppose you throw one ball upward and another down from a building. Which ball will have greater displacement when it reaches the ground?
9. A ball is dropped from the top of a building and a second ball is thrown upward. Which ball will have greater speed when they pass each other?
10. A ball is launched on earth, and then it is launched on the moon with the same velocity and angle of projection, where will it have
    a. Greater range b. Greater height
11. A woman walking on a road at 3km/h encounters rain falling vertically at 18 km/h. at what angle should she hold the umbrella to protect herself from the rain?

12. For a certain projectile, the range is twice the maximum height attained. For what angle of projection is this happening

Unit 3: Problems

Average and Instantaneous velocity

13. If a body moved distance S_1 with speed V_1 and distance S_2 with speed V_2. Show that its average speed is

$$V_{av} = v_1 v_2 \left(\frac{s_1 + s_2}{s_1 v_2 + s_2 v_1} \right)$$

14. The position of a body as a function of time is given by $S(t) = 2t^2 - 5t - 20$. What is the instantaneous velocity of the body at $t = 2$ sec

Average and Instantaneous Acceleration

15. A car traveling at 5m/s starts to speed up. After 3 sec. its velocity has increased to 11m/s. what is the acceleration of the car?
16. The velocity of a particle as a function of time is given by $v(t) = 3t^2 - 4t$, What is the instantaneous acceleration of the particle at $t = 1$ sec.

Motion with constant Acceleration

17. An Aeroplane taxis onto the runway going at 10m/s. If it can accelerate steadily at $3m/s^2$ and its take off speed is 90 m/s. what length of runway will it need?
18. A city bus starts from rest and accelerates at $2m/s^2$ till it reaches a speed of 20 m/s. It then travels for 20 sec at constant speed until the brakes are applied and stop the bus in an additional 5 sec.
    a. How long is the bus in motion
    b. What is the average velocity of the bus?

Freely Falling Bodies

19. A stone dropped from rest down a deep well, and 3 sec later a splash is heard as it hits the water below.
    a. how fast will it be moving as it hits the water?
    b. what will its average speed over the three seconds?
    c. how deep is the well?
20. A ball is thrown at a velocity of 15 m/s vertically down
    a. what height will the ball reach
    b. what is its time of flight?
21. A ball dropped from a bridge strikes the ground in 5 sec. What is
    a. The speed with which it strike the ground
    b. The height of the bridge

22. A body standing on the edge of 180m cliff dropped a tone and one second later he threw another stone down ward from the same height. If the stones touch the ground at the same time. Find
    a. the time elapsed by each stone
    b. initial velocity of the second stone
    c. final velocities of the stones when they strike the ground
23. A ball is thrown up with 10m/s. One second later, a stone is thrown upward with 20m/s. Find
    a. the time the stone takes to overtake the ball
    b. the height where the stone and the ball meet up

Graphical representation of Motion

24. Abeba walks to school. She walks 1 km in 15 min. She meets her friend Makeda, they talk for 5 min and then carry on walking. They walk 800m in 10 min
    a. Draw a $\vec{s}-t$ graph to show Abeba's journey
    b. What was the average velocity of Abeba
    c. When was she walking the fastest?
25. A car starts from rest and moves for 2hrs until its velocity reaches 50 km/hr which is maintained for 1 hr. In the next 1hr, its velocity is uniformly decreasing until it becomes zero. It then continues Moving in the opposite direction for 2 hrs until its velocity is 75 km/hr south.
    a. Draw the $\vec{v}-t$ graph of the motion
    b. Calculate the total distance moved

Projectile Motion

26. An aircraft flying at a steady velocity of 70m/s east ward at a height of 800m drops a package on supplies.
    a. How long will it take for the package to reach the ground?
    b. How fast will it be going as it lands?
27. A stone is thrown with an initial velocity of 25 m/s at an angle of 30° from the ground.
    a. What will be its vertical velocity when it again reaches the ground?
    b. What will be the stone's horizontal velocity just before it lands?
    c. Workout the stone's range
28. A stone is thrown upward with a speed of 20 m/s at an angle of 37° from the ground. Find
    a. The position (x, y) of the stone after 2 sec.

b. The time taken to reach the maximum height
c. The maximum height of the stone
d. The range of the stone

29. A projectile is fired with a horizontal velocity of 12m/s from the top of a cliff 180m high
a. How long will it take to hit the ground?
b. How far from the cliff will it land?
c. With what velocity will it land?

Uniform Circular Motion

30. A tire 0.5m in radius rotates at a constant speed of 10.5m/s. Find the acceleration of a small stone embedded in the tire.
31. A satellite moves in a circular orbit 640 km above the Earth's surface. If the period of its motion is 98min. What is
a. the speed of the satellite
b. the acceleration at that height
32. a) What is the centripetal acceleration of a body on the equator owing to the rotation of the earth?
b) What is the period of rotation of the earth if the centripetal acceleration is $10m/s^2$?

Relative Velocity

33. A river flows at 2m/s. The velocity of ferry relative to the shore is 4m/s at right angle to the current. What is the velocity of the ferry to the current?
34. An elevator is moving upward at a constant speed of 4m/s. A piece of iron falls from the coiling of the elevator. What is the velocity of the iron at that instant to a man in the elevator?

Unit 3: Explanations to Questions

1. Yes
2. Yes. A body thrown up will change direction of motion after a maximum height, while its acceleration is still g.
3. No.
4. Yes. While slowing down
5. The direction is changing. The velocity is varying
6. Both have the same acceleration, $g = 9.8\ m/s^2$
7. Same speed both have
8. Both have the same displacement
9. The have the same displacement
10. a. $R=\frac{u^2\sin^2\theta}{g}$, R ↑ when g ↓ Greater R on the moon

b. $h_{max} = \frac{u^2 \sin^2\theta}{2g}$, $h_{max} \uparrow$ when $g \downarrow$. Grater h_{max} on the moon

11. $\tan\theta = \frac{v_w}{v_r} = \frac{3}{18} \Rightarrow \theta = 9.5°$ from the vertical

12. $R = 2h_{max} = \frac{2u^2 \sin\cos\theta}{g} = \frac{2u^2 \sin^2\theta}{2g} \Rightarrow \tan\theta = 2, \theta = 63°$

Unit 3: Solution to Problem

13. $V_{av} = \frac{S_T}{t_T} = \frac{s_1 + s_2}{t_1 + t_2}$, $t_1 = \frac{S_1}{V_1}$ and $t_2 = \frac{S_2}{V_2}$. substitute these

14. $v = \frac{ds}{dt} = 4t - 5$, at $t = 2sec$, $v_{inst} = 3m/s$

15. $a = \frac{v-u}{t} = \frac{11-5}{3} = \frac{3}{3} = 1m/s^2$

16. $a = \frac{dv}{dt} = 6t - 4$, at $t = 1sec$, $a_{inst} = 2m/s^2$

17. $v_2 = u_2 - 2as \Rightarrow s = 1.3$ km

18. a. $t = t_1 + t_2 + t_3 = 10 + 20 + 5 = 35$ sec

b. $V_{av} = \frac{\Delta s}{\Delta t} = \frac{1}{2}a_1 t_1^2 + vt_2 + vt_3 + \frac{1}{2}a^2 \frac{t_3^2}{\Delta t} = 15.7m/s$

19. a. $v = gt = (10)(3) = 30$ m/s

c. $Y = \frac{1}{2}gt^2 = 5(3)^2 = 45m$

b. $V_{av} = \frac{u+v}{2} = \frac{0+30}{2} = 15m/s$

20. a. $h = \frac{u^2}{2g} = \frac{(15)^2}{2(10)} = 11.25m$ b. $t = \frac{u}{g} = \frac{15}{10} = 1.5sec$

21. a. $v = gt = (10)(5) = 50$ m/s

b. $y = \frac{1}{2}gt^2 = 5(5)^2 = 125m$

22. a. $y = ut_1 + 5t^2 \Rightarrow 180 = 5t^2$ or $t_1 = 6sec$ and $t_2 = 5sec$

b. $y = u_2 t_2 + 5t_2^2 \Rightarrow 180 = 5u_2 + 5(5)^2 \Rightarrow u_2 = 11m/s$

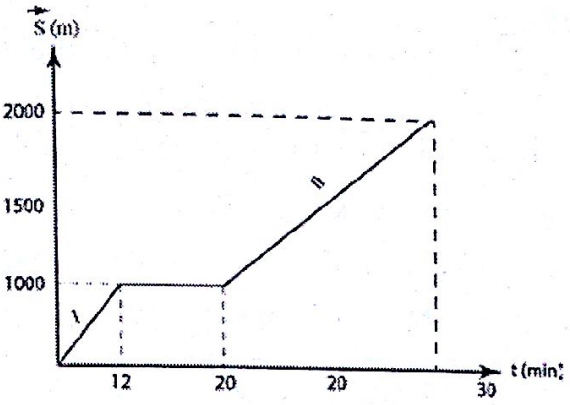
c. $v_1 = gt_1 = (10)(6) = 60m/s$ and $v_2 = u_2 + gt_2 = 11 + (10)(5) =$ 61m/s

23. a. $y_b = y_s \Rightarrow u_b t_b - \frac{1}{2} g t_b^{\,2} = u_s t_s^{\,2} - \frac{1}{2} g t_s^{\,2}$, but $t_s = t_{b-1}$

Thus, $t_s = 0.25$ sec

b. $y_s = y_b = u_s t_s - \frac{1}{2} g t_s^2 = 20(0.25) - 5(0.25)^2 = 4.7\text{m}$

24. a.



b.

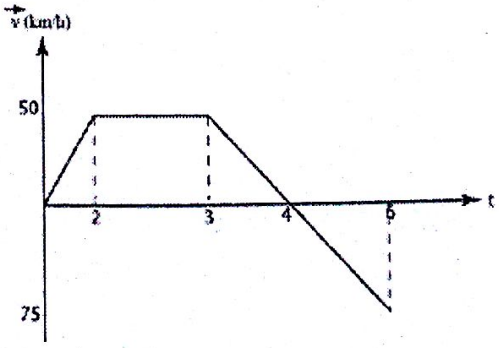
$\vec{V}_{av} = \frac{\Delta s}{\Delta t} = \frac{1800}{1800} = 1\text{m/s}$

c. $\vec{V}_{avI} = \frac{1000}{900} = 1.11\text{m/s}$

$V_{avII} = \frac{800}{600} = 1.33\text{m/s}$

Which is fast

25. a.



b. $A = A_1 + A_2 + A_3 + A_4 = \frac{1}{2}(2)(50) + (1)(50) + \frac{1}{2}(1)(50) + \frac{1}{2}(2)(75)$

$S = A = 200$ km

26. a. $y = \frac{1}{2} g t^2 \Rightarrow 800 = 5t^2 \Rightarrow t = 12.6\text{sec}$

b. $v = ui + (gt)j = (70i - 126j)$ m/s $= 144$ m/s

27. a $v_y = u\sin\theta = -12\text{m/s}$, b. $v_x = u\cos\theta = 21.7\text{m/s}$ c. $R = 54$ m

28. a. $x = (u\cos\theta)(t) = 32m$, $y = (u\sin\theta)t - \frac{1}{2}gt^2 = 4m$

b. $t_{max} = \frac{u\sin\theta}{g} = 1.2sec$, c. $h_{max} = \frac{u^2\sin^2\theta}{2g} = 7m$

d. $R = \frac{u^2\sin^2\theta}{g} = 38.4m$

29. a. $t = \sqrt{\frac{2y}{g}} = 6sec$ b. $x = ut = 72\ m$ c. $v = \sqrt{v_x^2 + v_y^2} = 61m/s$

30. $a_c = 220.5m/s^2$

31. a. $v = \frac{2\pi r}{T} = 7.5km/s$ b. $a_c = g = 8m/s^2$

32. a. $0.034\ m/s^2$ b. 84 min

33. $(-4, 2) = 4.5$ m/s, 63° to the current

34. $V = 0 - 4 = -4m/s$

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Unit 4

Dynamics

All of us are familiars with the word force as we use it in our everyday life to stop motion, to start motion, to change direction or to change size or to change shape of a body.

Dynamics: Is the study of cause of motion, and come from the Greek Word 'dynamikos' meaning force or power.

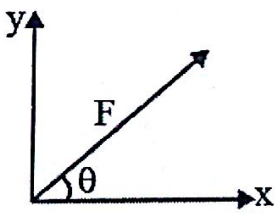
4.1 The Force Concept

Force: Is a pull or a push exerted on a body which changes the state of motion, or shape and size of a body.

- Force is a vector quantity measured in Newton (N) and causes a body to move, to stop, to change direction of motion or to change size or shape of the body.

Resolving a Force

Since force is a vector quantity, it can be decomposed into components.



i) A force in xy plane can be resolved as

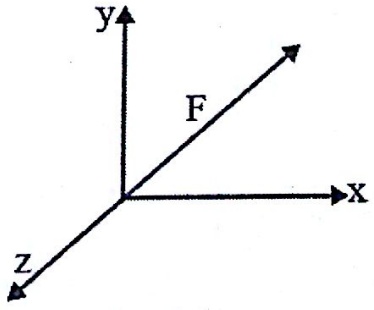
$$\vec{F} = F_x i + F_y j$$

The magnitude of the force is,

$$F = \sqrt{F_x^2 + F_y^2}$$

The angle the force make from the horizontal is,

$$\tan\theta = \frac{F_y}{F_x} \Rightarrow \theta = \tan^{-1}\left(\frac{F_y}{F_x}\right)$$



ii) A force in space, that is xyz plane can be resolved as:

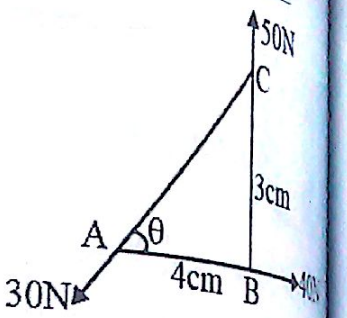
$$\vec{F} = F_x i + F_y j + F_z k$$

And, its magnitude is

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2}$$

Illustrative Example

1. Triangle ABC has sides AB = 4cm, BC = 3cm and forces 40N, 50N and 30N act along sides AB, BC and CA respectively. Determine the resultant of these systems of forces.



Solution: Resolving forces horizontally

$F_x = -30\cos\theta + 40 = -30\left(\frac{4}{5}\right) + 40 = 16N$

Resolving forces vertically

$F_y = -30\sin\theta + 50 = -30\left(\frac{3}{5}\right) + 50 = 32N$

$\Rightarrow F_y = \sqrt{F_x^2 + F_y^2} = \sqrt{(16)^2 + (32)^2} = 35.8N$

4.2 Basic Laws of Dynamics

Newton's law of motions are the basic laws of dynamics. It was in 168 that a great English Physicist Sir Issac Newton Summarized the basic principles which govern the motion of a body in the form of three laws. These laws are known as *Newton's Laws of motion* and are as follows.

First Law or Law of Inertia, states that; "A body will continue in its state of rest or uniform motion unless a net force acts on it."

☛ The first law implies that, in the absence of external force, a body at rest will remain at rest and a body moving with constant speed in a straight line will continue moving at that speed forever.

That is,

- A book lying on a table remain there forever, and
- A ball given a constant velocity will move forever.

☛ **Important points to remember**

- **Inertia:** Is the natural tendency of a body to remain at rest or in motion at a constant velocity. The mass of a body is measure of inertia. Therefore, the bigger the mass the greater the inertia.
- Newton's first law is also known as the *law of inertia* which is appropriate since the literal meaning of the word inertia is "Laziness". Speaking loosely, we can say that

matter is "Lazy". In that it will not change its motion unless forced to do so. If an object is at rest, it won't start moving on its own. If an object is already moving with constant velocity, it won't alter its speed or direction, unless a force causes the change.

- According to Newton's first law, being at rest and moving with constant velocity are actually equivalent. Because in both cases, the net force on an object is zero.
- Newton's laws are valid only in inertia frame of reference we can assume the Earth as an inertia frame if we neglect its rotation. And a body moving with constant velocity can be taken as an inertial frame. However, an accelerating body is a non – inertia frame and hence, Newton's laws are not valid there.

Conceptual Examples

2. In a certain scientific film, a spaceship moving in vacuum stops when its engine dies. Can this event happen in reality?
   Explanation: No. in vacuum, there is no external force such as friction, drag force or viscosity. Therefore, the spaceship cannot stop.
3. When a car stops, suddenly, the passengers tend to move forward and when a car makes a sharp turn, the passenger tend to slide to one side of the car. Why is this happening?
   Explanation: This is due to inertia. The passenger's body wants to be in its original state of motion.
4. Is it possible to have motion in the absence of a force?
   Explanation: Yes. Motion requires no force.
5. While dusting a carpet, we give a sudden jerk or beat it with a stick. Explain how dust particles are removed, from the carpet in this process.
   Explanation: Before the carpet was beaten, the carpet and the dust particles were at rest. When the carpet is given a sudden jerk, it moves away and due to inertia of rest dust particles remain in their position.

Newton's Second Law or Law of Acceleration: states that; "The acceleration of a body is directly proportional to the net force acting on the body and inversely proportional to the mass of the body."

Mathematically, we can write it as,

$$\boxed{F = ma}$$

In vector form, we can have,

$$\boxed{\vec{F} = m\vec{a} = m(\vec{a}_x + \vec{a}_y + \vec{a}_z)}, \quad F_x = ma_x,\ F_y = ma_y,\ F_z = ma_z$$

$$\vec{F} = \vec{F}_x i + \vec{F}_y j + \vec{F}_z k \text{ and } \vec{a} = a_x i + a_y j + a_z k$$

☛ **Important point to remember**

- The direction of the acceleration is always in the direction of the net force.
- When solving problems involving forces and Newton's second law, it is essential to begin by making a sketch that indicates each and every external force acting on the given body. This type of sketch is referred to as a *free – body diagram*.
- The second law can also be stated as "the rate of change" of momentum is directly proportional to the force, $F = \Delta P / \Delta t$.

Conceptual Examples

6. If gold were sold by weight, would you prefer to buy it in Dalule or in Ras Dashen? Why?

   Explanation: In Ras Dashen. Because, g is smaller there.

7. Is it possible for an object to be moving in one direction. While the net force acting on it is in another direction? Explain

   Explanation: Yes. Consider a ball thrown upward.

Illustrative Examples

8. A certain force gives mass m_1 an acceleration of 12 m/s^2 and mass m_2 an acceleration of 3.3. m/s^2. What acceleration will this force give to the combined mass?

   Solution: $m_1 a_1 = m_2 a_2 \Rightarrow m_2 = \frac{a_1}{a_2}(m_1) = \frac{(12)}{3.3} m_1$

$$a = \frac{F}{m_1 + m_2} = \frac{12m_1}{m_1 + \frac{12}{3.3}m_1} = \frac{(12)(3.3)m_1}{3.3m_1 + 12m_1} = 2.5 m/s^2$$

9. A force of unknown magnitude acts on a 2 kg mass and gives it an acceleration of 3 m/s^2. What acceleration will this force give to a 1 kg mass? How large is the force?

Solution: $m_1a_1 = m_2a_2,\ a_2 = \frac{m_1}{m_2}(a_1) = \frac{2}{1}(3) = 6m/s^2$ and,

$$F = m_1a_1 = m_2a_2 = (1)(6) = 6N$$

10. $F_1 = 5i - 3j + 2k$ and $F_2 = 7i + 24j - 14k$ are applied on a body of mass 3kg. What is the magnitude of the acceleration produced?

Solution: $\vec{a} = \frac{\vec{F}}{m} = \frac{(5+7)i+(-3i+24)j+(2-14)k}{3} = 4i + 7j - 4k$

$$a = \sqrt{a_x^2 + a_y^2 + a_z^2} = \sqrt{4^2 + 7^2 + (-4)^2} = 9m/s^2$$

11. A 136N force acts on a 4kg body in the direction of 8i + 15j. Find the acceleration of the body.

Solution: The unit vector in the force direction is

$$\vec{U}_F = \frac{8i+15j}{\sqrt{8^2+15^2}} = \frac{8i+15j}{17} \Rightarrow \vec{F} = |F|U_{\vec{F}} = 136\left(\frac{8i+15j}{17}\right)$$

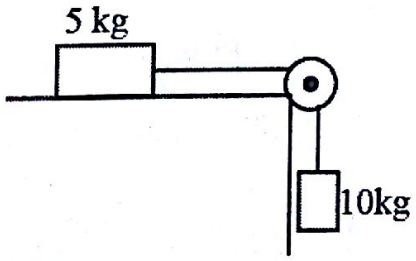
$\vec{F} = 64i + 120j$, and $\vec{a} = \frac{\vec{F}}{m} = \frac{64i+120j}{4} = 16i + 30j$ or $a = 34\ m/s^2$

12. A body moves at constant velocity of (3i – 4j)m/s when two forces, $F_1 = (2i - 6j)N$ and F_2 are acting on it. Find F_2.

Solution: $\vec{F}_1 + \vec{F}_2 = m\vec{a} = 0$, $\vec{a} = 0$, since $\vec{v}$ = constant

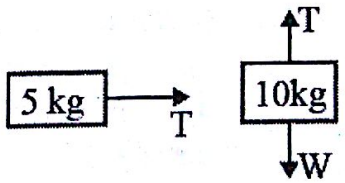
$$\Rightarrow \vec{F}_2 = -\vec{F}_1 = (2i - 6j) = (-2i + 6j)N$$

13. A body of mass 5 kg rests on a frictionless table and is connected to a cable that passes over a pulley onto a hanging body of mass 10 kg. Find



a) Acceleration of each body.
b) Tension in the cable.

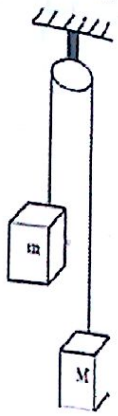
Solution: a) Free body diagram, FBD



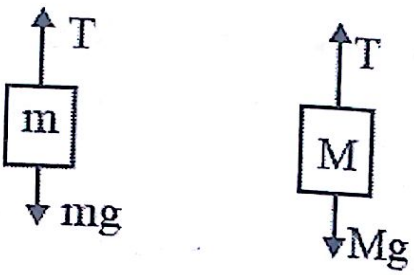
$$T = 5a$$
$$10g - T = 10a$$
$$\Rightarrow 10g - 5a = 10a \text{ or } 15a = 10g$$
$$\Rightarrow a = \frac{10g}{15} = \frac{100}{15} = 6.7m/s^2$$

b) $T = 5a = 5(6.7) = 33.5N$

14. Two blocks connected by a cord pass over a frictionless pulley. If m = 1.5kg and M = 3.5 kg, Find:
   a) The acceleration
   b) The tension in the cord



Solution: a) FBD

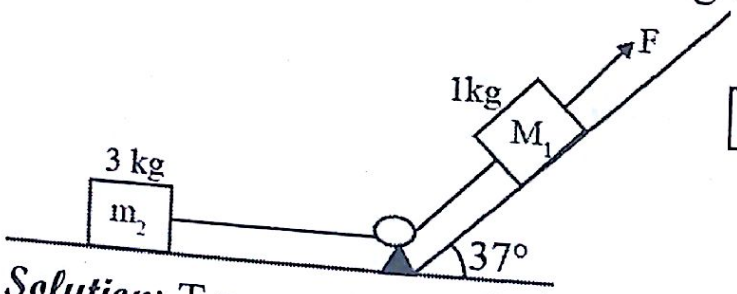


$$T - mg = ma, \text{ and } Mg - T = Ma$$

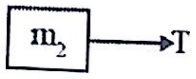
$$\Rightarrow a = \frac{(M-m)}{M+m}g; a = \left(\frac{3.5-1.5}{3.5+1.5}\right)(10) = 0.4 m/s^2$$

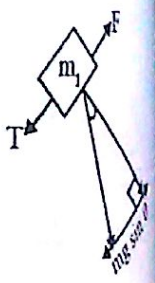
b) $T - mg = ma \Rightarrow T = m(a+g) = 1.5(0.4+10) = 15.6N$

15. Assuming the surfaces and the pulley frictionless, and F = 15N What is the tension in the connecting cord?



FBD





Solution: $T = m_2a \Rightarrow a = T/m_2$, and

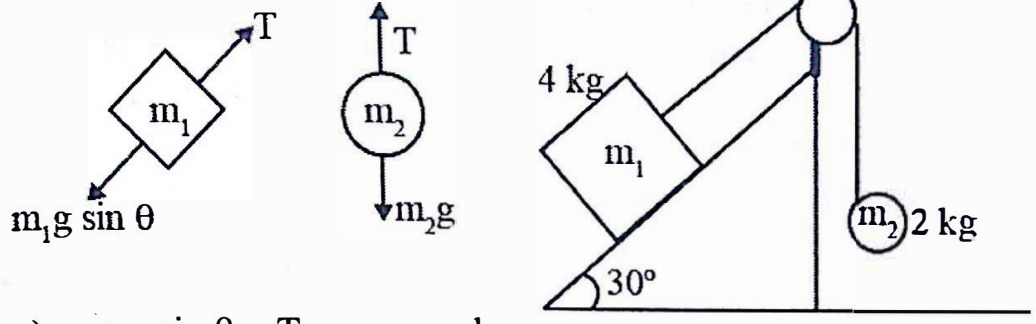
$$F - m_1 g\sin\theta - T = m_1 a$$

$$\Rightarrow F - m_1 g\sin\theta - T = m_1\left(\frac{T}{m_2}\right) \text{ or } T\left(1+\frac{m_1}{m_2}\right) = F - m_1 g\sin\theta$$

$$T = \frac{F - m_1 g\sin\theta}{1 + m_1/m_2} = \frac{15-(1)(10)(0.6)}{1+1/3} = 6.75N$$

16. Assuming that the surface and the pulley are frictionless. Find:
   a) The acceleration of the blocks.
   b) The tension in the cord.

Solution: FBD



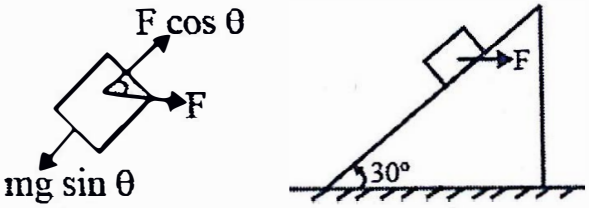
a) $m_1 g \sin\theta - T = m_1 a$, and
$T - m_2 g = m_2 a$

$$\Rightarrow a = \frac{m_1 g \sin\theta - m_2 g}{m_1 + m_2} = \frac{(4)(10)\sin 30° - (2)(10)}{2+4} = 0 m/s^2$$

b) $T - m_2 g = m_1 a \Rightarrow T = m_2 g = (2)(10) = 20N$

17. A 100 kg box is pulled up a frictionless inclined plane of $\theta = 30°$. What should be the force F to move the box up at constant speed?

Solution:



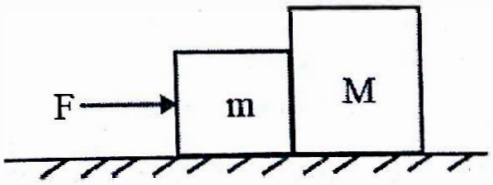
$\vec{v}$ = constant, $\vec{a} = 0$, $F\cos\theta - mg\sin\theta = 0$

Or, $$F = \frac{mg\sin\theta}{\cos\theta} = \frac{(100)(10)}{\cos 30°}\sin 30° = 577N$$

18. Two blocks m and M with masses 2 kg and 6 kg respectively are in contact on a smooth horizontal surface. If a horizontal force of 10N pushes them, what is;

a) The acceleration of the system.

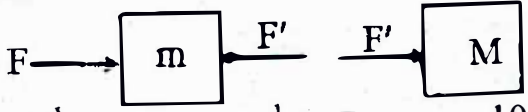
b) The force that block m exerts on block M.



Solution:

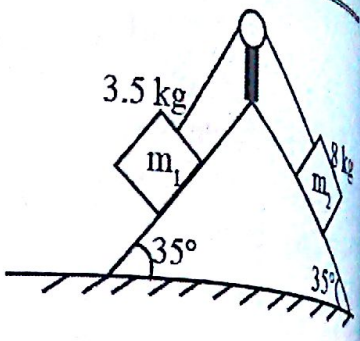
a) $F = ma = (m + M)a \Rightarrow a = \frac{F}{m+M} = \frac{10}{2+6} = 1.25 m/s^2$

b) FBD



$F - F^1 = (m)(a) \Rightarrow F^1 = F - ma = 10 - (2)(1.25) = 7.5N$

19. Two blocks of mass 3.5 kg and 8 kg are connected by a mass less string passing over a frictionless pulley. If the inclines are frictionless. Find the;



a) Acceleration of each block.
b) Tension in the string.

Solution:

a) FBD

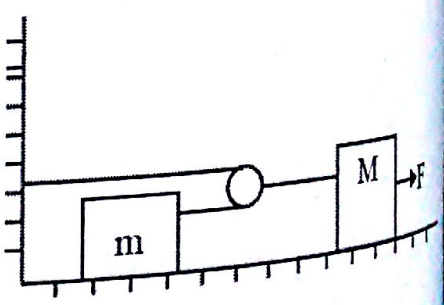
$m_2 g \sin\theta - T = m_2 a$, and

$T - m_1 g \sin\theta = m_1 a$

$$\Rightarrow a = \frac{m_2 g \sin\theta - m_1 g \sin\theta}{m_1 + m_2} = \frac{(10)(0.5)(8-3.5)}{11.5} = 2 m/s^2$$

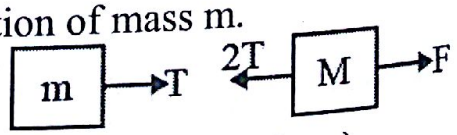
b) $T - m_1 g \sin\theta = m_1 a \Rightarrow T = m_1 a + m_1 g \sin\theta = 24.5N$

20. Assuming that the surfaces are frictionless and given that; $m = 0.2$ kg, $M = 0.3$ kg and $F = 0.4N$. Find:



a) The tension in the cord.
b) The acceleration of mass m.

Solution: a) FBD

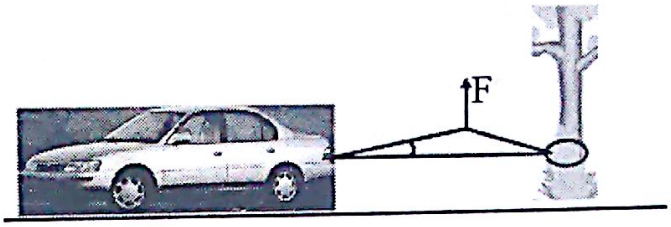


$$F - 2T = Ma_1 \Rightarrow F - 2T = M\left(\frac{1T}{2m}\right) \Rightarrow F - 2T = \frac{MT}{2m}$$

$$T = \frac{F}{\frac{M}{2m} + 2} = \frac{0.4}{\frac{0.3}{0.4} + 2} = 0.145N$$

b) $T = ma_2 \Rightarrow a_2 = \frac{T}{m} = \frac{0.145}{0.2} = 0.73 m/s^2$

21. A car is stuck in a mud. How could the driver pull out this car from the mud? Explain



Explanation: A knowledge of the parallelogram law may help the driver to pull his car out of the mud if it gets stuck. For instance, a rope is attached to the car and tied tightly to a nearby tree. If the rope makes 0.5° to the horizontal and the driver exerts a 200N force at the center, then the tension in the rope become 11500N, which is enough to pull the car out of mud.

22. Two masses 2 kg and 3 kg are attached to the end of a light string passing over a pulley. What are the tension and acceleration interms of g? (UEE 2001)

A. $\frac{12}{5}g, \frac{g}{5}$ B. $\frac{7}{8}g, \frac{g}{5}$ C. $\frac{21}{8}g, \frac{g}{8}$ D. $\frac{7}{8}g, \frac{g}{5}$

Solution: For the 2kg mass

$T - mg = ma$

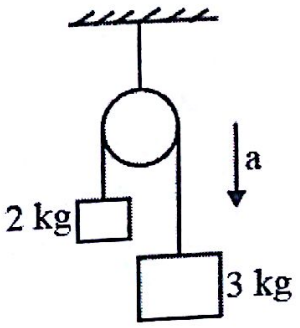
$T - 2g = 2a$ ..... (*)

For the 3 kg mass

$mg - T = ma$

$3g - T = 3a$ ..... (**)

From the two equations, $T = \frac{12}{5}g, a = \frac{g}{5}$

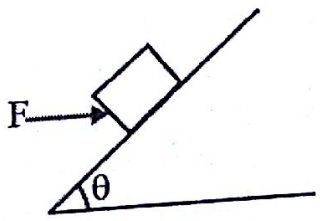


Answer: A

23. A 100kg box is pushed at constant speed up the smooth 30° inclined plane. What horizontal force is required? (UEE 2002)

A. 577.33N C. 1732.35N

B. 1000.35N D. 1154.70

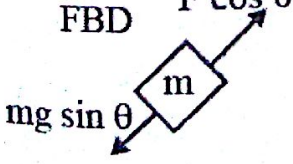


Solution: $F \cos\theta = mg \sin\theta$

$F = mg \tan\theta$

$F = (100)(10) \tan 30° = 577.35N$

FBD



Answer: A

24. Which pair of forces cannot give a resultant of 5N? (UEE: 2003)

A. 1N and 4N C. 3N and 4N

B. 1N and 6N D. 1N and 7N

Solution: The resultant of any two vectors A and B lies in the range $|A - B| \leq R \leq |A + B|$.

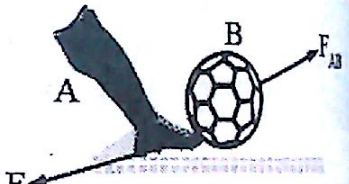
Answer: D

Newton's Thrid law or Law of Action and Reaction: States that "Whenever one body exerts a force on a second body, the second body exerts an equal and opposite force on the first."

- The third law implies that for every action, there is an equal and opposite reaction.

For example, when spring balance A exerts a 5N force on spring balance B, spring B is also exerting 5N on A.

That is if body A exerts a force on body B, then body B exerts a force on body A that is equal in magnitude and opposite in direction.



When you kick a ball it kicks you with the same force.

$$F_{AB} = -F_{BA}$$

- **Important points to remember**
  - Forces always exist in pairs, there is no isolated or single force.
  - Action and reaction forces act on different bodies. If they were to act on the same body, they would add up to zero. Since they are equal and opposite.
  - We can consider either force as the action and the other as the reaction.

Conceptual Examples

25. If a Bajaji collides head on with a Truck, which one experiences a greater acceleration?

Explanation: They experience equal forces but in opposite directions and because the bajaji has smaller mass, it experiences greater acceleration.

26. Assume you are walking on a mountain with a big rock in your hand. If you are on the verge of falling off the mountain, how could you save your life?

Explanation: Throw the rock forwards, as a result, the reaction will drive you back.

Illustrative Example

27. Which one of the following is true about action and reaction forces (UEE: 2003)
    A. They act on the same body.
    B. They are equal in magnitude but need not have the same line of action.
    C. They act on two different bodies.
    D. They are not equal in magnitude but have the same line of action.

Answer: C

4.3 Frictional Force

If you place a ball on a horizontal surface and give it a kick, it will move for a short distance and come to rest. According to Newton's 1st law, the ball should have continued to move with a uniform velocity. But this is not what you observe. The external force that opposes motion is what we call friction.

Friction: Is a resistive force that opposes the motion of a surface over another surface.

☛ Friction is a *surface phenomenon*. That is the origin of friction is due to surface irregularity. Even a highly polished surface when viewed under a microscope reveals a series of hills and valleys. As a result the irregularities interlock when surfaces tend to make relative motion on each other.

Commonly, we focus on two types of frictions called *static friction* and *kinetic friction*.

Static Friction (f_s): Is the force of friction between two surfaces that are in contact and at rest relative to each other.

Mathematically, we have

$$f_s = \mu_s F_N$$

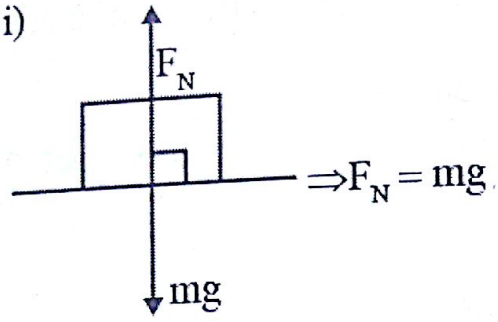
Where; μ_s (read as "mew" "s") is called *coefficient of static friction*.

☛ F_N is called the *normal force*

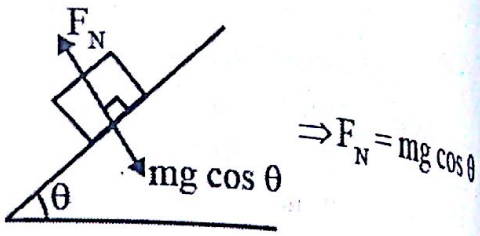
Normal force (F_N), Is a force that is always acting perpendicular to the surfaces in contact.

For instance,

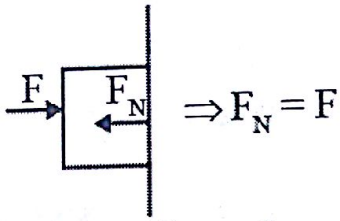
i)



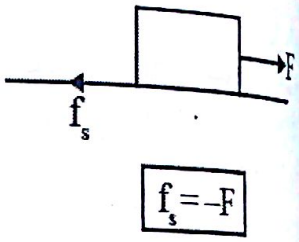
ii)



iii)



☛ **Remember**, the magnitude of static friction is equal to applied force tending to cause the relative motion and its direction is always opposite to that of the applied force.



Note also that, the maximum static friction is called *Limiting Friction* and it arises when the body is on the verge of motion.

Kinetic friction (f_k): Is the friction force between two surfaces that are in relative motion to each other.

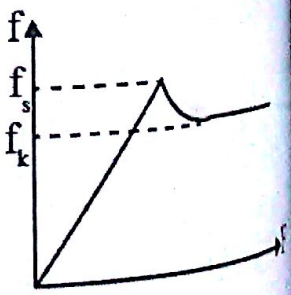
Mathematically, we have

$$f_k = \mu_k F_N$$

Where: μ_k is called the *coefficient kinetic friction*

☛ **Important points to remember**

- Kinetic friction is usually less than static friction.
- Friction doesn't depend on the area of the surfaces in contact.
- Friction depends on the nature of the surfaces in contact that is their roughness or smoothness and the normal force.
- If a body is at rest on a horizontal surface and no external force is applied, the force of friction is zero.



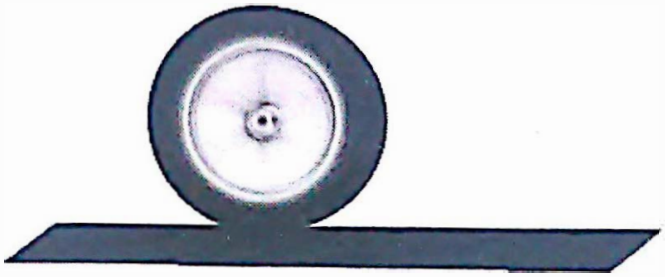
The variation of friction with the applied force can be visualized graphically as shown. From the graph, it can be seen that, it requires a larger force to start motion than it does not maintain a constant velocity.

Conceptual Examples

28. A car moves with its tires rolling. Is the friction between the tires and the road static or kinetic? Explain

Explanation: It is static. If you say why? It is because, the tire can be thought of as a series of feet arranged in a circle, each of which is momentarily in static contact with the ground.

To summarize, if a car skids, the friction acting on it is kinetic, if its wheels are rolling, the friction is static. OK!



29. When you walk on the road, the friction force pulls you back. Is this true or false? Explain

Explanation: It is false. Because, when you walk, your feet push the ground backward and friction pushes you forward. This is the actual case happening.

30. Walking on horizontal slippery ice can be much more tiring than walking on ordinary road. Why?

Explanation: In slippery road, we can't walk freely. Backward motion is there.

31. A woman in an elevator lets go off her briefcase but it does not fall to the floor. How is the elevator moving?

Explanation: The elevator must be accelerating down ward with an acceleration $\vec{a}$ such that $\vec{a} > g$.

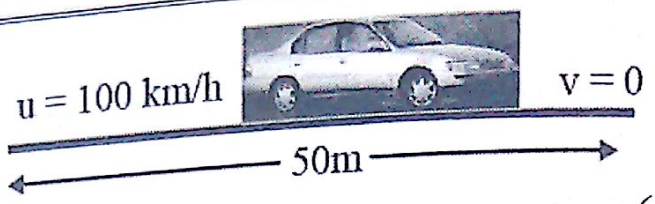
32. When you go up hill, it becomes harder to move your body forward. However, when you go downhill, it becomes easier to go forward. Why? Explain

Explanation: When you go up a hill, part of your weight (mg sin θ) pulls you down, and it become harder to move yourself. But, in the downhill movement part of your weight is in the direction of motion and hence pushes forward.

Illustrative Examples

33. A car of mass 1500 kg is stopped from an initial speed of 100 km/h in a distance of 50m. If all four wheels contribute equally to the braking, what is the breaking force at each wheel?

100



Solution: $v^2 = u^2 + 2as \Rightarrow a = \dfrac{u^2}{2(5)} = \dfrac{\left(\frac{100}{3.6}\right)^2}{2(5)} = -7.72 m/s^2$

$$4F = ma,\ F = \frac{ma}{4} = \frac{(1500)(7.72)}{4} = 2890N$$

34. A boy slides down a snow mountain starting from rest. If it takes him 3 sec to slide 20m down the 40° slope. What is the coefficient of kinetic friction between the snow and his shoe?

Solution: $s = \dfrac{1}{2}at^2$

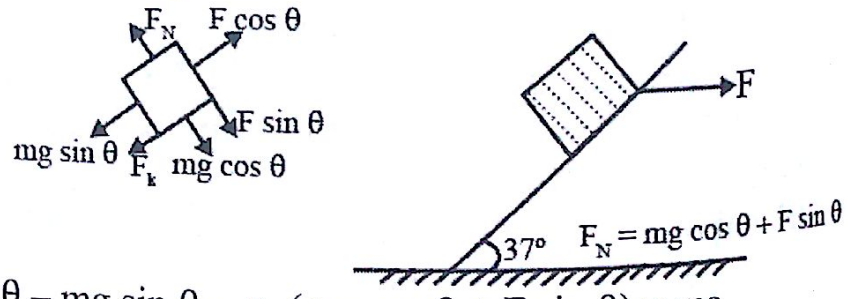
$$a = \frac{2s}{t^2} = \frac{2(20)}{(3)^2} = 4.4 m/s^2$$

$mg \sin\theta - \mu_k mg \cos\theta = ma$

$$\Rightarrow \mu_k = \frac{-a + g\sin\theta}{g\cos\theta} = \frac{10 \times 0.6 - 4.4}{10 \times 0.8} = 0.2$$

35. A horizontal force of 50N acts on a 5 kg block placed on an inclined plane. If $\mu_k = 0.3$. What acceleration will the block get?

Solution: FBD



$F\cos\theta - mg\sin\theta - \mu_k(mg\cos\theta + F\sin\theta) = ma$

$a = 2.2\ m/s^2$

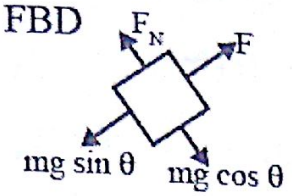
36. A block weighing 100N rests on a plane inclined at 20° to the horizontal. If $\mu_s = 0.25$, $\mu_k = 0.15$.

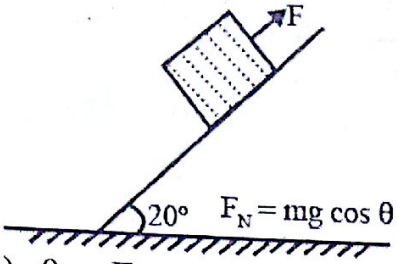
a) What is the minimum force that prevent the block from sliding?

b) What is the minimum force that will start the block to move up?

Solution:

FBD

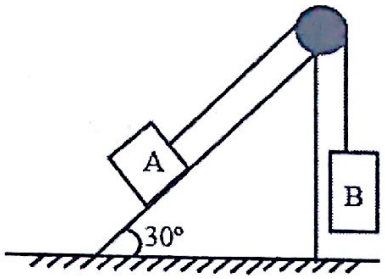




a) $F - mg \sin \theta + \mu_s (mg \cos \theta) = 0 \Rightarrow F = mg \sin \theta - \mu_s mg \cos \theta$
$F = (100)(\sin 20°) - (0.25)(100 \cos 20) = 10.7N$

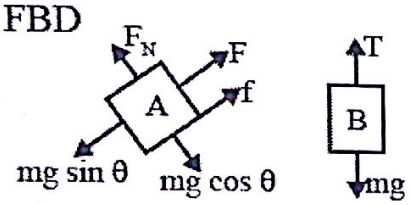
b) $F = mg \sin \theta + \mu_s (mg \cos \theta) = 57.7N$

37. Two blocks are connected over a frictionless pully. If block A = 10 kg, $\mu_k = 0.2$, $\theta = 30°$ and that block A slides down the incline at constant speed, what is the mass of block B.



Solution:

FBD



For block A, $mg \sin \theta - \mu_k mg \cos \theta = 32.7N$

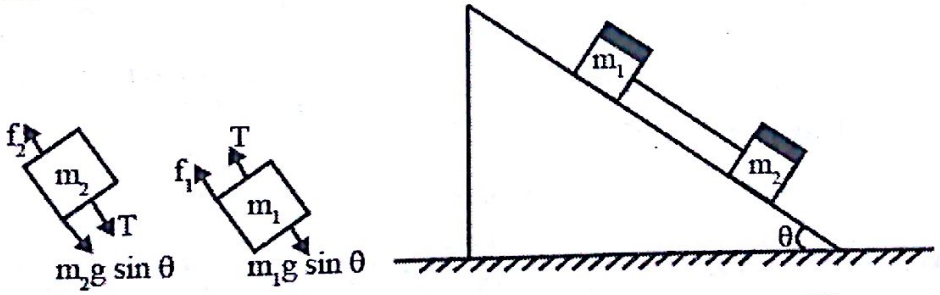
For block B, $T - mg = 0 \Rightarrow mg = T$, $m_B = \dfrac{32.7}{10} = 3.3kg$

38. Two masses $m_1 = 1.65$ kg and $m_2 = 3.3$ kg attached by a mass less rod if $\theta = 30°$, $\mu_{k_1} = 0.226$ and $\mu_{k_2} = 0.113$. Compute

a) The tension in the rod.

b) The common acceleration of the masses as they travel down.

Solution:



For mass m_1, $m_1 g \sin \theta - T - \mu_{k_1} m_1 g \cos \theta = m_1 a$ . . . (*)

For mass m_2, $T + m_2 g \sin \theta - \mu_{k_2} m_2 g \cos \theta = m_2 a$ . . . (**)

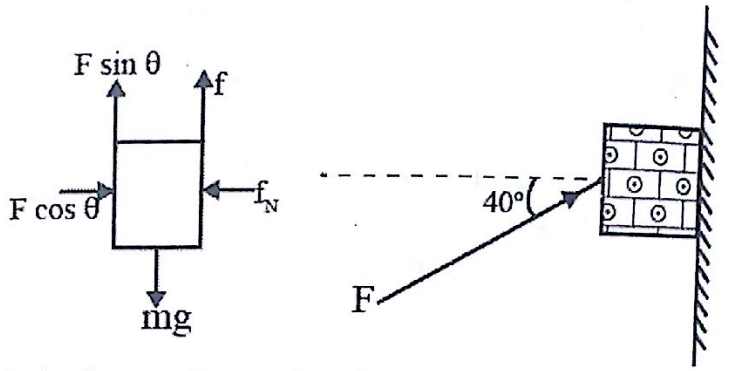
$g \sin\theta (m_1 + m_2) - g \cos\theta (\mu_{k_1} m_1 + \mu_{k_2} m_2) = a (m_1 + m_2)$

$a = [g \sin\theta (m_1 + m_2) - g \cos\theta (m_1 \mu_{k_1} + \mu_{k_2} m_2)] / m_1 + m_2$

$= 3.7 \text{ m/s}^2$ and $T = m_2 a + \mu_{k_2} m_2 \text{ co } \theta - m_2 g \sin\theta = 25.5\text{N}$

39. A block of mass 8 kg is pushed up against a wall by a force F that makes a 40° with the horizontal. Assuming $\mu_s = 0.3$. What possible value of F will hold the block in position?

Solution:

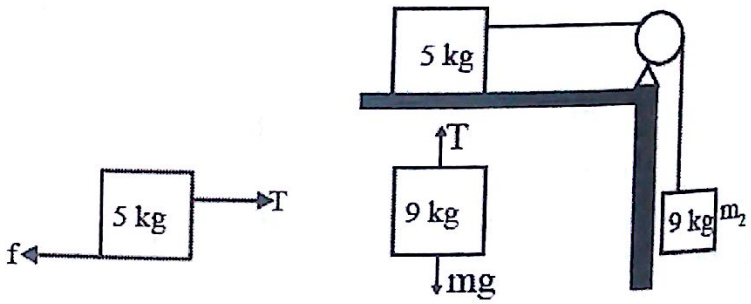


$mg - F \sin\theta - \mu_s F \cos\theta = 0$

$$F \sin\theta + \mu_s F \cos\theta = mg \Rightarrow F = \frac{mg}{\sin\theta + \mu \cos\theta} = 92\text{N}$$

40. A 9 kg mass is connected to a 5 kg mass. If $\mu_k = 0.2$ find the tension in the string.

Solution:



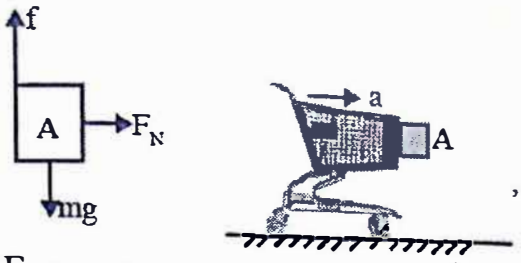
$T - \mu m_1 g = m_1 a \ldots (*)$

$m_2 g - T = m_2 a \ldots (**)$

$$T = \frac{m_2 g(1+\mu_k)}{\frac{m_2}{m_1}+1} = 38.6\text{N}$$

41. A block is placed against the vertical front of a cart. What acceleration must the cart so that block A will not fall. The coefficient of static friction between the cart and the block is μ_s.

Solution:



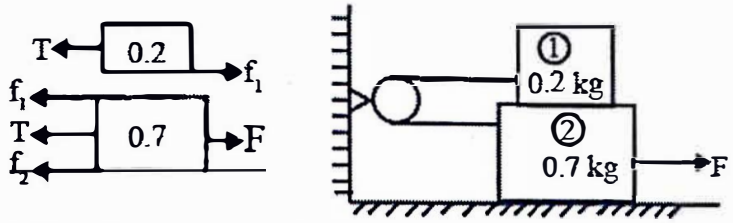
$, F_N = ma$

$f - mg = 0$; $\mu_s F_N = mg$; $\mu_s ma = mg$; $a = g/\mu_s$

If the block is not to fall, the friction force (f), must balance the weight (w) of the block. Thus, $f = mg = w$.

42. How large must F be to give the 0.7 kg block an acceleration of 3 m/s^2. Take $\mu_s = 0.15$ in all surfaces.

Solution:



For block 1 $T - f_1 = m_1 a \Rightarrow T - \mu m_1 g = m_1 a$ . . . (*)

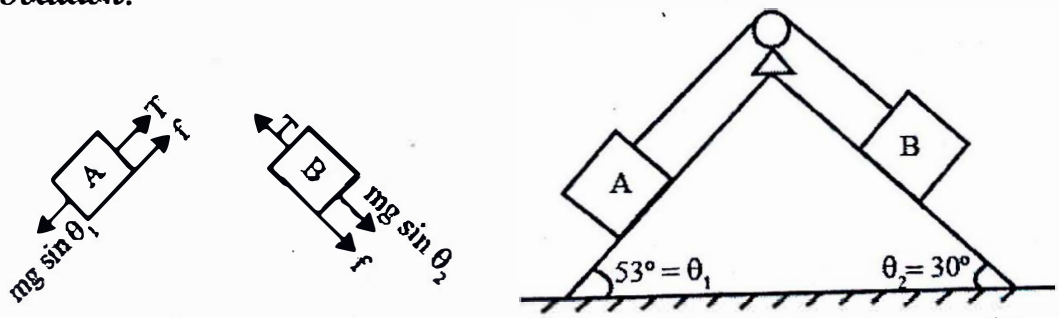
For block 2 $F - f_1 - f_2 - T = m_2 a$

$\Rightarrow F - \mu m_1 g - \mu m_2 g - T = m_2 a$ . . . (**) adding the two equation; we have $F - 2\mu m_1 g - \mu m_2 g = (m_1 + m_2)\, a$

$F = (m_1 + m_2)a + 2\mu m_1 g + \mu m_2 g = (0.9)(3) + 2(0.15)(0.2)(10) + (0.15)(0.9)(10) = 4.65N$

43. Two blocks have equal masses, and the coefficient of friction is 0.3, what is the acceleration of the blocks.

Solution:



For block A, $mg \sin\theta_1 - T - \mu\, mg \cos_1\theta_1 = ma$ . . . . . (*)

For block B, $T - mg \sin\theta_2 - \mu mg \cos\theta_2 = ma$ . . . . . (**)

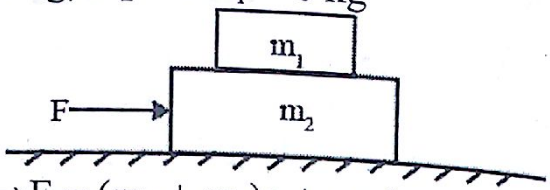
Add the two equations,

$mg \sin\theta_1 - \mu mg \cos\theta_1 - mg \sin\theta_2 - \mu\, mg \cos\theta_2 = 2ma$

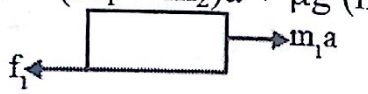
$a = g(\sin\theta_1 - \mu\cos\theta_1 - \sin\theta_2 - \mu \cos\theta_2)/2 = -0.7\ m/s^2$

44. Mass, $m_1 = 2$ kg rests on top of mass $m_2 = 3m_1$. If the coefficient of friction between surfaces is 0.3.
a) What force will give the masses an acceleration of 3 m/s^2?
b) What minimum coefficient of friction is required so that m_1 will not slip under the acceleration of 8 m/s^2?

Solution: $m_1 = 2$ kg, $m_2 = 3m_1 = 6$ kg



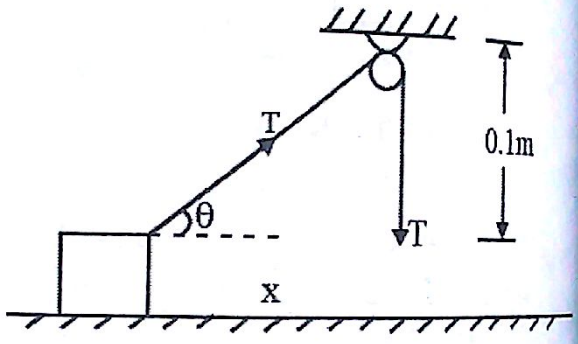
a) $F - f = ma \Rightarrow F = (m_1 + m_2)a + \mu g\,(m_1 + m_2) = 48N$



b) m_1 not to slip,

$$f_1 = m_1 a,\ \mu_s m_1 g = m_1 a \Rightarrow \mu_s = \frac{a}{g} = \frac{8}{10} = 0.8$$

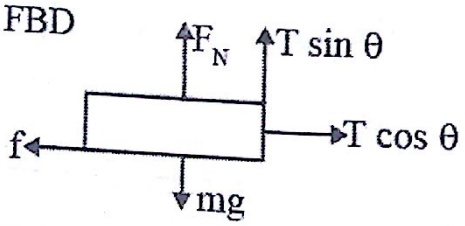
45. A block of 2.2 kg is pulled by a rope passing over a pulley. If the tension in the rope is 10N, $\mu_k = 0.2$ and the pulley is 0.1m above the block. Find the acceleration of the block when x = 0.4m.

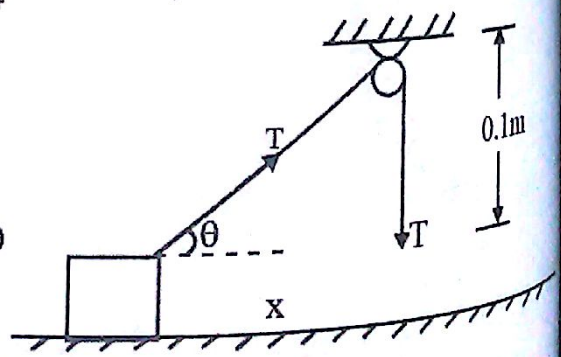


Solution: For x = 0.4 & y = 0.1

$$\tan\theta = \frac{y}{x} = \frac{0.1}{0.4} \Rightarrow \theta = 14°$$

FBD



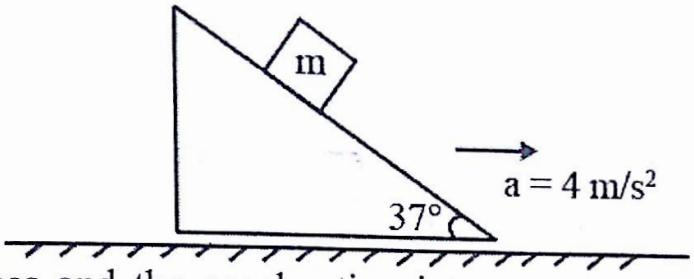


$T\cos\theta - f = ma,\ F_N = mg - T\sin\theta$

$$a = \frac{T\cos\theta - \mu(mg - T\sin\theta)}{m} = 2.6 m/s^2$$

46. A block rests on an inclined plane as shown in the figure.
   a) What friction is needed to keep the block in position as the inclined accelerate to the right at 4 m/s^2?
   b) What minimum μ_s is required for the above care to happen?
   c) Which direction will the block slide if friction is ignored?

Solution: FBD

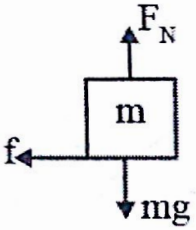


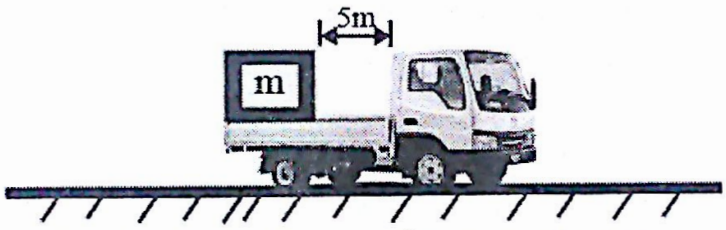
a) Resolving the forces and the acceleration into components that are parallel and perpendicular to the plane.
$mg \sin\theta - f = (m \cos\theta)$ (a) and $F_N - mg\cos\theta = (m \sin\theta)(a)$
$\Rightarrow f = 2.8N$, $f_N = 10.4N$

b) $\mu_s = \dfrac{f}{F_N} = 0.27$ c) down

47. Assuming $\mu_s = 0.3$, determine the minimum stopping distance of the truck that is moving at a speed of 20 m/s. If the box on the bed of the truck is not to slide forward.

Solution: FBD



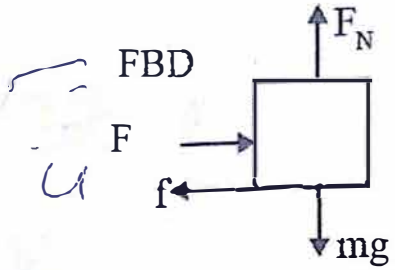


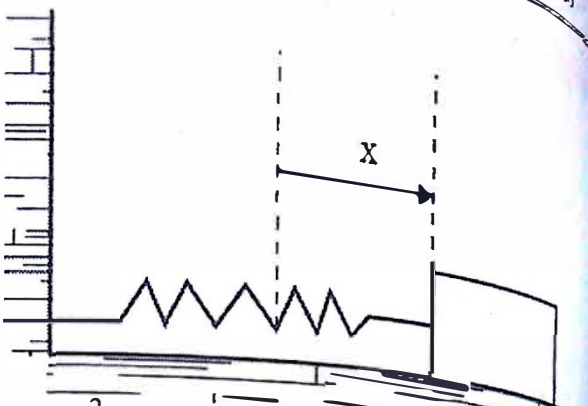
The box decelerate with the truck. Hence, $f = -ma \Rightarrow \mu mg = -ma$
$\Rightarrow a = -3$ m/s^2 and $v^2 = u^2 + 2as$, $v = 0$, $u = 20$ m/s

$$s = \frac{u^2}{2a} = 66.7m$$

48. A **nonlinear** spring has a force deflection relationship given by $F = 150x + 400x^2$. If a 5 kg block attached to the spring is released from rest at, $x = 100$mm, what acceleration will it attain take $\mu k = 0.25$.

Solution:



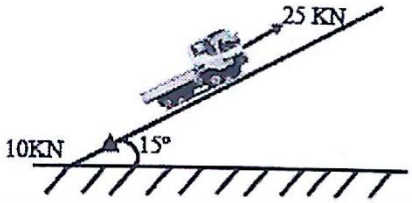


$F - f = ma$, $F = 150x + 400x^2 = 150\,(0.1) + 400\,(0.1)^2 = 19$

$F - \mu_k mg = ma$

$$a = \frac{F - \mu_k mg}{m} = \frac{19 - 12.5}{5} = 1.3\ m/s^2$$

49. A truck of mass 8 tones is given in the figure. Will the truck move up the incline.

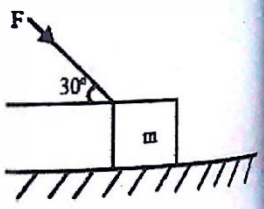


Solution:

$F - f - mg \sin\theta = 25000 - 10000 - 8000 \sin 15° = -5.3N$

NO, it cannot

50. A force of 400N pushes a 25 kg box on a floor. Starting from rest, the box attains 2 m/s in 4 sec. Determine the coefficient of friction.

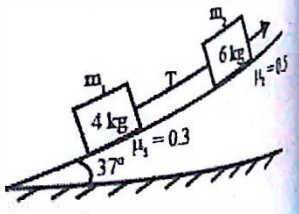


Solution:

$F \cos\theta - \mu\,(mg + F \sin\theta) = ma$,

$$a = \frac{\Delta v}{\Delta t} = 0.5 m/s^2 \text{ and, } \mu = 0.7$$

51. If the given masses have an acceleration of 2 m/s^2. Find the tension in the cord and the applied force.



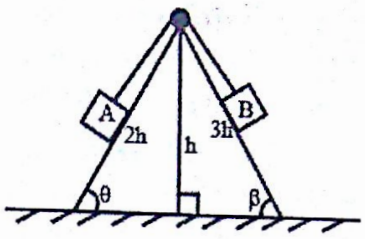
Solution:

For m_1, $T - m_1 \sin 37° - \mu_1 m_1 g \cos 37° = m_1 a \Rightarrow T = 42N$

and, for m_2, $F - T - m_2 g \sin 37° - \mu_2 m_2 g \cos 37° = m_2 a$

$\Rightarrow F = 114N$

52. For the two blocks A and B, if $m_A = 2m_B$, find the acceleration of the blocks. Assume frictionless surface.



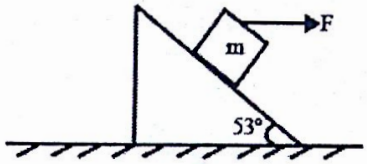
Solution: For A, $m_A g \sin\theta - T = m_A a$,

$$\sin\theta = \frac{1}{2}$$

For B, $T - m_B g \sin\beta = m_B a$, $\sin\beta = \frac{1}{3}$

$\Rightarrow a = 2.2\ m/s^2$

53. A block of mass 2 kg placed on a rough inclined plane is acted by a horizontal force of 10N as shown. If $\mu_k = 0.4$. What is the acceleration of the block?

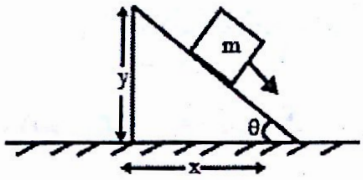


Solution:

$F\cos\theta + mg\sin\theta - \mu(mg\cos\theta - F\sin\theta) = ma$

$\Rightarrow a = 10.2\ m/s^2$

54. Calculate the acceleration of a block sliding down an incline plane having equal height and base if $\mu = 0.4$.

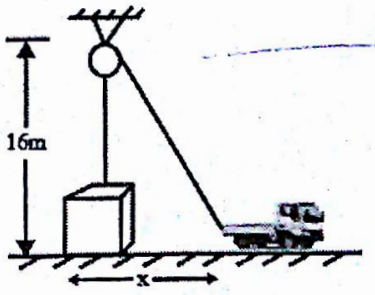


Solution: $mg\sin\theta - \mu mg\cos\theta = ma \Rightarrow a = 3\sqrt{2}\,m/s^2$

55. A packing crate of weight 50N is placed on a plane inclined at 35° from the horizontal. If $\mu_s = 0.65$ will the create slide down the plane?

Solution: $mg\sin\theta - \mu mg\cos\theta = 28.7 - 26.6 = 2.1N$ Yes

56. The pickup is used to lift a 40 kg box as shown. If the pickup has reached a constant velocity of 5 m/s. When x = 12m, compute the corresponding tension in the rope.



Solution:

$$T\sin\theta = mg \Rightarrow T = \frac{mg}{\sin\theta} = \frac{(40)(10)}{\left(16/20\right)} = 500\,N$$

57. A block slides on a rough 45° incline. The coefficient of friction is μ_k. What is the ratio of acceleration when the block accelerates down the incline to the acceleration when the block is projected up the incline? (UEE: 2003)

A. $\dfrac{\mu_k - 1}{\mu_k + 1}$ C. $\dfrac{1-\mu_k}{1+\mu_k}$

B. $\dfrac{\frac{\mu_k}{\sqrt{2}} - 1}{\mu_k + 1}$ D. $\dfrac{1+\mu_k}{1-\mu_k}$

Solution: For downward, $mg \sin\theta - \mu_k mg \cos\theta = ma_1 \ldots (*)$

F upward, $-mg \sin\theta - \mu_k mg \cos\theta = ma_2 \ldots (**)$

Taking ratio, $\dfrac{a_1}{a_2} = \dfrac{\mu_k - 1}{\mu_k + 1}$ **Answer: A**

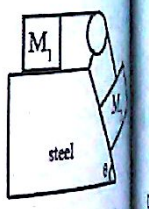
58. A 4 kg block is given an initial speed of 8 m/s at the bottom of 30° incline. If the frictional force that retards its motion is 15N and if the block is directed up the incline, how far will it move before it stops? (UEE 2004)

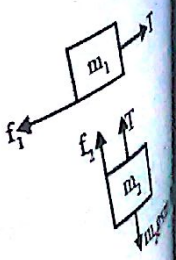
A. 3.66m B. 4.40m C. 3.35m D. 4.67m

Solution: $-mg \sin\theta - f = ma \Rightarrow a = -35/4\ m/s^2$

$$s = \frac{v^2 - u^2}{2a} \Rightarrow s = 3.66m$$ **Answer: A**

59. An aluminum block of mass m_1 = 3 kg and a copper block of mass m_2 = 5 kg are connected by a light string over a frictionless pulley. They sit on a steel surface where θ = 53°. The coefficient of kinetic friction between the steel surface and each block is 0.3. If the blocks are released from rest, what is the tension T in the string?





A. 29.4N C. 12.25N

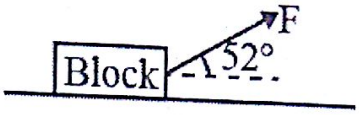
B. 17.25N D. 8.6N

Solution: $T - f_1 = m_1 a \ldots (*)$

$m_2 g \sin\theta - T - f_2 = m_2 a \ldots (**)$

$\Rightarrow T = 17.21N$ **Answer: B**

60. A force F of magnitude 20N is applied to a block of mass 2 kg that lies on a rough, horizontal surface. The coefficient of kinetic friction between the block and surface is 0.4. What is the magnitude of the acceleration of the block? (UEE 2007)



A. 10 m/s^2 B. 5.2m/s^2 C. 4m/s^2 D. 2.8 m/s^2

Solution: $F\cos\theta - \mu_k(mg - F\sin\theta) = ma$

$\Rightarrow a = 5.2\ m/s^2$

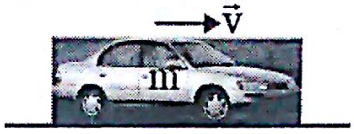
Answer: B

4.4 Linear Momentum and its Applications

It is usually difficult to stop a moving body of considerable mass. The quantity of motion possessed by such moving body is called *momentum*.

Momentum ($\vec{P}$) : Is the product of the mass of a moving body with its velocity.

The linear momentum of a body of mass m moving with a velocity of $\vec{v}$ in a straight line is mathematically given by, $\boxed{\vec{P} = m\vec{v}}$, its unit is kg m/s.



☛ **Important points to remember**

- Momentum is a vector quantity.
- When a body is moving with a velocity of

$\vec{v} = v_x i + v_y j + v_z k$

$\vec{P} = m\vec{v} = m(v_x i + v_y j + v_z k) = P_x i + P_y j + P_z k$

Impulse (J): Is the product of an average force and the time interval for which the force acts.

From Newton's 2nd law,

$$F = ma = \frac{m\Delta v}{\Delta t}$$

$\boxed{F\Delta t = m\Delta v = \Delta P}$ ← Impulse

☛ **Important points to remember**

- Impulse is equals to the change in momentum.
- The same impulse may result from a small force acting for a longer time or a larger force acting for a very short time.

Conceptual Examples

61. A Karatist can break a bundle of bricks with the side of his bare hand. How is this possible?

Explanation: $F\Delta t = m\Delta v \Rightarrow F = \frac{m\Delta v}{\Delta t}$. Thus, as the contact time is very small, the impact force is very large.

62. A glass dropped onto a concrete floor is more likely to break than when it is dropped into a pond of water. Why?

Explanation: In the former case, the glass changes its momentum in a very short time and hence experience a very large force while in the later case, it changes its initial momentum through a long time and thus feels small impact force.

Illustrative Examples

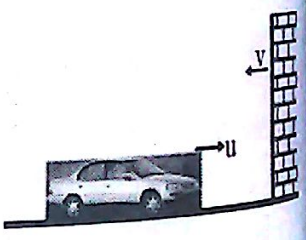
63. a) What is the momentum of a 10,000 kg truck moving at speed of 12 m/s?

b) What speed should a 2000 kg taxi have to attain the same momentum as the truck?

Solution: a) $P = mv = (10{,}000)(12) = 120{,}000$ kg m/s

b) $P = mv \Rightarrow v = \frac{p}{m} = \frac{120{,}000}{2000} = 60\text{m/s}$

64. A car of mass 1500 kg collides with a wall and rebounds. The initial and final velocities of the car are; 15 m/s and 3 m/s respectively and if the collision lasts for 0.15 sec. Find



a) The impulse delivered to the car, and

b) The average force exerted on the car.

Solution: a) $I = F\Delta t = m\Delta v = 1500\ (-3 - 15) = -27000$ kg m/s

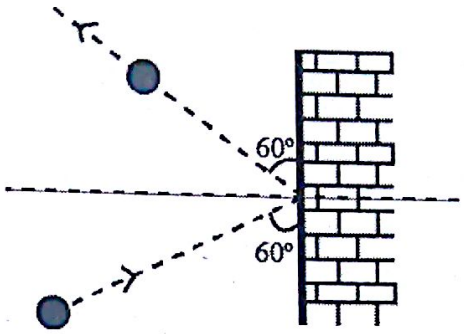
b) $F\Delta t = m\Delta v \Rightarrow F = \frac{m\Delta v}{\Delta t} = \frac{27{,}000}{0.15} = 180\text{kN}$ to left

65. A ball of mass 0.15 kg is dropped from a height of 1.25m. If the ball rebounds to a height of 0.96m, what impulse it experience?

Solution: $I = m\Delta v = mv - mu = m\,(v - u) = 0.15\,(4 + 5) = 1.35$ kg m/s

$v^2 = u^2 + 2gh_1 \Rightarrow u = 5$ m/s and $v = \sqrt{2gh_2} = 4\text{m/s}$

66. A 3 kg ball strikes a wall at 10 m/s at an angle of 60°. It bounces off the wall with the same speed and angle. If the ball is in contact with the wall for 0.2 sec, what is the average force exerted by the wall on the ball?



Solution: $F\Delta t = m\Delta v$

$$F = \frac{m\Delta v}{\Delta t} = \frac{m}{\Delta t}(-u\sin 60 - u\sin 60)$$

$$= \frac{3}{0.2}(2(10)(0.86)) = 258N$$

$u_x = u \sin 60$

$v_x = - u \sin 60$

$u_y = u \cos 60$

$v_y = u \cos 60$

$\Delta v_y = 0$

67. A ball of mass 4 kg falling to the ground with a velocity of $(5i - 12j)$ m/s bounces off the ground with $(12i + 16j)$ m/s. If the contact time is 0.2 sec. Find the:

a) Impulse of the ball.

b) Force it experience.

Solution: a) $I = m\Delta v = m(v - u) = 4[(12 - 5)i + (16 - (-12)j)]$

$= 28i + 112j$ kg m/s $= (28, 112)$ kg m/s

b) $$F = \frac{I}{\Delta t} = \frac{28i + 112j}{0.2} = 140i + 560j = (140, 560)N$$

68. A ball of mass 0.25 kg moving with a speed of 20 m/s bounces off a wall with 90% of its initial speed along the same line of motion. What is the change in momentum of the ball?

Solution: $\Delta P = m(v - u) = m(0.9u - u) = -0.1$ mu

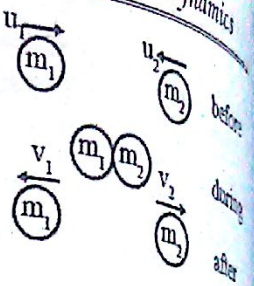
$= -(0.25)(20)(0.1) = -0.5$ kg m/s

4.5 Law of Conservation of Linear Momentum

If the net force acting on an object is zero, its momentum is conserved. Therefore, the law of conservation of linear momentum states that;

> "The total linear momentum of an isolated system remains constant." $\sum P_i = \sum P_f$

Now, consider two bodies with masse m_1 and m_2 and velocities u_1 and u_2 before impact and v_1 and v_2 after impact respectively, then we have.



$$\boxed{m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2}$$

Important points to remember

- Isolated system is a system which is not interacting with its surrounding.
- Conservation of momentum is another way of stating Newton's 1st law.
- Conservation of momentum is an alternative and more general statement of Newton's 3rd law.

Conceptual Example

69. If two objects, where one is initially at rest collide is it possible for both to be at rest after the collision? Explain.

Explanation: Yes! Consider a piece of mud that is thrown onto a wall and stick to the wall.

Illustrative Example

70. A 75 kg man is riding a 39 kg bike at a speed of 2.3 m/s. If he jumps off, what is the resulting speed of the cycle?

Solution: $\sum P_i = \sum P_f \Rightarrow 2.3\ (75 + 39) = 39v \Rightarrow \Delta v = 4.4$ m/s

71. A 15 kg fish swimming at 1.1 m/s in *Lake Tana* suddenly gets a 4.5 kg fish initially at rest. Find the speed of the large fish just after it eats the small fish.

Solution: $\sum P_i = \sum P_f \Rightarrow m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$$m_1v_1 = (m_1 + m_2)v \Rightarrow v = 0.85 \text{ m/s}$$

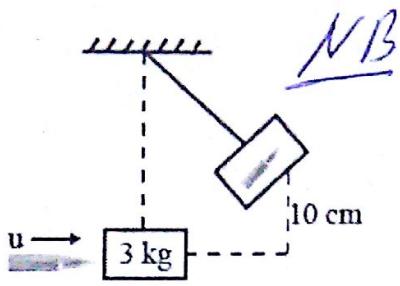
72. A hunter has a rifle that can fire 0.06 kg bullets with a muzzle velocity of 900 m/s. A 40 kg leopard springs at him at a speed of 10 m/s. How many bullets must the hunter fire into the leopard to stop it?

Solution: $P_b = P_\ell \Rightarrow nm_bv_b = m_\ell v_\ell \Rightarrow n = \dfrac{m_\ell v_\ell}{m_b v_b} = \dfrac{(40)(10)}{(0.06)(900)}$

$$n = 7.4 \approx 8 \text{ bullets}$$

73. a bullet of 15 g is fired horizontally into a 3 kg block of wood suspended by a cord. The bullet sticks in the block and it to swing 10cm above its initial level compute the initial speed of the bullet.

Solution: $v^2 = 2gh \Rightarrow v = \sqrt{2gh}$



$v = 1.4$ m/s

$m_1u_1 = (m_1 + m_2)$

$$u_1 = \frac{(m_1 + m_2)}{m_1} v$$

$$u_1 = \frac{(0.015 + 3)}{0.015}(1.4) = 282 m/s$$

74. A 50 g bullet is shot at a speed of 150 m/s into a 2 kg block initially at rest. If the block together with the bullet slides 2.7m before stop. Find the
   a) speed of the block just after impact
   b) the friction force that stops the system

Solution: a) $m_1u_1 = (m_1 + m_2)\, v$

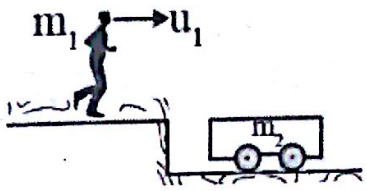
$$\Rightarrow v = \frac{m_1u_1}{m_1 + m_2} = \frac{(0.05)(150)}{(0.05 + 2)} = 0.37 m/s$$

b) $v^2 = 2as \Rightarrow a = \frac{v^2}{2s} = \frac{(0.37)^2}{2(2.7)} = -0.025 m/s^2$

$F = ma = (2.005)(0.025) = 0.05N$

75. A 60 kg person running at an initial speed of 4 m/s jumps onto a 120 kg cart that is initially at rest. Find the velocity of the person and the cart after impact.

Solution: $m_1u_1 = (m_1 + m_2)\, v$



$$v = \frac{m_1u_1}{m_1 + m_2} = \frac{(60)(4)}{120 + 60}$$

$v = 1.3$ m/s

76. A 1 kg ball is dropped form a 4m height on to a floor. If the ball bounces to 2.5m above the floor, Find;
   a) The momentum transferred to the floor.
   b) The average acceleration of the ball if the contact time between the ball and the floor is 0.15 sec.

Solution: a) $u^2 = 2gh_1 \Rightarrow u = \sqrt{2gh_1} = 8.9 m/s$, $\Delta P = m(v - u)$

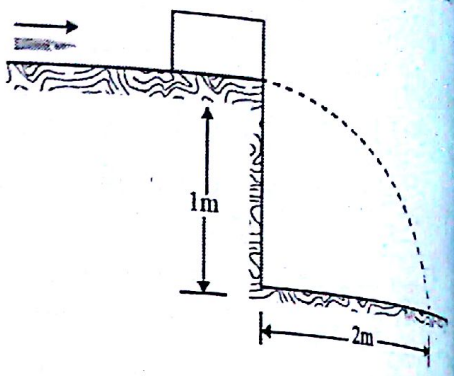
$v^2 = 2gh_2 \Rightarrow v = \sqrt{2gh_2} = 7 m/s$, $\Delta P = 15.9$ kg m/s

b) $\vec{a}_{av} = \dfrac{\Delta v}{\Delta t} = 106\ m/s^2$

77. An 8 g bullet is fired into a block of mass 2.5 kg initially at rest at the edge of a frictionless table of height 1m. The bullet remains in the block and lands 2m from the bottom of the table. What was the initial speed of the bullet?

Solution:

$m_1u_1 = (m_1 + m_2)v$



$$t = \sqrt{\frac{2y}{g}} = \sqrt{\frac{2(10)}{10}}$$

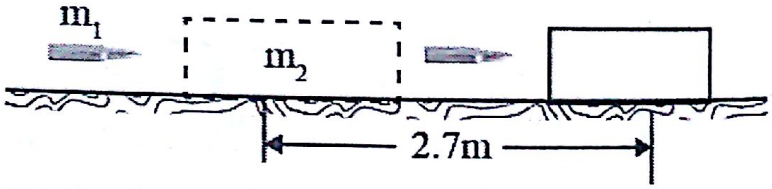
$t = 0.45$ sec

$$v = v_x = \frac{x}{t} = 4.44 m/s$$

$$u_1 = \frac{(m_1 + m_2)}{m_1} v_2 = \left(\frac{2.508}{0.008}\right) 4.4 = 1375 m/s$$

78. A 60 g bullet is fired horizontally with a velocity of 600 m/s into a 3 kg block of soft wood initially at rest. If the bullet emerges from the block with a speed of 400 m/s and the block is observed to slide 2.7m before coming to rest. What is the coefficient of kinetic friction between the block and the surface?

Solution:



$m_1u_1 = m_1v_1 + m_2v_2 \Rightarrow v_1 = 2.3$ m/s

$$v_2^1 = v_1^2 + 2as\ ,\ \text{but } v^1 = 0 \Rightarrow a = \frac{-v_1^2}{2s} = -\frac{(2.3)^2}{2(2.7)} = -1 m/s^2$$

$$f = ma \Rightarrow \mu F_N = ma \Rightarrow \mu_k = \frac{ma}{F_N} = \frac{3(1)}{30} = 0.1$$

4.6 Elastic and Inelastic Collisions

We now turn our attention to collision, which is a situation in which two objects strike one another, and in which the net external force is zero.

Now, it may seem surprising at first, but the fact that the momentum of a system is conserved during collisions does not necessarily mean the system's kinetic energy is conserved. In fact most, or even all, of a system's kinetic energy may be converted to other forms during a collision, while at the same time, not one bit of momentum is lost.

Collision: Is an event where two or more bodies make physical contact for a very short time and experience impulsive forces.

Type of collisions

A. Based on Conservation of Kinetic energy

Based on what happens to the kinetic energy, collisions are of two types, elastic and inelastic.

Elastic collision: Is a collision in which both momentum and kinetic energy of the system are conserved.

So here, $kE_i = kE_f$

Inelastic Collision: Is a collision in which the momentum is conserved, but the kinetic energy is not conserved.

In this case, $kE_i \neq kE_f$

☛ **Important points to remember**

- For both, elastic and inelastic collisions, momentum is conserved.
- When two bodies collide and stick together, the collision is said to be *perfectly inelastic collision*.
- Kinetic energy may change during a collision. Usually, it decreases due to losses associated with sound, heat, vibration and deformation. Sometimes it increases, if the collision sets off an explosion.

Newton's experimental law of impact

It states that, the ratio of the relative speed of two objects in direct impact is constant. This constant is called *coefficient of restitution* e and is given by:

$$\text{coefficient of restituion} = -\frac{\text{Relative velocity of separation}}{\text{Relative velocity of appraoch}}$$

$$e = -\frac{v_2 - v_1}{u_2 - u_1}$$

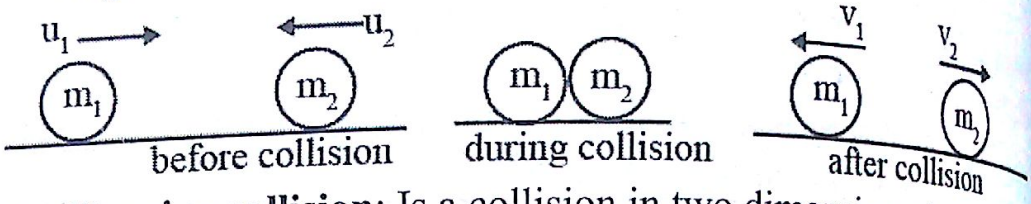
Where v_1 and v_2 are final velocities of the two interacting bodies after impact and u_1 and u_2 are initial velocities before impact.

The experiment also revealed the following collision classification.

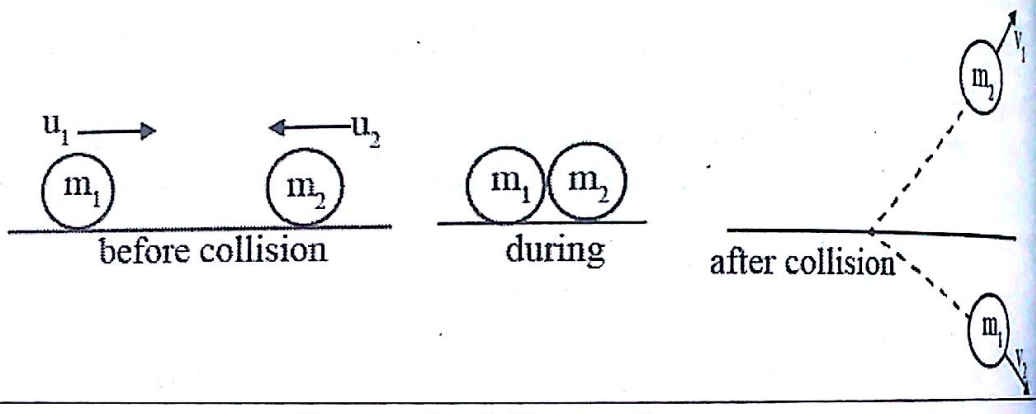
e	<1	0	1	>1
Type	Inelastic	Perfectly inelastic	Elastic	Explosive collision

B. Based on Dimension

1. **Head on Collision**: Is a collision in one dimension, where the objects rebound on straight line path that coincides with the original direction of motion.



2. **Glancing collision**: Is a collision in two dimension where the objects rebound in the same plane but not necessary in the same direction as the original path.



Conceptual Examples

80. In a perfectly inelastic collision between two identical objects under what condition is all the initial kinetic energy of the system transformed to other form of energy?

Explanation: When objects are moving towards each other with the same initial speeds.

81. In a perfectly elastic collision between two bodies, does the kinetic energy of each body change as a result of the collision?

Explanation: Not necessarily! but it should be $\sum k_{E_i} = \sum k_{E_f}$.

82. Consider a completely inelastic collision between a car and a truck. Which vehicle loses more kinetic energy as a result of the collision?

Explanation: The car. Even if the vehicles experience the same impulsive forces during the collision, the change in kinetic

energy of the car is more pronounced than the truck due to its small mass.

83. Does a large force always produce a larger impulse on a body than a smaller force? Explain

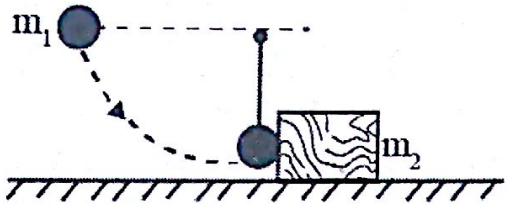
Explanation: No! it depends on the time interval for which the forces interact with the contact surfaces.

Illustrative Examples

84. Two masses each of 2 kg have velocities $u_1 = 15i + 30j$ $u_2 = -10i + 4j$, $v_1 = -5i + 20j$. What is the final velocity of m_2, after their collision?

Solution: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$, so we have;

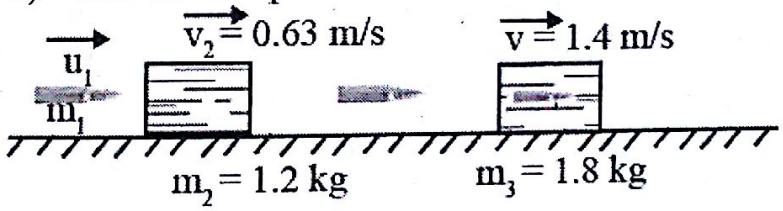
$2(15i + 30j) + 2(-10i - 5j)$
$= 2(-5i + 20j) + 2(v_2)$
$(15i - 10i) + (30j + 5j)$
$= -5i + 20j + v_2$
$v_2 = 10i + 15j$



85. A 3.5 g bullet is fired on to two blocks resting on a smooth surface. If the bullet passes through the first block and giving it a 0.63 m/s and then embedded itself in the second block and move with 1.4 m/s together with it. What was?

a) The speed of the bullet immediately after it emerges from the first block, and

b) The initial speed of the bullet?

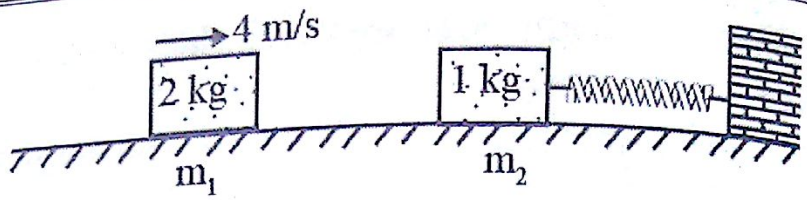


Solution:

a) $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 \Rightarrow m_1u_1 = m_1v_1 + (1.2)(0.63)$
$\Rightarrow 0.0035u_1 = 0.0035v_1 + 0.756 \ldots (*)$
and $m_1v_1 = (m_1 + m_3)v \Rightarrow 0.0035v_1 = 1.8035 \Rightarrow v_1 = 515$ m/s

b) Substituting the value of v_1 into equation (*), we get;
$0.35 \quad u_1 = 0.0035(515) + 0.756 \Rightarrow u_1 = 731$ m/s

86. A 1 kg block connected to a spring of k = 200 N/m rests on a frictionless surface. If a 2 kg block moving at 4 m/s collides with the 1 kg block and sticks to it, what is the maximum compression of the spring then?



Solution: $m_1u_1 + m_2u_2 = (m_1 + m_2)\,v$

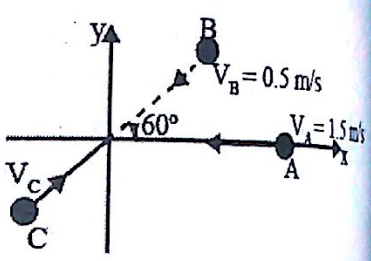
$$\Rightarrow v = \frac{m_1u_1}{m_1 + m_2} = \frac{(2)(4)}{(1+2)} = 2.67\text{m/s}$$

Kinetic energy transformed to Elastic potential energy.

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow (m_1 + m_2)v^2 = kx^2$$

$$x = \sqrt{\frac{(m_1u_1)v^2}{k}} = \sqrt{\frac{8(2.7)^2}{200}} = 54\text{m/s}$$

87. Spheres A, B and C have masses 0.02, 0.03, and 0.05 kg respectively. If they collide at the origin and move together with 0.5 m/s in the x – axis. What was the speed of the sphere C?



Solution: $m_Av_{Ax} + m_Bu_{Bx} + m_Cu_{Cx} + m_Au_{Ay} + m_Bu_{By} + m_Cu_{Cy} = (m_A + m_B + m_C)v_x + (m_A + m_B + m_C)v_y$

$\Rightarrow -1.5\,(0.02) + (-0.86)(0.5)(0.03) + (0.5)u_{Cy} = 0 \Rightarrow u_{Cy} = 0.26$

88. A collision occurs between a 2 kg body moving with (–4i - 5j) m/s and a 4 kg body moving with (6i – 2j) m/s. After the collision the bodies stick together. What is their common velocity?

Solution: $m_1u_1x + m_2u_2x = (m_1 + m_2)v_x$

$2(-4) + 4(6) = (2 + 4)v_x \Rightarrow v_x = 2.7$ m/s

and, $m_1v_1{}^y + m_2v_2{}^y = (m_1 + m_2)v_y$

$2(-5) + 4(-2) = (2 + 4)v_y \Rightarrow v_y = -3$ m/s

Therefore, $\vec{v} = (2.7i - 3j)$ m/s

89. An object at rest explodes into three pieces of equal mass. One moves east at 20 m/s, a second moves south east at 30 m/s. What is the velocity of the third piece?

Solution: $P_i = P_f = 0$

$\rightarrow$ $m_1v_{1x} + m_2v_{2x} + m_3v_{3x} = 0$, $m_1 = m_2 = m_3 = m$

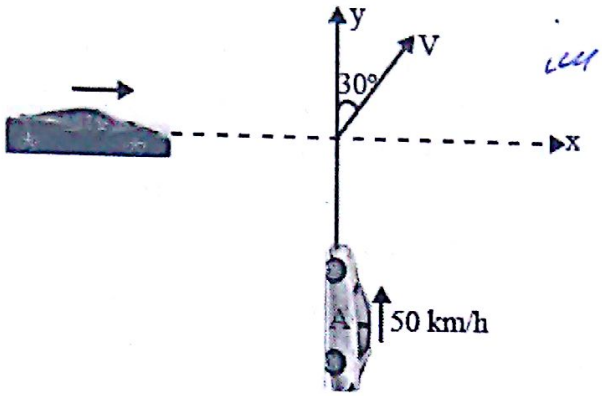
$v_{3x} = -v_{1x} - v_{2x} = -20 - 30\,(\cos 45°) = -41.2$ m/s

$\rightarrow$ $m_1v_{1y} + m_2v_{2y} + m_3v_{3y} = 0$

$-v_{3y} = 0 - 30\ (\sin 45°) = -21.3$ m/s

$v_3 = (-41.2i + 21.3j)$ m/s $\Rightarrow v = 46.4$ m/s 27°NW

90. Two cars collide at right angles in the intersection. Car A was mass of 1200 kg, car B has mass of 1600 kg. The cars move with common velocity after impact. If car A was moving at 50 km/h, what was the velocity of car B before impact?



Solution: $P_{xi} = P_{xf}$

$M_B u_B = (m_A + m_B)\ v \sin 30°$

$1600 u_B = 2800(v)(0.5)$... (*)

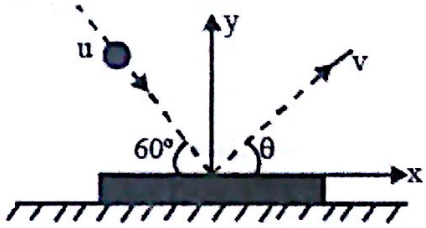
$P_{yi} = P_{yf}$

$M_A u_A = (m_A + m_B)\ v \cos 30°$

$1200\ (50) = (2800)(v)(0.86)$

. . . (**) from equations (*) & (**), $u_B = 21.7$ km/h

91. A steel ball strikes a heavy steel plate with a velocity of 24 m/s at an angle of 60° with the horizontal. If the ratio, $\frac{v_y}{u_y} = 0.8$. Compute the final velocity and direction of motion with which the ball rebounds.



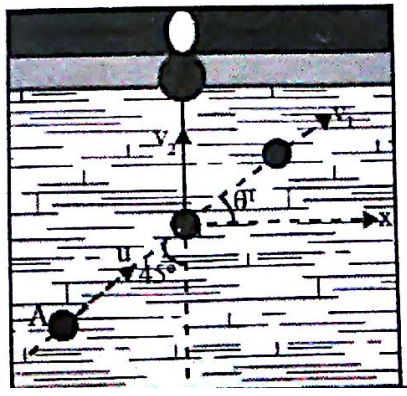
Solution: In the x direction, $mu_x = mv_x$

$\Rightarrow v_x = u_x = u \cos\theta = 24 \times 0.5 = 12$ m/s . . . (*)

$$\frac{v_y}{u_y} = 0.8 \Rightarrow \frac{v \sin\theta}{24(\cos 30°)} = 0.8 \Rightarrow v_y = 16.63 \ldots (**)$$

$$\tan\theta = \frac{v_y}{v_x} = \frac{16.63}{12} = 1.39 \Rightarrow \theta = 54°,\ v = 20.5 \text{ m/s}$$

92. In a pool game, the cue ball A must strike the eight ball in a position shown to send it to the pocket. If the balls have equal masses and $\frac{v_1 \sin\theta^1}{u_1 \sin\theta} = 0.9$. Find the rebound angle θ^1 of ball A.



Solution:

In the x – direction $mu_x = mv_x$

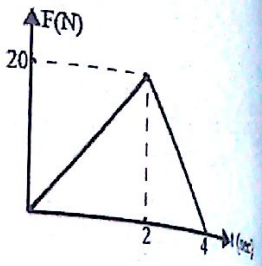
$u\cos 45 = v_1 \cos\theta^1$

From $\frac{v\cos\theta}{u\sin\theta} = 0.9 \Rightarrow v_1 \sin\theta^1 = (0.7u)(0.9)\ldots(*)$

and $v_1 \cos\theta^1 = 0.7\ u\ldots(**)$ taking ratio of equation (*) to equation (**) $\tan\theta^1 = 0.9$

$\theta^1 \approx 42°$

93. A force F acts on a ball initially at rest on a smooth surface for a time t. The variation of F with t is shown. The momentum of the ball after 4 sec is:



A. 10 B. 20 C. 30 D. 40

Solution: $\Delta P = F\Delta t$ = Area under $\vec{F}$ - t

$$\Delta P = \frac{1}{2}Ft = \frac{1}{2}(4)(20) = 40 \text{ sec}$$

Answer: D

94. A rocket moving in space at 3×10^3 m/s and fuel is ejecting in a opposite direction at 5×10^3 m/s relative to the rocket. What is the speed of the rocket as its mass is reduced to half its mass before ignition?

A. 7×10^3 m/s B. 5×10^3 m/s C. 4×10^3 m/s D. 2×10^3 m/s

Solution:

$$P_i = P_f \Rightarrow Mv_1 = \frac{m}{2}v_2 \Rightarrow (3\times10^3)m = \left(\frac{m}{2}+\frac{m}{2}\right)(v+5\times10^3)$$

$v = 2 \times 10^3$ m/s

Answer: D

95. A hard ball moving on a horizontal frictionless surface with a speed of 5 m/s in the positive x direction strikes a stationary hard ball of the same mass. After the collision, the first ball moves with a speed of 4 m/s along a direction that makes an angle of 37° with its initial direction of motion. What is the struck ball's speed and the smallest angle between its direction of motion and the positive x –axis after the collision (UEE: 2007)

A. 3m/s and 53° C. 4m/s and 53°

B. 3 m/s and 37° D. 4 m/s and 37°

Solution: From conservation of momentum

$m_1v_1 + m_2v_2 = m_1v_1 + m_2v_2,\ m_1 = m_2 = m$

$m[v_x, 0] + 0 = m[v_{1x}, v_{1y}] + m[v_{2x}, v_{2y}]$

$(5, 0) = (4\cos 37°, 4\sin 37°) + (v_{2x}, v_{2y})$

$\Rightarrow 5 = 4\cos 37° + v_{2x} \Rightarrow v_{2x} = 1.8$ m/s

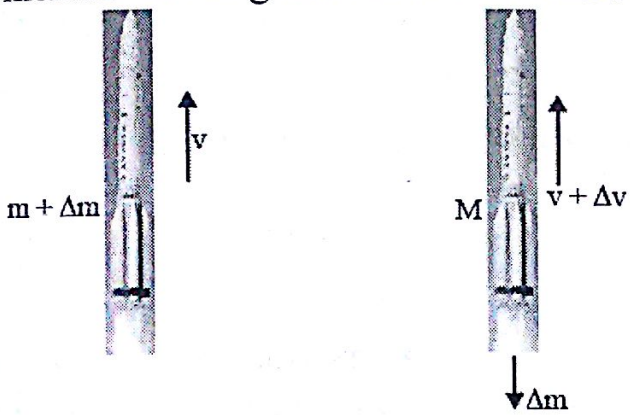
and $0 = 4\sin 37° + v_{2y} \Rightarrow v_{2y} = -2.4$m/s

$v_2 = \sqrt{v_{2x}^2 + v_{2y}^2} = 3$m/s and $\tan\theta = \dfrac{-2.4}{1.8} \Rightarrow \theta = 53°$ **Answer: A**

4.7 Momentum Conservation in a variable Mass system

Rocket propulsion and rain drop collection are typical and interesting variable mass systems. And in both cases we cannot use Newton's second law directly as the system changes mass through time.

- **Note that**: As a rocket is fired, it ejects exhausted gas continuously and decreases in mass. This in turn causes the rocket to lose momentum as the gases leave the rocket.



a) at time t b) at time $t = \Delta t$

Observe that, the initial mass of the rocket and fuel is $m + \Delta m$, at time t and the rocket speed is v. And at time $t + \Delta t$, the rocket mass has been reduced to M and an amount of fuel ΔM has been ejected, and hence the rocket speed increases by Δv.

$$\boxed{F = u\frac{\Delta m}{\Delta t} = ma}$$

Where: F is the force on the rocket, thrust

u is the velocity of the exhaust gas

$\dfrac{\Delta m}{\Delta t}$ is the rate of change of mass of the rocket

→ As rain drops combine with other droplets of water, the mass of the drops increases and hence causes the momentum of the rain drop to increase.

$$\boxed{F = ma + u\frac{\Delta m}{\Delta t}}$$

Where, F is the force acting on the raindrop.
u is the velocity of the extra mass.
is the rate of increase in mass of the rain drop.

Illustrative Examples

96. A rocket has a total mass of 1×10^5 kg and a bumout mass of 10^4 kg. The rocket exhausts all its fuel in 4 min, burning the fuel at a steady rate with an exhaust velocity of 4.5×10^3 m/s.
a) What thrust does the engine develop at lift off?
b) What is the initial acceleration of the rocket?

Solution: $\Delta m = M_f - M_i = 1 \times 10^4 - 1 \times 10^5 = -9 \times 10^{+4}$ kg

$$\frac{\Delta m}{\Delta t} = \frac{-9\times10^4}{2.4\times10^2} = -3.75\times10^2 \text{kg/sec}$$

a) $F = u\dfrac{\Delta m}{\Delta t} = (4.5 \times 10^3)(3.75 \times 10^2) = 1.69 \times 10^6 N$

b) $F - mg = ma \Rightarrow a = \dfrac{F-mg}{m} = \dfrac{1.69\times10^6}{1\times10^5} - 10 = 6.9 m/s^2$

97. An open truck of mass 24000 kg is moving along a friction less track with 4 m/s. If it is raining hard, what is the speed of the truck after it collected 3000 kg of rain water?

Solution: $m_1u_1 = (m_1+m_2)\ v \Rightarrow (24000)(4) = (27000)v \Rightarrow v = 3.5 m/s$

98. A rocket is designed to produce 15,500N of thrust a liftoff. If the speed of the ejected full is 2500 m/s, what is the rate at which the fuel must be burned?

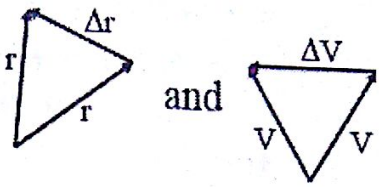
Solution: $\dfrac{\Delta m}{\Delta t} = \dfrac{\text{thrust}}{u} = \dfrac{15,000}{2,500} = 6.2 m/s$

4.8 Dynamics of Circular Motion

A. Horizontal Circle

When a body is moving in a horizontal circle, its speed remains constant, but the direction of motion is changing and hence, its velocity changes resulting in an acceleration called *centripetal acceleration* since it points towards the center of the circle.
The centripetal acceleration of a body in a circular motion can be obtained by considering the position and velocity vectors as shown.

This is, from similarity of triangles theorem, since they are isosceles triangles $\frac{\Delta v}{v}=\frac{\Delta r}{r}$, divide both sides by $\frac{1}{\Delta t}$



$$\frac{\Delta v}{\Delta tv}=\frac{\Delta r}{\Delta tr}\Rightarrow\frac{a}{v}=\frac{v}{r}$$

$$\boxed{a_c=\frac{v^2}{r}}\longleftarrow \text{Centripetal Acceleration}$$

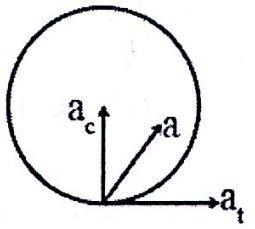
- For one cycle, the speed of the body is given by $\boxed{v=\frac{2\pi r}{T}}$

And, the force that keeps the body in circular path is called *centripetal force* and is given by;

$$\boxed{F_C=ma_C=\frac{mv^2}{r}}$$

And, when the body is moving in a changing speed, we have tangential acceleration, which arises from a change in speed.

Where, $\boxed{a_t=\frac{\Delta v}{\Delta t}}$



The total acceleration is then, $\vec{a}=\vec{a}_c+\vec{a}_t$

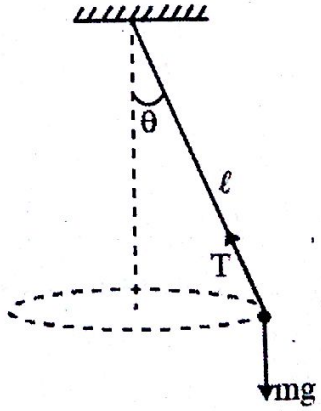
$$\boxed{a=\sqrt{a_t^2+a_c^2}}$$

Illustrative Examples

99. A ball of mass 0.5 kg is connected at the end of a 1.5m string. If the ball is whired in a horizontal circle, and if the string can tolerate a maximum tension of 50N, what maximum speed can the ball have before the sting breaks?

Solution: $F_C=\frac{mv^2}{r}\Rightarrow v_{max}=\sqrt{\frac{(F)(r)}{m}}=\sqrt{\frac{(50)(1.5)}{0.5}}=12.2\text{m/s}$

100. Consider a conical pendulum with an 80 kg bob on a 10m wire making an angle of 5° with the vertical. Determine:
   a) The horizontal and vertical components of the forces of the wire.
   b) The radial acceleration of the bob.



Solution: a) $T\cos\theta = mg = (80)(10) \Rightarrow T = 803\ N$

Horizontal $\rightarrow T\sin\theta = (803)(\sin 5) = 70N$

Vertical $\rightarrow T\cos\theta = (803)(\cos 5°) = 800N$

b) $a = \frac{F_C}{m} = \frac{T\sin\theta}{m} = 0.9 m/s^2$

101. If the coefficient of static friction for tires on road is 0.25, at what maximum speed can a car turn round a level curve of 47.5m radius without slipping?

Solution: $f_s = f_C$, $\mu_s mg = \frac{mv^2}{r}$

$$v = \sqrt{\mu_s gr} = \sqrt{(0.25)(10)(47.5)} = 10.9 m/s$$

B. Vertical Circle

When a body moves in a vertical circle, the forces acting on it are centripetal force, normal force and gravitational force.

Moreover, when a body is moving in a horizontal circle, its speed remains constant, but when it is moving in a vertical circle, its speed varies. The speed decreases as the body moves towards the top of the circle and increases as it moves towards the bottom of the circle.

Now, consider a body of mass m fixed at a rope of length L which is equivalent to the radius r when the body rotates in a vertical circle.

At any time then, the tension in the rope is given by

$$\boxed{T - mg\cos\theta = \frac{mv^2}{r}}$$

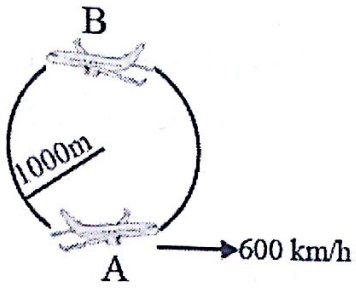
☛ We can see that, the tension has maximum value at the bottom and minimum value at the top of the circle.

Illustrative Examples

102. A pilot flies an airplane at a constant speed of 600 km/h in a vertical circle of radius 1000m. What is the force exerted by the seat on the 80 kg pilot at point A and point B?

Solution: FBD at A,

$$F_{NA} - mg = \frac{mv^2}{r}$$



$$F_{NA} = mg + \frac{mv^2}{r}$$

$$= (80)(10) + \frac{(80)(167)^2}{1000} = 3031N$$

FBD, at B

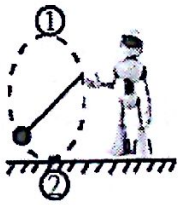
$$F_{NB} + mg = \frac{mv^2}{r}$$

$$F_{NB} = \frac{mv^2}{r} - mg = \frac{80(167)^2}{1000} - (80)(10) = 1431N$$

103. A Robot swings a small 50 g ball attached at the end of 1m string and the ball moves in vertical circle.

a) What is the minimum speed of the ball in position 1?

b) If this speed is maintained, what will be the tension in the string at position 2?



Solution: For minimum speed, T = 0

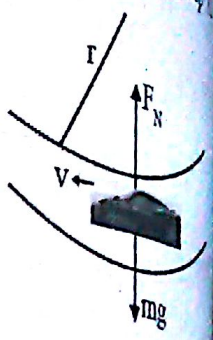
a) At 1: $T - mg\cos\theta = \frac{mv^2}{r}$, $\theta = 180°$, at position 1 (top)

$$mg = \frac{mv^2}{r} \Rightarrow v = \sqrt{10}m/s$$

b) At 2: $T - mg\cos\theta = \frac{mv^2}{r}$, $\theta = 0°$ at position 2 (bottom)

$$T = mg + \frac{mv^2}{r} = (0.05)(10) + (0.05)\frac{\left(\sqrt{10}\right)^2}{1} = 1N$$

104. While a car is moving at a speed of 17 m/s, it encounter a dip in the road of radius 65m. What is the normal force exerted by a car seat on a 80 kg passenger when the car is at the bottom of the dip?



Solution: $F_N - mg = \frac{mv^2}{4}$

$$F_N = mg + \frac{mv^2}{r} \Rightarrow F_N = 1140N$$

☛ **Observe that,** at the bottom, the normal force is greater than the weight of the passenger since it must supply the centripetal force. As a result, the passenger feels heavier than usual. As a result, the 80 kg passenger feels as if his mass has increased to 116kg.

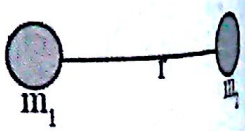
4.9 Newton's Law of Universal Gravitation

In addition to his famous three laws of motion, Newton also formulated the law of universal gravitation which explains that, every particle in the universe exerts an attractive force on every other particle.

> **Newton's Law of Universal Gravitation** is stated as "Everybody in the universe attracts every other body with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them."

In mathematical notation, it may be written as,

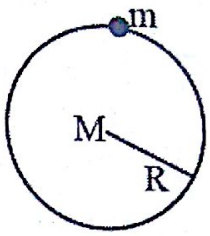
$$F = G\frac{m_1 m_2}{r^2}$$



Where G is a constant known as *universal gravitational constant*, and is equal to, $G = 6.67 \times 10^{-11} Nm^2/kg^2$

Variation of Acceleration due to Gravity

Objects fall downward because of gravity on the earth's surface at about an acceleration of 9.81 m/s^2. However, nothing has been said about why g is 9.81 m/s^2. The reason is fascinating, as we will now see.



Consider a body of mass m is on the surface of the earth of mass M and radius R. Now, the force of gravity is balanced with the weight. Therefore, $mg = \frac{GmM}{R^2}$

$$\boxed{g = \frac{GM}{R^2}}$$

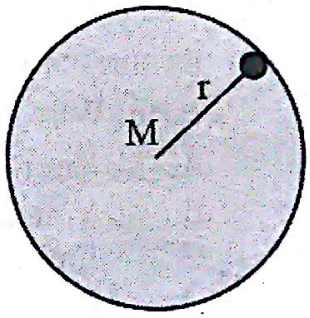
For, $M = 6 \times 10^{24}$ kg and R = 6400 km

$g = 9.81\ m/s^2$

- The acceleration due to gravity varies, with factors such as altitude, depth, latitude, etc.

Illustrative Examples

105. Compute the mass of the earth, assuming it to be a sphere of radius 6370 km.



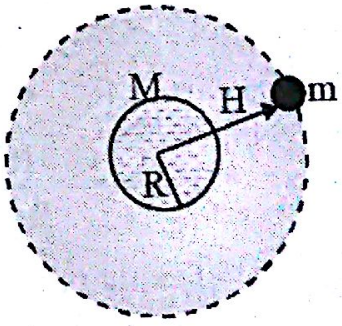
Solution: Let M be mass of the earth and m mass of a body on the earth's surface.

Gravitational force = Weight

$$\frac{G_m M}{r^2} = mg$$

$$\Rightarrow M = \frac{gr^2}{G} = \frac{(10)(6.37\times10^6)^2}{6.67\times10^{-11}} = 6\times10^{24}\,kg$$

106. The moon orbits the earth in a circular path of radius 3.8×10^8m. It takes about 27 day to complete one revolution. Compute the mass of the earth.



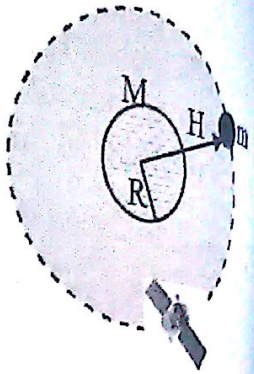
Solution: Gravitational = Centripetal force

$$\frac{GmM}{r^2} = \frac{mv^2}{r}$$

$$v^2 = \frac{GM}{r}, M = \frac{v^2 r}{G}, v = \frac{2\pi r}{T}$$

$$M = \frac{4\pi^2 r^3}{GT^2} = \frac{4(3.14)^2(3.8\times10^8)^3}{(6.67\times10^{-11})(2.3\times10^6)^2} = 6\times10^{24}\,kg$$

107. A satellite orbits the Earth at a height of 1000 kg above the Earth's Surface. What is the orbital speed of the satellite?



Solution: $F_C = F_g$

$$\frac{mv^2}{r} = \frac{GmM}{r^2},\ r = h + R = 7.4\times10^6 m$$

$$v = \sqrt{\frac{GM}{r}} = 7.4\times10^3\ m/s$$

108. Uniform circular motion is? (UEE 2003)

A. Motion with constant velocity.
B. Motion in one dimension.
C. Motion with acceleration of constant magnitude.
D. Motion in which net force is constant.

Answer: C

109. The planet Jupiter is about 318 times as massive as Earth, yet on its surface you would weigh only about 2.4 times your weight on earth. Why is this happened? (UEE 2005)

A. Your mass is 132.5 times less on Jupiter.
B. You are 132.5 times more weightless there.
C. Jupiter is significantly farther from the sun.
D. Jupiter's radius is about 11.5 times the Earth's radius.

Solution: The acceleration due to gravity on Jupiter is 2.4 times that of the earth. i.e., $g_{jupiter} = 24g$

$$g = \frac{GM_e}{R_e^2}\ldots(*),\ \text{and}\ g_{jup} = \frac{GM_{Jup}}{R_{Jup}^2}, M_{Jup} = 318M_e$$

$$g_{jup} = \frac{G(318)M_e}{R_{Jup}^2}\ldots(**)$$

Taking the ratio of equation (**) to equation (*)

$R_{Jup} = 11.5R_e$

110. A women is rotating a bucket of water in a vertical circle of radius 0.9m. The mass of bucket and water is 5 kg. What is the bucket's minimum speed at the top of the circle if no water is to spill out? (UEE: 2007)

Answer: D

A. 0 B. 1m/s C. 3m/s D. 9 m/s

Solution: $\frac{mv^2}{r} = mg \Rightarrow v = \sqrt{gr} = \sqrt{(10)(0.9)} = 3$ m/s

Answer: C

111. A highway curve is banked, inclined in such a way that a car travelling at a speed of 13.5 m/s can round the curve without skidding, in the absence of friction. If the banking angle is 30°, what is the radius of the curve? (UEE 2007)

A. 10.5m B. 21.0m C. 31.7m D. 36.7m

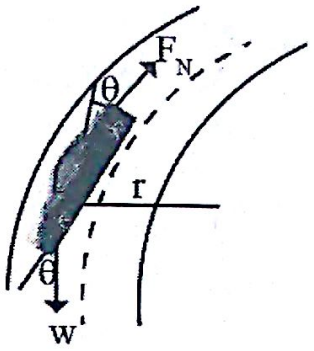
Solution: $\sum F_y = 0$

$F_N \cos\theta = mg \ldots (*)$

$\sum F_x = \frac{mv^2}{r}$

$F_N \sin\theta = \frac{mv^2}{r} \ldots (**)$

Taking ratio, (**) to (*)



$\tan\theta = \frac{v^2}{rg}$, or $r = \frac{v^2}{g\tan\theta} = \frac{(13.5)^2}{(10)(0.58)} \simeq 31.7$ m **Answer: C**

Additional Examples

Laws of motion

112. When a body is stationary

A. There is no force acting on it.
B. The force acting on it is not in contact with it.
C. The combination of forces acting on it balance each other.
D. The body is in vacuum. **Answer: C**

113. Why an athlete runs some distance before taking a long jump? This is because by doing so

A. The athlete acquires larger inertia of motion.
B. The athlete overcome inertia of rest.
C. The athlete gets inertia of direction.
D. The athlete acquires kinetic energy. **Answer: A**

114. A force of 6N acts on a mass of 1kg which acquires a velocity 3 m/s. The time for which the force acts is

A. 26s B. 6s C. 5s D. 2s

121. When horse starts running all of a sudden, the rider on the horse back falls back because

A. He is taken aback.
B. He is afraid.
C. Due to inertia of rest, the upper part of his body remain at rest.
D. Due to inertia of motion, the lower part of his body comes in motion.

Hint: Think of Newton's first law. **Answer: C**

122. A force of 5N is applied to one end of a spring balance and a force of 5N is applied simultaneously at the other end in the opposite direction. The reading of the balance will be

A. 2.5N B. 3N C. 5N D. 0N

Hint: Think of Newton's third law. **Answer: C**

123. A thief steals box of weight W and jumps from the third floor of a building. During jump, he experiences a weight of

A. 1W B. 3W C. 1.5W D. zero

Hint: Think of weightless ness **Answer: D**

124. A curtain hanging in a moving Aeroplane hits the face of the passenger sitting behind. At this particular time, the aeroplane is

A. Accelerating forward.
B. Accelerating backward.
C. Moving forward with uniform speed.
D. Being braked upon.

Hint: The acceleration of the curtain equal but opposite the plane acceleration. **Answer: A**

125. An empty plastic box of mass 5 kg is observed to accelerate up at the rate of $g/6$ when placed deep inside water. What mass of sand should be put inside the box so that it may accelerate down at the rate of $g/6$?

A. 1kg B. 1.5kg C. 2kg D. 2.5kg

Solution: From F = ma

$F - mg = ma$ . . . 1st case

$$F - 5g = 5\left(g/6\right) = \frac{5}{6}g \Rightarrow F = \frac{35}{6}g$$

$Mg - F = ma \ldots 2^{nd}$ case

$(m + 5)\,g - F = (m + 5)\left(g/6\right)$

$\Rightarrow gm + 5g - F = g/6\, m + 5/6\, g,\ F = 5/6\, g + 5g$

$gm + 5g - \left(5/6\, g + 5g\right) = g/6\, m + 5/6\, g$

$gm - 5/6\, g = g/6\, m + 5/6\, g$

$5\,gm/6 = \frac{10}{6} g \Rightarrow m = \frac{10}{5} = 2kg$

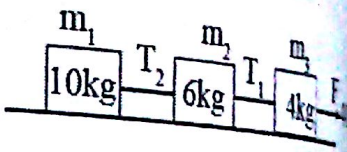
126. A 40N force pulls a system of three masses on a horizontal frictionless surface. Tension T_1 is

A. 40N B. 32N C. 10N D. 5N

Solution: $F - T_1 = m_3 a$

$T_1 - T_2 = m_2 a$

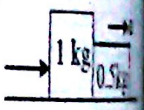
$T_2 = m_1 a$



$$\Rightarrow a = \frac{F}{m_1 + m_2 + m_3} = \frac{40}{4 + 6 + 10} = 2m/s$$

$\therefore 40 - T_1 = 4(2)$ or $T_1 = 32N$

127. A 1 kg block and a 0.5kg block move together on a horizontal frictionless surface. If each block exerts force of 6N on the other. Their acceleration is:



A. $3\ m/s^2$ B. $6m/s^2$ C. $9m/s^2$ D. $12m/s^2$

Solution: $6 = 0.5a \Rightarrow a = 12\ m/s^2$

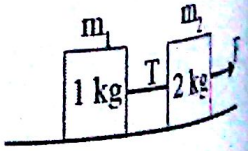
128. Two blocks of masses 1 kg and 2 kg rest on a smooth horizontal table. When the 2 kg block is pulled by a certain force F, the tension T in the string is 1.5N. The value of F is

A. 1.5N B. 2.5N C. 3.5N D. 4.5N

Solution: $T = m_1 a,\ a = T/m_1$

and $F - T = m_2 a$

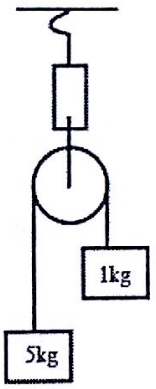
$$\Rightarrow F = m_2 a + T = m_2\left(\frac{T}{m_1}\right) + T$$



$F = \frac{2}{1}(1.5) + 1.5 = 4.5N$ **Answer: D**

129. A smooth pulley of negligible weight is suspended by a spring balance. Weight of 1 kg and 5 kg are attached to the opposite ends of a string passing over the pulley, and move with acceleration due to gravity. During their motion, the spring balance reads a weight of

A. 6 kg
B. Less than 6 kg
C. More than 6 kg
D. May be more or less than 6 kg



Explanation:

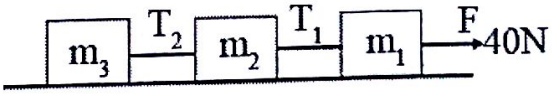
Spring balance reading is interms of

$$\text{Newton} = \frac{4m_1m_2}{m_1 + m_2} = \frac{4(5)(1)}{5+1} = \frac{10}{3}$$

Answer: B

130. Three blocks of masses $m_1 = 10$ kg, $m_2 = 6$ kg and $m_3 = 4$ kg are connected by a string on a smooth surface. If they are pulled by a force of 40N, then tension T_1 will be

A. 20N B. 40N
C. 10N D. 32N



Solution: $F - T_1 = m_1a$

$T_1 - T_2 = m_2a$

$T_2 = m_3a \Rightarrow T_1 = F - m_1a = 40 - (10)(2) = 20N$

$\Rightarrow a = \frac{F}{m_1 + m_2 + m_3} = \frac{40}{10+6+4} = 2m/s^2$ **Answer: A**

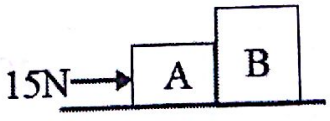
131. On a smooth plane surface, two blocks A and B are accelerated by applying a force 15N on A. If mass of B is twice that of A, the force on B is?

A. 30N B. 15N C. 10N D. 5N

Solution: Common acceleration,

$m_B = 2mA$

$15 = (m_A + m_B)\, a \Rightarrow a = \frac{15}{3mA}$



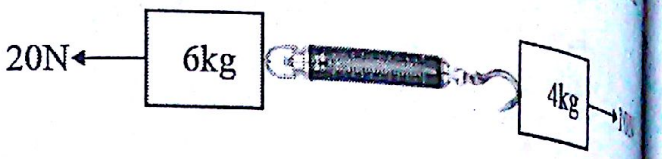
The force on m_B is,

$$F = (m_B)a = (2m_A)\left(\frac{15}{3m_A}\right) = 10\,N$$

Answer:

132. A newton meter is attached to two blocks of masses 6 kg and kg. Forces of 20N and 10N are applied on the blocks. What is the meter reading?

A. 10N
B. 20N
C. 6N
D. 14N



Solution: $20 - T = 6a$

$T - 10 = 4a$

$\Rightarrow a = 1\ m/s^2$ and $T = 14N$

Answer:

133. A monkey of mass 20 kg is holding a vertical rope. The rope will not break when a mass of 25kg is suspended from it but will break if the mass exceeds 25kg. What is the maximum acceleration with which the monkey can climp up the rope?

A. $5m/s^2$ C. $25m/s^2$
B. $10m/s^2$ D. $2.5m/s^2$

Solution: $T - W = ma$

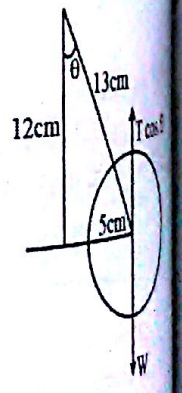
$25g - 20g = 20a,\ g = 10\ m/s^2$

$5g = 20a \Rightarrow a = 2.5\ m/s^2$

Answer:

134. A uniform sphere of weight W and radius 5 cm is being held by a string. The tension in the string is

A. $\frac{12}{5}W$ C. $\frac{13}{12}W$
B. $\frac{13}{5}W$ D. $\frac{12}{13}W$



Solution: $T\cos\theta = W$

$$T = \frac{W}{\cos\theta} = \frac{W}{\frac{12}{13}} = \frac{13}{12}W$$

Answer:

135. A 0.5 kg ball thrown up in air with velocity 30 m/s reaches the highest point in 2.5 second. If g = 10m/s^2 then the average air resistance encounter by the ball during upward motion is

A. Zero B. 0.5N C. 1N D. 1.5N

Solution: $v = u + at \Rightarrow 0 = 30 + 2.5a$ or $a = -12$ m/s^2. This is total retardation due to air resistance and gravity. The retardation due to air resistance is then, $a = 12 - 10 = 2$ m/s^2

$\therefore$ Air resistance $= (0.5)(2) = 1$N **Answer: C**

136. Two carts joined by a string are pulled by a 20N force on a frictionless surface, what force is exerted by the string on the 3 kg cart?

A. 10N B. 20N C. zero D. 15N

Solution: Combined acceleration

$20 = (3 + 1)a \Rightarrow a = 5$ m/s^2

The force of the string on the 3 kg cart is, $(3)(5) = 15$N

Answer: D

137. When a horse pulls a cart, the force that helps the horse to move forward is the force exerted by

A. The cart on the horse
B. The ground on the horse
C. The ground on the cart
D. The horse on the ground

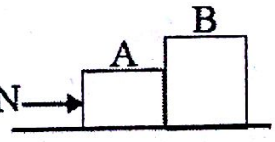
Answer: B

138. A car accelerates on a horizontal road due to the force exerted by

A. The engine of the car
B. The driver of the car
C. The earth
D. The road

Answer: D

139. Two blocks A and B of masses 4 kg and 12kg are placed on a smooth plane surface. A force of 16N is applied on A as shown. The force of contact is



A. 4N B. 8N C. 12N D. 16N

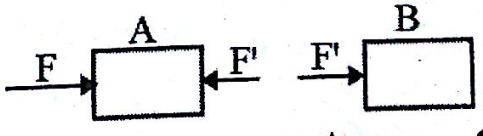
Solution: Common acceleration, $a = \dfrac{F}{m_A + m_B} = \dfrac{16}{4+12} = 1\text{m/s}^2$

Let F' is force of contact,

$F - F' = m_A a$

$\Rightarrow F' = F - m_A a$

$= 16 - (4)(1) = 12$N



Answer: C

Friction

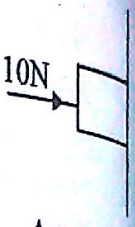
140. A body of mas 2 kg is placed on a horizontal surface having coefficient of static friction 0.5. If the force applied on the body is 2.5N, then the frictional force acting on the body will be?

A. 8N B. 10N C. 20N D. 2.5N

Solution: $f_s = (0.5)(2)(10) = 10N$

Since the applied force is less than the force of friction, the force of friction is equal to the applied force. **Answer: D**

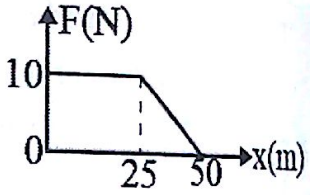
141. A horizontal force of 10N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is



A. 2N B. 20N C. 50N D. 100N

Solution: Weight of block = $\mu F = (0.2)(10) = 2N$ **Answer: A**

142. An object of mass 5 kg is acted upon by a force that varies with the position of the object as shown. If the object starts out from rest at the point x = 0, then its speed at x = 25m is



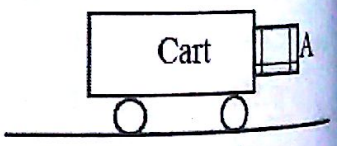
A. 5m/s C. 20m/s

B. 10m/s D. 28m/s

Solution: $a = \frac{10}{5} = 2m/s^2$

$v^2 - u^2 = 2as \Rightarrow v^2 = 2(2)(25) = 100$ or $v = 10$ m/s **Answer: B**

143. If μ is coefficient of friction between cart and block, what acceleration must the cart have in order that the block A will not fall?



A. μg C. μ/g

B. g/μ D. $\mu + g$

Solution: $\mu ma = mg \Rightarrow a = g/\mu$ **Answer: B**

144. With what maximum acceleration can fireman slide down a rope whose breaking strength is two-third of his weight?

A. $\frac{2}{3}g$ B. g C. $g/3$ D. Zero

Solution: $mg - \frac{2}{3}mg = ma \Rightarrow a = g/3$ **Answer: C**

145. A bullet of mass 25g moving with velocity 2m/s is stopped within 5 cm of a target. The average resistance of the target is

A. 1N B. 2N C. 4N C. 8N

Solution: $v^2 = u^2 + 2as \Rightarrow \Rightarrow a = \frac{u^2}{2s} = \frac{(2)^2}{2(0.05)} = -40m/s^2$

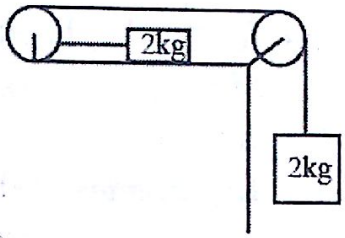
$$f = ma = \left(\frac{25}{1000}\right)(40) = 1N$$

Answer: A

146. Two blocks each of mass 2 kg are connected by a string passing over two pulleys. Assume all the surfaces are frictionless. The tension in the string is

A. 2g Newton C. g Newton

B. $g/2$ Newton D. $g/3$ Newton



Solution: For the hanging block

$$2g - T = 2a \ldots\ldots\ldots (*)$$

For the block on the horizontal surface

$$T = 2a \ldots\ldots\ldots\ldots (**)$$

From the two equations, T = g newton **Answer: C**

147. A glass plate can tolerate a maximum weight of 12 kg. A 10 kg block is placed on the glass plate and this system is moved vertically upwards with a uniformly increasing acceleration. The glass plate breaks at a certain value of acceleration. The value of this acceleration is

A. g B. $g/2$ C. $g/3$ D. $g/5$

Solution: 12g = m (g + a), for upward motion

$$12g = 10(g + a) \Rightarrow a = g/5$$

Answer: D

148. A force of 200N is required to push a car of mass 500kg slowly at constant speed on a level road. If a force of 500N is applied, the acceleration of the car in m/s^2 will be

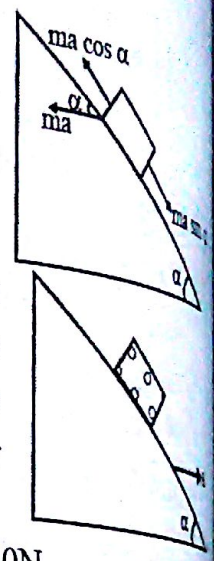
A. Zero B. 0.2 C. 0.6 D. 1.0

Solution: $a = \frac{F_{net}}{m} = \frac{500 - 200}{500} = 0.6m/s^2$ **Answer: C**

149. A block is kept on a frictionless inclined surface with angle of inclination α. The incline is given an acceleration a to keep the block stationary. Then the acceleration a is equal to

A. g tan α C. g cosec

B. g D. $\alpha \frac{g}{\tan \alpha}$



Solution: ma cos α = mg sin α

a = g tan α

Answer: A

150. A woman of 50 kg mass is standing in an elevator. Suddenly, the cable of the elevator break and the elevator begins to fall freely. The force exerted by the floor of the elevator on the woman is?

A. 0N B. 500N C. 100N D. 5N

Explanation: When a body is falling freely, it is a condition of weightlessness.

Answer: A

151. The apparent weight of a person of mass m in an elevator is 2mg. The elevator is moving

A. Up with an acceleration of $\frac{g}{2}$

B. Up with an acceleration of g.

C. Up with an acceleration of 2g

D. Down with an acceleration of g.

Solution: 2 mg = m(g + a) ⇒ a = g

Answer: B

152. An elevator is moving up with an acceleration $\frac{g}{5}$. The apparent weight of a 60 kg man standing in the lift is?

A. 480N B. 720N C. 600N D. 1000N

Solution: Apparent weight = mg + ma = 720N

Answer: B

153. The ratio of the weight of a man in a stationary lift and in a lift accelerating downwards with a uniform acceleration a is 3:2. The acceleration of the lift is:

A. $\frac{g}{3}$ B. $\frac{g}{2}$ C. g D. 2g

Solution: When stationary, weight = mg

When accelerating down, apparent weight = mg – ma

Since, $\frac{mg}{mg-ma}=\frac{3}{2}\Rightarrow a=\frac{g}{3}$ **Answer: A**

154. A man weighs 80 kg. He stands in a lift which is moving upwards with a uniform acceleration of 5 m/s^2. What would be the reading on the scale?

A. Zero B. 400N C. 800N D. 1200N

Solution: Apparent weight = mg + ma = 1200N **Answer: D**

155. A bag weighs 49N when the elevator is at rest. What will be weight of the bag if the left is moving down with an acceleration of 5 m/s^2?

A. 49N B. 24N C. 74N D. 15N

Solution: Apparent weight = $4g-\left(\frac{49}{10}\right)(5)=24\,N$ **Answer: B**

156. A girl in a lift will weigh more when the lift

A. Is falling freely
B. Is moving down with an acceleration $a < g$
C. Is moving down with an acceleration $a > g$
D. Is about to move up

Explanation: Only an upward acceleration increases apparent weight. **Answer: D**

Momentum and Impulse

157. The momentum of the system is conserved

A. Always C. Only in the absence of external force
B. Never D. Only when an external force acts

Answer: C

158. A bomb at rest suddenly disintegrates into two pieces of equal mass. The fragmented masses will move in

A. Opposite directions with equal speeds.
B. Opposite directions with equal velocities.
C. Opposite directions with unequal velocities.
D. Same direction with same velocity.

Hint: The momentum is conserved, which is zero. **Answer: A**

159. A machine gun mounted on a 2000kg car on a horizontal frictionless surface fires 10 bullets per second. If 10g be the mass of each bullet and 500 m/s the velocity of each bullet, then the acceleration of the car will be?

A. $\frac{1}{10}$m/s² B. $\frac{1}{20}$m/s² C. $\frac{1}{40}$m/s² D. $\frac{1}{60}$m/s²

Solution: Care acceleration $= \frac{\text{force on the car}}{\text{mass of the car}}$

$$a = \frac{MnV}{M} = \frac{(0.1)(10)(500)}{2000} = \frac{1}{40}\text{m/s}^2$$

Answer: C

160. An object at rest in space suddenly explodes into three parts of same mass. The momentum of the two parts are 2pi and pj. The momentum of the third part will have a magnitude of

A. $P\sqrt{3}$ B. $P\sqrt{5}$ C. P D. 2P

Solution: $P = \sqrt{(2P)^2 + P^2} = P\sqrt{5}$

Answer: B

161. A particle of mass moving with velocity v strikes a stationary particle of mass 2m and strikes to it. The speed of the system will be

A. $v/2$ B. 2v C. $v/3$ D. 3v

Solution: $(2m + m)v_2 = mv \Rightarrow v_2 = v/3$

Answer: C

162. A bomb explodes in air when it has a horizontal speed of 200 km/h. It breaks into two pieces A and B of mass ratio 1: 2. If A goes vertically at speed 400 km/h, the speed of B will be?

A. 200km/h B. 250km/h C. 300km/h D. 500 km/h

Solution: $(3m)100 = 2mv_x$ or $v_x = 150$ km/h

Again, $(m)(400) = 2mv_y$ or $v_y = 200$ km/h

$$v = \sqrt{v_x^2 + v_y^2} = 250\text{km/h}$$

Answer: B

Rocket Propulsion

163. A 6000 kg rocket is set for vertical firing. If the exhaust speed is 1000 m/s, the gas ejected per second to supply the thrust needed to overcome the weight of the rocket is

A. 6 kg/s B. 60kg/s C. 600kg/s D. 6000kg/s

Solution: $v\frac{\Delta m}{\Delta t} = mg$ or $\frac{\Delta m}{\Delta t} = \frac{mg}{v}$

$$\frac{(6000)(10)}{1000} = 60\text{kg/s}$$

Answer: B

164. A rocket with a lift – off mass 10^5kg is blasted upward with an initial acceleration of 5 m/s^2. The initial thrust of the blast is

A. 1.5×10^2N C. 1.5×10^5N

B. 1.5×10^3N D. 1.5×10^6N

Solution: Thrust = mg + ma = 1.5×10^6N **Answer: D**

165. In a rocket, fuel is ejected with a velocity of 6 km/s. If the fuel burns at a rate of 1 kg/s, then the force exerted by the exhaust gas on the rocket is

A. 6000N B. 600N C. 60N D. 6N

Solution: $F = v\Delta m/\Delta t = (6000)(1) = 6000N$

Answer: A

Collisions

166. A body of mass m moving with velocity v makes a head on elastic collision with another body of mass 2m which is initially at rest. The loss of kinetic energy of mass m is?

A. $1/2$ of its initial kinetic energy.

B. $1/9$ of its initial kinetic energy.

C. $8/9$ of its initial kinetic energy.

D. $1/4$ of its initial kinetic energy.

Hint: Fractional loss of kinetic energy = $\frac{4\times 2}{(1+2)^2} = \frac{8}{9}$

Answer: C

167. A mass of 20kg moving with 10 m/s collides with another stationary mass of 5 kg. As a result of collision, the two masses stick together. The kinetic energy of the composites mass will be?

A. 600J B. 800J C. 1000J D. 1200J

Solution: $(20)(10) = 25v \Rightarrow v = 8$ m/s

$$k_E = \frac{1}{2}mv^2 = \frac{1}{2}(25)(8)^2 = 800J$$ **Answer: B**

168. A ball hits the floor and rebounds after collision. Which of the following is correct?

A. The total momentum of the ball and the earth is conserved.

B. The total energy of the ball and the earth is conserved.

C. The mechanical energy of the ball remains the same.
D. The momentum of the ball just after collision is zero.

Answer:

169. Two perfectly elastic particles P and Q of equal mass travelling along the line joining them with 15 m/s and 10 respectively. After collision, their velocities respectively in will be?

A. 0, 25 B. 5, 20 C. 10, 15 D.

Explanation: For elastic collision in one dimension, between equal masses, their velocities are inter changed.

Answer:

170. The ratio of the velocity of separation to the velocity of appr of two bodies is called

A. Coefficient of elasticity
B. Coefficient of contact
C. Coefficient of veloc
D. coefficient of restitu

Hint: $e = \frac{v_2 - v_1}{u_1 - u_2}$

Answer:

171. A mass of 10 g moving with a velocity of 100 cm/s strik pendulum bob of mass 10g. The two masses stick together. T maximum height reached by the system is

A. Zero B. 5 cm C. 2.5cm D. 1.25

Solution: $(10)(100) = (10 + 10)v \Rightarrow v = 50$ cm/s

And, $(2m)gh = \frac{1}{2}(2m)v^2$ or $h = \frac{v^2}{2g} = 1.25$cm

Answer:

172. Two bodies of masses 3 kg and 2 kg collide head on. T relative velocities before and after collision are 15 m/s and 5 respectively. The loss of kinetic energy of the system is

A. 120J B. 100J C. 80J D. 240J

Solution: $e = \frac{5}{15} = \frac{1}{3}$

Loss of $k_E = \frac{1}{2}\frac{m_1 m_2}{m_1 + m_2}(v_{1i} - -v_{2i})^2(1 - e^2) = 120$J

Answer:

173. Which of the following collisions is an elastic collision?

A. Two bodies collide and there is a rise in temperature.
B. Two bodies collide and sound of collision is heard.

C. Two steel balls collide and their kinetic energy conserved.
D. A man jumps on to a moving cart.

Answer: C

174. A 1 kg block moving with a velocity of 4 m/s collides with a stationary 2 kg block. The lighter block comes to rest after the collision. The loss of kinetic energy of the system is

A. 1J B. 2J C. 3J D. 4J

Solution: $(1)(4) = (2)v \Rightarrow v = 2$ m/s

Loss of $k_E = \frac{1}{2}(1)(4)^2 = -\frac{1}{2}(2)(2)^2 = 4J$

Answer: D

175. A body falls on the ground from a height of 10m and rebounds to a height of 2.5m. The ratio of the velocity before to after collision is

A. 2: 1 B. 1: 2 C. 4: 1 D. 3: 1

Solution: Before collision, $v_1^2 = (2g)(10) = 20g$

After collision, $0^2 = v_2^2 - 2g(2.5) \Rightarrow v_2^2 = 5g$

$\frac{v_1^2}{v_2^2} = \frac{4}{1} \Rightarrow \frac{v_1}{v_2} = \frac{2}{1}$

Answer: A

176. The quantities remaining constant in an elastic collision are

A. Momentum, kinetic energy and temperature.
B. Momentum, and kinetic energy but not temperature.
C. Momentum, and temperature but not kinetic energy.
D. Momentum, but neither kinetic energy nor temperature.

Answer: A

177. What percentage of k_E of a moving body is transferred to a stationary body when collides head on. Assume their mass are equal.

A. 25% B. 50% C. 75% D. 100%

Solution: $\frac{4x}{(1+x)^2} = \frac{(4)3}{(1+1)^2} = 1$

Answer: D

178. A metal ball hits a wall and does not rebound. But, a rubber ball of same mass on hitting the wall with same velocity rebounds. It can be concluded that:

A. Metal ball suffers greater change in momentum.
B. Rubber ball suffers greater change in momentum.
C. The initial momentum of metal ball is greater.

D. Both suffer same change in momentum.

Explanation: Change in momentum of rubber ball is 2mv but that of metal ball is mv.

Answer: B

Gravitation

179. An artificial satellite moving in a circular orbit around the earth has a total energy E_o. Its potential energy is?

A. $-E_o$ B. $1.5E_o$ C. $2E_o$ D. E_o

Solution: $E_o = -\frac{GMm}{2r} = \frac{1}{2}\left(-\frac{GMm}{r}\right) = \frac{u}{2} \Rightarrow u = 2E_o$

Answer: C

180. The period of revolution of a certain planet in an orbit of radius R is T. Its period of revolution in an orbit of radius 4R will be:

A. 2T B. $2\sqrt{2}T$ C. 4T D. 8T

Solution: $\frac{T'}{T} = \left(\frac{R'}{R}\right)^{3/2} = (4)^{3/2} = 8 \Rightarrow T' = 8T$

Answer: D

181. Acceleration due to gravity is g on the surface of Earth. Then the value of acceleration due to gravity at a height of 32 km above the surface of Earth is?

A. 0.99g B. 0.8g C. 1.01g D. 0.9g

Solution: $g' = g\left(1 - \frac{2h}{R}\right) = g\left(1 - \frac{2(32)}{6400}\right) = 0.99g$

Answer: A

182. If the distance between two point masses is doubled, the gravitational attraction between them

A. Is doubled
B. Becomes four times
C. Is reduced to half
D. Is reduced to a quarter

Answer: D

183. The gravitational force of attraction between two bodies, each of mass 1 kg and separated by 1m is

A. 6.67×10^{-11}N
B. 6.67×10^{-18}N
C. 6.67N
D. 1N

Hint: $F = \frac{Gm_1m_2}{r^2}$

Answer: A

184. Two spheres of masses m and M are situated in air and the gravitational force between them is F. If the space between them is filled with a liquid of specific gravity 3. The gravitational force will now be

A. 3F B. F C. $F/3$ D. $F/9$

Explanation: Gravitational force does not depend on the medium between the masses. **Answer: B**

185. If the mass of one of the two bodies is doubled and the distance between them is also doubled, the gravitational force between them

A. Remains unchanged
B. Gets doubled
C. Becomes half
D. Becomes zero

Hint: $F = \frac{Gm_1m_2}{r^2}$, for $m_1 = 2m_1$ and $r' = 2r$

$F' = \frac{F}{2}$ **Answer: C**

186. Average density of the earth

A. Is inversely proportional to g.
B. Is directly proportional to g.
C. Is a complex function of g.
D. Does not depend on g.

Explanation: $g = \frac{GM}{R^2} = \frac{G}{R^2}\left(\frac{4}{3}\pi R^2\right)\rho \Rightarrow \rho \sim g$ **Answer: B**

187. Imagine a new planet having the same density as that of earth but it is 3 times bigger than the earth in size. If the acceleration due to gravity on the surface of earth is g and that on the surface of the new planet is g' then

A. $g' = 3g$ B. $g' = 9g$ C. $g' = 27g$ D. $g' = \frac{g}{9}$

Solution: $g = \frac{Gm}{R^2} = \frac{G}{R^2}\left(\frac{4}{3}\pi R^2\right)\rho = \frac{4}{3}\pi GR\rho$

ρ is given to be constant

$\rho \sim R$ and $g' = 3R \Rightarrow g' = 3g$

Answer: A

188. Mass remaining constant, the radius of the earth shrinks by 1%. The acceleration due to gravity on the earth's surface would
A. Increase by 2% C. Decrease by 1%
B. Increase by 1% D. Decreases by 0.5%

Solution: $g = \frac{GM}{R^2}, \frac{\Delta g}{g} \times 100 = 2\frac{\Delta R}{R} \times 100 = 2 \times 1\% = 2\%$

Answer: A

189. If the Earth stops rotating, the apparent value of g on its surface will
A. Increase everywhere C. Remain the same everywhere
B. Decrease everywhere D. Increase or remain the same

Explanation: $g' = g - R\omega^2 \cos^2\phi$. At the poles, there will be no effect.

Answer: D

190. If the Earth shrinks, what will be the effect on the value of acceleration due to gravity g?
A. It will decrease C. It will become zero.
B. It will increase D. It will become infinite

Explanation: $g = \frac{GM}{R^2}$. If the Earth shrinks, its mass remains unchanged and its radius decreases. So, the value of acceleration due to gravity increases.

Answer: B

191. A body weighs 700N on the surface of the Earth. How much will it weigh on the surface of a planet whose mass is $\frac{1}{7}$ and radius of half that of the Earth?
A. 200N B. 400N C. 50N D. 300N

Solution: $g' = G\left(\frac{m}{7}\right)\left(\frac{2}{R}\right)^2 \Rightarrow g' = \frac{4}{7}g$

$$mg' = \frac{4}{7}mg = 400N$$

Answer: B

192. At sea level, a body will have minimum weight at
A. Pole B. equator C. 42° south D. 37° north

Solution: $g' = g - R\omega^2 \cos^2\phi$, $mg' = mg - mR\omega^2\cos^2\phi$
ϕ is least at the equator. So, $mR\omega^2 \cos^2\phi$ is maximum here.

Answer: B

193. Spot the wrong statement. The acceleration due to gravity g decreases if
A. We go down from the surface of the earth towards its center.
B. We go up from the surface of the Earth.
C. We go from the equator towards the poles on the surface of the earth.
D. The rotation velocity of the Earth is increased. **Answer: C**

194. If the Earth stops rotating about its axis, the acceleration due to gravity will remain unchanged at
A. Equator C. Latitude 60°
B. Latitude 45° D. Poles **Answer: D**

195. What keeps an Earth satellite moving in its orbit?
A. Gravitational attraction between satellite and Earth.
B. Ejection of gases from the satellite.
C. Burning of fuel.
D. Gravitational attraction of sun. **Answer: A**

196. Choose the correct statement from the following. Weightless of an astronaut moving in a satellite is due to
A. Zero g B. No gravity C. Zero mass D. Free fall
Answer: D

197. The tidal waves in the sea are primarily due to
A. The gravitational effect of moon on earth.
B. The gravitational effect of sun on earth.
C. The gravitational effect of Venus on earth.
D. The atmospheric effect of earth itself. **Answer: A**

198. If both the mass and radius of Earth decreases by 1%, the value of acceleration due to gravity will decrease by nearly
A. 1% B. 1.5% C. 2% D. 2.5%

Solution: $g' = G\dfrac{0.99M}{(0.99R)^2} = 1.01\dfrac{GM}{R^2} = 1.01g$

$$\frac{g'}{g} - 1 = 0.01 \quad \text{or} \quad \frac{g'-g}{g} \times 100 = 1$$

Answer: A

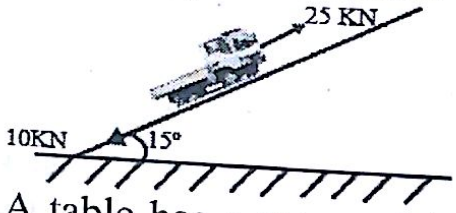
Unit 4: Conceptual Questions

1. If a body experiences no acceleration, can we conclude that no force is acting on it? Explain?
2. A big boy and a small boy stand facing each other on frictionless floor. If they push each other, who would experience a larger force?
3. If you are in an elevator that accelerates upward, you will feel heavier. Does this mean that your weight has increased?
4. What is wrong with the statement "Because the car is at rest, no force is acting on it."
5. In a frictionless world, which activity could you do?
   a) Driving a car
   b) Walking
   c) Climbing a ladder
   d) Writing a letter to your friend
6. A net force of 4N acts on a body initially at rest for 0.25 sec, and gives it a final speed of 5 m/s. How could a net force of 2N produce the same speed?
7. The moon is accelerating towards the Earth. Why isn't it getting closer to us?

Unit 4: Problems

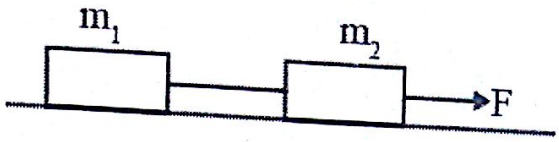
Basic laws of Dynamics

8. A truck of mas 8 tones is given a force of 25kN while it is retarded by 10kN force. Will it climb the plane?



9. A table has a mass of 150 kg. If $\mu_s = 0.45$ and $\mu_k = 0.4$ between the table legs and the floor.
   a) What force is needed to start the table moving?
   b) What force is needed to keep the table moving?
10. A person is dragging a 15 kg box along the ground by a rope 30° to the horizontal with a force of 58N, at a constant velocity. What is the coefficient of friction?
11. What force is required to give a body of mass 10 kg an acceleration of 2 m/s^2 up the 30° smooth incline plane?
12. A jet of 20,000 kg generates 160kN of force. If 40kN of drag force opposes its motion, how much time will it take to reach a speed of 33 m/s, from rest?

13. A 12kg box is released from the top of a 5m long incline making $40°$ with the horizontal. If a 60N friction impend the motion of the box.
    a) What will be the acceleration of the box?
    b) How long will it take to reach the bottom of the plane?
14. If F = 20N, $m_1 = m_2 = 3$ kg and acceleration = 0.5 m/s^2
    a) What is the tension in the cord?
    b) What is the friction acting?



Conservation of Linear Momentum

15. A ball of mass 4 kg falls on to a floor with a velocity of $\begin{bmatrix} 4 \\ -9 \end{bmatrix}$ m/s and bounces off with $\begin{bmatrix} 4 \\ 9 \end{bmatrix}$ m/s in 0.15 sec. What is
    a) Impulse of the ball.
    b) The force exerted by the floor on the wall.
16. A 0.2kg stone is dropped from a height of 2m. What is the impulse of the ball?
17. Particle A = 2kg moving at (6, -2) m/s collides with particle B = 3 kg moving at (10, 8) m/s. What is their common velocity?
18. Mass A of 0.8 kg moving at 5 m/s collides head on with mass B of 1.2 kg moving in opposite direction with 4 m/s. After collision, A moves with 4 m/s. Find the velocity of B after collision.

Elastic and Inelastic Collisions

19. Two particles, 5 kg and 2 kg collide and come to rest. If the 5 kg move at (8, −9) m/s before collision, what was the velocity of the 2 kg before collision?
20. A pool ball of mass 100g and velocity (5, -4) m/s collides with a stationary ball of the same mass. After the collision, one of the balls move at (2, -3) m/s. What is the velocity of the other ball?
21. A 400 g ball moving at 4 m/s collides with an unknown mass initially at rest. If the ball rebounds with half of its initial speed and assume perfectly elastic collision. Find:
    a) the velocity of the unknown mass
    b) the unknown mass
22. A gun 10 kg is fired a bullet 30 g with a speed of 330 m/s. What is the recoil velocity of the gun?

Momentum conservation in a variable mass system

23. A rocket of mass 850 kg consumes fuel at the rate of 2.3 kg/s. The speed of exhaust gas relative to the rocket is 2800 m/s. Find:
    a) the thrust provided by the rocket engine
    b) the initial acceleration of the rocket
24. A body of mass 50 kg explodes and splits into three pieces. The first piece has a mass of 10 kg and moves at (-18, 23) m/s. The second has a mass of 18 kg and moves at (25, -12) m/s. What is the velocity of the third piece?
25. A body explodes and splits into three pieces has a mass of 12 kg and a velocity of (30, -10 m/s), the second piece has a mass of 3.25 kg and velocity (-17, 10) m/s, the third piece has a velocity of (35.5, 40) m/s. Find:
    a) The mass of the third piece.
    b) The total mass of the body before the explosion.
26. A rock explodes into three pieces. The two pieces go off at right angles to each other. And $m_1 = 2$ g flies with 12 m/s, $m_2 = 4$ g flies with 8 m/s. If the third piece flies off with 5 m/s, what is its mass?

Dynamics of uniform circular motion and Law of Gravitation

27. Calculate the velocity of a space craft which orbits the moon in a circular path of 80 km altitude.
28. A child swings a ball of mass 0.6 kg in a horizontal circle such that the cord makes 53° with the vertical. What centripetal force acts on the ball?
29. A small particle of mass 0.2 kg rotates in a vertical circle at the end of a 1.3 m rope at a 4 m/s speed. Find:
    a) The tension in the rope as the particles makes 60° to the vertical.
    b) The tension as the particle reaches the bottom of its path.
    c) The tension as the particle reaches the top of its path.
    d) The critical speed of the particle.
30. A person is trying to ride a bike inside a pipe of 8m diameter in a vertical circle. If the mass of the bike and rider is 400 kg and moves at constant speed of 5 m/s.
    a) What is its acceleration at the bottom?
    b) What is the force on the bike at an angle of 30° up from the bottom?

c) What is the minimum velocity at the top for the bike and rider to move in a circle?

31. Assuming that the mass of the moon is $\frac{1}{81}$ of the mass of the earth, at what distance from the earth would the weight of a man be zero if the earth & the moon are 4×10^5 km apart?

32. At what point above the earth's surface would the acceleration due to gravity be $\frac{g}{9}$?

33. A satellite revolves in a circular path at a height of 60,300 km above the earth's surface.
    a) What is the orbital speed of the satellite?
    b) Find its period.
    c) What is the radial acceleration of the satellite?

Unit 4: Explanation to Questions

1. No. A body will not accelerate if No net force acts on it.
2. They experience the same forces. Think of Newton's 3rd law.
3. No. The extra force comes from the elevator floor reaction. Your weight is constant as long as you are near the Earth's surface.
4. The car is at rest, because the forces acting on it are balanced.
5. None are possible in the absence of friction.
6. $F_1\Delta t_1 = F_2\Delta t_2 \Rightarrow \Delta t_2 = 0.5$ sec
7. Yes, the moon is accelerating towards the earth but, its acceleration is balanced by the centripetal and gravitational forces.

Unit 4: Solution to Problems

8. $F - f - mg\sin\theta = -5.3$N. NO, the truck cannot climb.
9. a) $F = \mu_s F_N = \mu_s mg = 675$N b) $F = \mu_k F_N = \mu_k mg = 600$N
10. $F\cos\theta = f_k \Rightarrow \mu_k(mg - F\sin\theta) = F\cos\theta$ or $\mu_k = 0.4$
11. $F - mg\sin\theta = ma \Rightarrow F = ma + mg\sin\theta = 30$N
12. $F - f = ma \Rightarrow a = 6$ m/s^2 and, $v = u + at$, or $t = 5.5$ sec
13. a) $mg\sin\theta - f = ma \Rightarrow a = 1.4$ m/s^2

    b) $s = ut + \frac{1}{2}at^2 \Rightarrow t = 2.7$ sec
14. a) For m_1, $T - f = m_1a$ and for m_2, $F - T - f = m_2a \Rightarrow T = 10$N
    b) $f = 8.5$N

15. a) $I = \Delta P = m(v-u) = (0.72)\text{kgm/s}$ b) $F = \frac{\Delta P}{\Delta t} = 0.48\text{N}$

16. $v^2 = u^2 + 2gh \Rightarrow v = 6.3\text{ m/s}$, $I = \Delta P = m(v-u) = 1.26\text{ kgm/s}$

17. $m_1 = (v_{1x}, u_{1y}) + m_2 (v_{2x}, u_{2y}) = m_1 + m_2 (v_x, v_y) \Rightarrow v = (8.4, 4)\text{m/s}$

18. $m_A u_A + m_B u_B = m_A v_A + m_B v_B$

19. $m_1 (u_{1x}, u_{1y}) + m_2(u_{2x}, u_{2y}) = 0 \Rightarrow v_2 = (-20, 22.5)\text{m/s}$

20. $m_1 (v_{1x}, v_{1y}) = m_1(v_{1x}, v_{1y}) + m_2(v_{2x}, v_{2y}) \Rightarrow v_2 = (3, -1)\text{m/s}$

21. a) $v = 2\text{ m/s}$ b) $m = 1.2\text{ kg}$

22. $m_g u_g + m_b u_b = m_g u_g + m_b u_b \Rightarrow v_g = -0.99\text{ m/s}$

23. $\vec{F} = m\vec{a} \Rightarrow \vec{a} = \left(\frac{F\cos\theta}{m}, \frac{F\sin\theta}{m}\right) = (18, 10.8)\text{m/s}$

24. a) $F = \frac{\Delta m}{\Delta t} u = (2.3)(2800) = 6440\text{N}$ b) $a = \frac{F}{m} = \frac{6440}{850} = 7.6\text{m/s}^2$

25. $P_i = P_f = 0 \Rightarrow m_1(v_1x, v_1y) + m_2(v_{2x}, v_{2y}) + m_3(v_{3x}, v_{3y}) = 0$ or $v_3 = (-12.3, 0.6)$

26. $P_1^2 + P_2^2 = P_3^2 \Rightarrow (m_1 v_1)^2 + (m_2 v_2)^2 = (m_3 v_3)^2 \Rightarrow m_3 = 8\text{ kg}$

27. $v^2 = \frac{gR^2}{R+H} = \frac{(16)\left(3476/2\right)}{\left(\frac{3476}{2}\right)+80}(1000) = 1641\text{m/s}$

28. $F_C = mg\tan\theta = 8\text{N}$

29. a) $T = m\left(\frac{v^2}{r} + g\cos\theta\right) = 0.2$

b) $T = m\left(\frac{v^2}{r} + g\right) = 0.2(12 + 10(0.5))$ c) $T = m\left(\frac{v^2}{r} - g\right)$

d) $\frac{mv^2}{r} = mg \Rightarrow v = \sqrt{rg} = \left(\sqrt{(1.3)(10)}\right) = 3.6\text{m/s}$

30. a) $a_c = \frac{v^2}{r} = 6.25\text{m/s}^2$ b) $T = m\left(\frac{v^2}{r} + g\cos\theta\right) = 895\text{N}$

c) $\frac{mv^2}{r} = mg$ or $v = 6.3\text{m/s}$

31. $\frac{GM_e m}{x^2} = \frac{GM_m m}{(d-x^2)} \Rightarrow x = 3.6 \times 10^5\text{km}$ above earth's surface

32. $g = \frac{GM}{R^2}$ and $g' = \frac{GM}{(R+h)^2}, g' = g/9$, or $h = 12800$ km

33. a) $\frac{mv^2}{R} = \frac{GmM}{R^2} \Rightarrow v = \sqrt{\frac{GM}{R}}, R = R_{eth}$

$R_e = 6400$ km, $M_e = 6 \times 10^{24}$g $\Rightarrow v = 7.71 \times 10^4$m/s

b) $T = \frac{2\pi R}{v} = \frac{(2\pi)(6.67 \times 10^4)}{7.7 \times 10^4} = 5.4 \times 10^3$ sec

c) $a_c = \frac{v^2}{r} = \frac{(7.7 \times 10^4)}{6.67 \times 10^4} = 8.9 \times 10^4$ m/s^2

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Unit 5

Work, Energy and Power

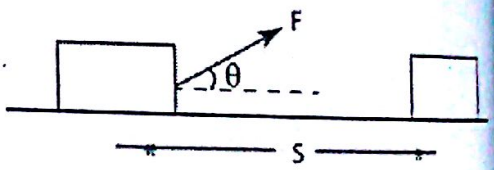
When a car accelerates, energy is transferred from the fuel in the engine to the motion of the car. That is, work is done to move the car. The rate at which this work can be done depends on the power of the engine

5.1 Work as a Scalar Product

In everyday language, any activity that requires muscular or mental effort is called "*work*". However, in physics the term work has a much more precise definition.

Work: Is the product of displacement and force in the direction of the displacement.

If a body is displaced by a constant force F that makes an angle θ with the horizontal as shown, the work done on it is obtained by



$$W = Fs\cos\theta$$

☞ Work is a scalar quantity and its SI unit is the Joule (J). Therefore, 1J = 1Nm

Work is the application of dot product. So if a force $\vec{F} = F_x i + F_y j + F_z k$ displaces the body to a position $\vec{S} = S_x i + S_y j + S_z k$ Then, the work done on the body is

$$W = \vec{F}\cdot\vec{S} = FS\cos\theta \quad \text{or} \quad W = F_xS_x + F_yS_y + F_zS_z$$

☞ **Important Points to remember**

- Work is positive when the force and the displacement are in the same direction or when $0^o < \theta \leq 90^o$
- Work is negative when the force and the displacement are in opposite direction or when $90^o < \theta \leq 180^o$

- Work is zero when the force is zero or when the displacement is zero or when the force and the displacement are at right angle to each other, $\theta = 90^{\circ}$

Conceptual Example

1. Athelte Haile G/Sellasi does an exercise in his home on a rotating belt as though he is running. Does he do work?
   Explanation: No! there is no displacement.
2. Is there any work done on a car moving with constant speed along a straight level road?
   Explanation: There is work done by friction.
3. When people are cold, they often rub their hands together to warm them up. How does doing this produce heat? Where did the heat come from?
   Explanation: The work done due to friction bring this. That is
   $$W = \Delta kE \Rightarrow fs = \frac{1}{2}m\left(v^2 - u^2\right) = \text{heat energy}$$
4. A 1.75m tall worker lifts a 50 kg cement bag by himself and goes 2 km carrying the bag along the horizontal. What is the work done by the man on the bag?
   Explanation:The minimum force needed to lift the bag is;
   $$F = w = mg = (50)\ (10) = 500N$$
   $$W = Fs\cos\theta = (500)\ (1.75) = 875J$$
   No work is done while walking the 2 km? But why?
5. When a particle rotates in a circle, a centripetal force of 1000N acts on it directed towards the center of rotation. If the radius of the circular path is 50m. What is the work done on the particle by this force?
   Explanation: $W = Fs\cos\theta$, here $\theta = 90^{\circ}$ and inward $W = 0J$ the displacement, in the force direction is zero.
6. A particle of mass 3 kg travels in uniform circular motion at speed 4 m/s. How much work is done on the particle by the centripetal force acting on it at any time?
   Explanation: $W = Fs\cos\theta$, $F = \frac{mv^2}{r}, \theta = 90^{\circ} \Rightarrow W = 0J$

Illustrative Example

7. A boy pulls a 20 kg carton across the floor at constant speed. If the coefficient of sliding friction between the carton and the floor is 0.6, how much work does the boy do in moving the carton 5m?

Solution: $F - f = ma$, $a = 0$

$$F = \mu F_N = \mu mg = (0.6)(20)(10) = 120N$$

$$W = Fs\cos\theta = (120)(5) = 600J$$

8. A man pulls a 40 kg box using a rope that makes 37° from the horizontal. If the man pulls the box 10m across the ground at constant speed and the coefficient of friction between the box and the ground is 0.4. What work is done on the box?

Solution: $F_x - f = ma$, $a = 0$

$$F\cos\theta = \mu F_N,\ F_N = mg - F\sin\theta$$

$$(F\cos\theta) = \mu(mg - F\sin\theta),\ F = \frac{\mu mg}{\cos\theta + \mu\sin\theta} = 154N$$

$$W = Fs\cos\theta = (154)(10)(0.8) = 1232J$$

9. A force of 20N is applied on a block of mass 4 kg and it displaces the block a distance of 3m from its original position. How much work is done on the block if:

a) The surface is frictionless.

b) The coefficient of friction is 0.3.

Solution: a) $W = Fs\cos\theta = (20)(3) = 60J$

b) $W = W_F + W_f = Fs + \mu mgs\cos 180°$

$= (20)(3) + (0.3)(4)(10)(3)(-1)) = 60 - 36 = 24J$

10. The force applied on a body is given by $\vec{F} = (2i - 3j + 4k)$ N and the displacement it produce is given by $\vec{S} = (i + 4j + 5k)$ m. What is the work done by this force?

Solution: $W = \vec{F}\cdot\vec{S} = (2i - 3j + 4k).(i + 4j + 5k) = 10J$

11. A force changes the velocity of a body from 10 m/s to 20 m/s in 5 sec. If the body has a mass of 5 kg. Find:

a) Work done in the 4 seconds.

b) Work done in the 4^{th} second.

Solution: a) $a = \frac{v - u}{t} = \frac{20 - 10}{5} = 2m/s^2$

$$F = ma = (5)(2) = 10N$$

$$S = ut + \frac{1}{2}at^2 = (10)(4) + (2)(4)^2 = 56m$$

$$W = Fs = (10)(56) = 560J$$

b) $S_n^{th} = u + a\left(t_n - \frac{1}{2}\right)$

$$S_n^{th} = 10 + 2\left(4 - \frac{1}{2}\right) = 17\,m$$

$$W = Fs = (10)(17) = 170J$$

12. A 2000 kg elevator rises from rest in the basement to the fourth floor a distance of 25m. As it passes the fourth floor, its speed is 3 m/s. If the friction is 500N, what is the work done by the lifting engine.

Solution: $W_F + W_f = P_E + k_E \Rightarrow W_F = mgh + \frac{1}{2}mv^2 + fh$

$$W_F = (2000)(10)(25) + \frac{1}{2}(2000)(3)^2 + (500)(25) = 521.5\ kJ$$

13. An object was thrown vertically up and comes to the thrower's hand in 6 sec. If the object was allowed to slide on a horizontal floor whose coefficient of frictions is 0.2. What is the maximum distance the object can slide?

Solution: $h = ut - \frac{1}{2}gt^2 \Rightarrow 0 = u(6) - \frac{1}{2}(10)t^2 \Rightarrow u = 30$ m/s and

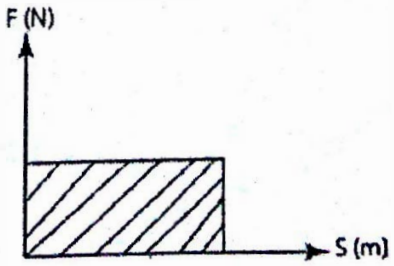
$$W = \Delta kE \Rightarrow \mu mgs = \frac{1}{2}mu^2 \Rightarrow S = 225m$$

5.2 Work Done by Constant and Variable Forces

Constant force: It a force having constant magnitude and direction in the given time interval. Example, Gravitational force

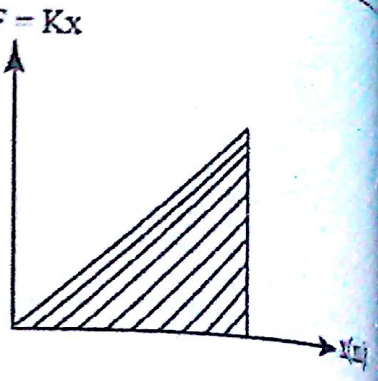
☞The work done by a constant force is equal to the area under the force – displacement graph

$W = Area = FS$



Variable Force: Is a force that constantly varies with time. Example spring force.

☞ The work done by a constantly varying force is equal to the area under the curve.

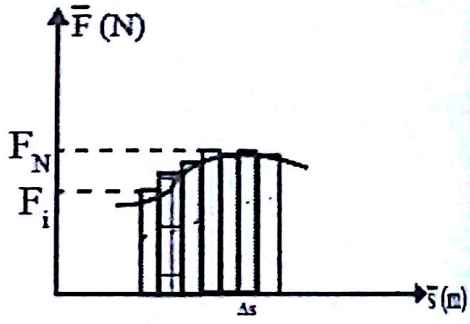


$$W = \text{Area} = \frac{1}{2}Kx^2$$

☞ When the force is neither constant nor varying constantly, the work done is obtained by

$W = F_1 \Delta S_1 + F_2 \Delta S_2 + ... + F_n \Delta S_n$

$$W = \sum_{i=1}^{n} Fi \, \Delta \, Si$$



Illustrative Example

14. A force applied on an object varies with the displacement according to the equation $F = (5x + 6)$ N/m. Calculate the total work done in displacing the object 4m on the x – axis starting from the origin.

Solution:

$$W = (F_{av})(s) = \frac{(F_1 + F_2)}{2}(S) = \left[\frac{(5(0)+6+5(4)+6}{2}\right](4) = 64$$

15. Calculate the work done by a force given by $F = (2 + x)\frac{N}{M}$ as x varies from $x = 0$ to $x = 6m$

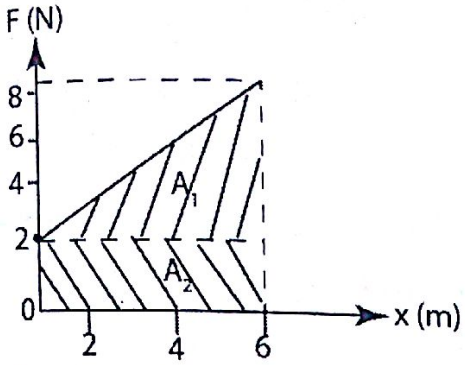
Solution: Method I, Analytically $W = (F_{av})(X)$

$$F_{av} = \frac{F_1 + F_2}{2} \text{ and } F_1 = (2 + 0) = 2N, \; F_2 = (2 + 6) = 8N$$

$F_{av} = \frac{2+8}{2} = 5$ N and x = 6m

W = (5) (6) = 30J

Method II Graphically, W = Area



$W = \text{Area} = A_1 + A_2$

$= \frac{1}{2}(6)(6) + (2)(6)$

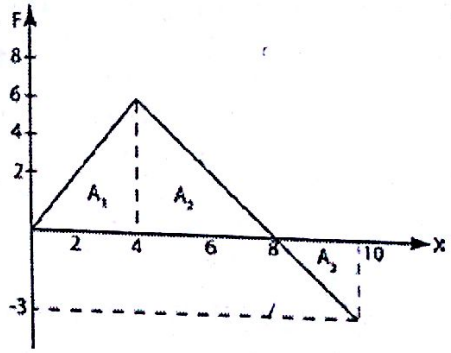
W = 30J

16. The force acting on a particle varies as shown. Find the work done by the particle as it moves from x = 0 to x = 10m.

Solution: $W = A_1 + A_2 + A_3 = \frac{1}{2}b_1h_1 + \frac{1}{2}b_2h_2 + \frac{1}{2}b_3h_3$

$= \frac{1}{2}(4)(6) + \frac{1}{2}(4)(6) + \frac{1}{2}(2)(-3)$

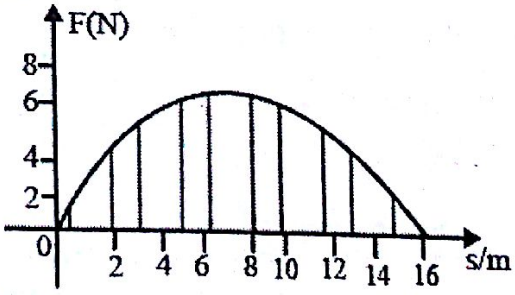
W = 21J



17. The graph shows the relationship between force and displacement. What is the work done in moving the object 16m. Assuming a semi – circle region.

Solution: Work = Area

$W = \frac{\pi r^2}{2} = \pi\frac{(8)^2}{2} = 32\pi J$



18. A force acts on a 3kg particle is in such a way that the position of the particle as a function of time is given by $x = 3t - 4t^2$. What is the work done by the force during the first 4 seconds? (UEE: 2001)

A. 1218 J C. 1248 J

B. 1248 J D. 1258 J

Solution: $a = \frac{d^2x}{dt^2} = -8 m/s^2$ and $F = ma = -24\,N$

$S = 3(4) - 4(4)^2 = -52m$

$W = Fs = (-24)(-52) = 1248J$

Answer: C

19. A body has moved from its initial position at origin along x – axis to a final positions x = 5m under the influence of a force given by $\vec{F} = (4x)i$ Newtons. What is the work done by the force in moving the body from its initial position to its final position? (UEE: 2003)

A. 100J B. 50 J C. 25 J D. 20J

Solution: $W = F_{av}\ S = \frac{4(0)+4(5)}{2}(5) = 50J$

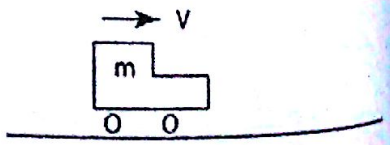
Answer: B

5.3 Kinetic Energy and Work – Energy Theorem

Energy, is in general the ability to do work and is measured by the amount of work a body can do. So, energy is measured in the same unit as work, namely, joule. Like work, energy is a scalar quantity. Energy can exist in various forms, such as heat, electrical, sound, light, chemical, nuclear, mechanical etc. Here we will focus on mechanical energy. Mechanical energy is of two types. Kinetic and potential

Kinetic energy: Is the energy possessed by a body because of its motion.

Consider a body of mass m moving with a speed of v. Then, its kinetic energy is obtained by

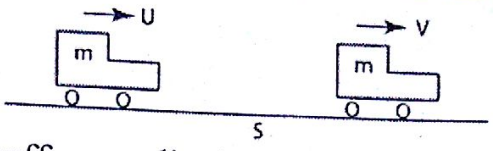


$$K_E = \frac{1}{2}mv^2$$

☞ A moving object can do work on another object when it strikes it. And, Work done and kinetic energy are related using the work energy theorem.

Work – Energy Theorem states that "work done by a net force is equal to the change in kinetic energy of the body"

Consider a body of mass m moving with an initial velocity u and let a constant force F acts on it for time t. The body acquires a velocity V and suffers a displacement S in the direction of the force.



Then we have.

$W = Fs,\ F = ma$

$W = mas,\ v^2 = u^2 + 2as$

$$W = m\frac{v^2 - u^2}{2}$$

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \quad \text{or} \quad Fs = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$\boxed{W = \Delta K_E}$ ← work – energy theorem

Conceptual Example

20. Can kinetic energy be negative? Explain

Explanation: Never! Because, $k_E = \frac{1}{2}mv^2$, here mass can never be less than zero and v^2 is also always positive.

21. What can be said about the speed of a particle if the net work done on it is zero?

Explanation: $W = \Delta kE = 0 \Rightarrow$ speed, constant

22. A car moving at a speed u skids a distance d after its brakes are applied. Estimate how far it will skid if its brakes are applied when its initial speed is 2u.

Explanation: $Fd = \frac{1}{2}mu^2$ .....(*)

$$Fd^1 = \frac{1}{2}m\,(2u)^2 = 4\left(\frac{1}{2}mu^2\right) \text{.....(**)}$$

$\Rightarrow Fd^1 = 4(Fd) \Rightarrow d^1 = 4d$, it skids four times

Illustrative Examples

23. A driver of a 1000 kg car traveling at 35 m/s applied on his brakes to avoid collision with a vehicle in front of him which come to rest because of congestion ahead. If the brakes offer a

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friction force of 8000N, at what minimum distance should the brakes be applied to avoid collision?

Solution: $Fs\cos 180° = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$$-fs = -\frac{1}{2}mu^2 \Rightarrow S = 76.6m$$

24. A police investigation measures skid mark 27m long in an accident investigation. If the mass of the car is 1000 kg and its stopping force is 8000N, what was the minimum speed of the car when the brakes are applied?

Solution: $fs\cos 180° = \frac{1}{2}mv^2 - \frac{1}{2}mu^2, v = 0$

$$-fs = -\frac{1}{2}mu^2 \Rightarrow u = \sqrt{\frac{2fs}{m}} = 20.8\,m/s$$

25. A 0.6 kg particle has a speed of 2 m/s at point A and a kinetic energy of 7.5J at point B. What is:
   a) Its kinetic energy at A.
   b) Its speed at point B.
   c) The total work done on the particle as its moves from A to B.

Solution: a. $k_E = \frac{1}{2}mv_A{}^2 = \frac{1}{2}(0.6)(2)^2 = 1.2\,J$

b. $k_E = \frac{1}{2}mv_B{}^2 \Rightarrow v_B = \sqrt{\frac{2kE}{m}} = \sqrt{\frac{2(7.5)}{0.6}} = 5m/s$

c. $W = \Delta kE = \frac{1}{2}mv_B^2 - \frac{1}{2}v_A^2 = 7.5 - 1.2 = 6.3J$

26. Ronaldinho, the Brazilian football artist, kicks a 0.42 kg soccer ball moving at 2 m/s with a constant force of 40N in the same direction as the ball's motion. Over what distance must his foot be in contact with the ball to increase the ball's speed to 6 m/s?

Solution: $Fs = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \Rightarrow S = \frac{m}{2F}(v^2 - u^2)$

$$s = \frac{0.42}{2(40)}\left(6^2 - 2^2\right) = 0.17m$$

27. A bullet of mass 50 g travelling at 80 m/s hits a sand bag and moves a distance of 10 cm before it stops. What force is exerted on the bullet by the sand bag?

Solution: $fs = \frac{1}{2}mu^2 \Rightarrow f = \frac{1}{2S}mu^2 = \frac{1}{2(0.1)}(0.05)(80)^2 = 1600N$

28. A 2 kg block initially at rest is pulled along a horizontal smooth surface by a constant horizontal force of 12N what is;
   a) The speed of the block after it travelled 3m.
   b) The acceleration of the block.
   c) The time taken by the block to travel 3m.

Solution: a. $Fs = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 \Rightarrow Fs = \frac{1}{2}mv^2, u = 0$

$$v = \sqrt{\frac{2Fs}{m}} = \sqrt{\frac{2(12)(3)}{(2)}} = 6m/s$$

b. $F = ma \Rightarrow a = \frac{F}{m} = \frac{12}{2} = 6m/s^2$

c. $s = ut + \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2(3)}{6}} = 1sec$

29. By what percent does the kinetic energy of a body increased, if its speed is increased by 20%?

Solution: $k_{E_i} = \frac{1}{2}mu^2$ .....(*)

$k_{E_f} = \frac{1}{2}mv^2$ ..... (**), $v = u + 0.2u = 1.2u$

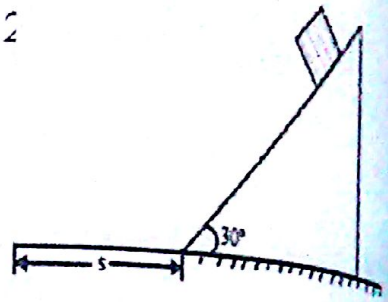
$\Rightarrow k_{E_f} = \frac{1}{2}m(1.2u)^2 = \frac{1}{2}m(1.44u^2)$ .....(***)

we have $k_{E_f} = 1.44\,k_{E_i}$

$$\Delta k_E = \frac{(k_{E_f} - k_{Ei})}{k_{E_i}}100\% = \frac{(1.44-1)}{k_{E_i}}k_{E_i} \times 100\% = 44\%$$

30. A bag of mass 50 kg slides down a 30° and 10m long incline at an acceleration of 2 m/s^2?
   a) What is the kinetic energy of the bag as it reaches the bottom of the incline?

b) How far will the bag slide horizontally, before coming to rest? Assume $\mu_k = 0.3$.



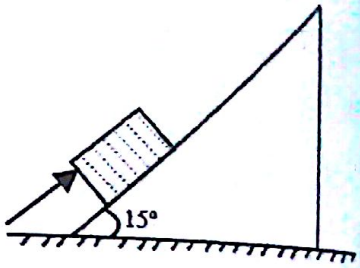
Solution: a. $k_E = \frac{1}{2}mv^2$, but $v^2 = u^2 + 2$

$$k_E = \frac{1}{2}m(2as) = \frac{1}{2}(50)(2)(2)(10)$$
$$= 1000J$$

b. $Fs = k_E \Rightarrow \mu mgs = k_E$, $s = \dfrac{kE}{\mu mg}$

$$s = \frac{1000}{(0.3)(50)(10)} = 6.7m$$

31. Calculate the work done in pulling up a block of mass 200 kg for 10m on smooth plane inclined at an angle of 15° with the horizontal.



Solution: $W = Fs \cos\theta$, $F = mg \sin\theta$

$= (mg \sin\theta)(s) = (200 \times 10 \times 0.26) = 520 \times 10 = 5.2$ kJ

32. A bullet moving at 300 m/s can penetrate a target up to 500mm before rest. What will be its speed of emergency if it is fired in to a 250mm thick target of the same resistance?

Solution: $W = \Delta k_E = \frac{1}{2}mv^2 - \frac{1}{2}mu^2, v = 0$

$$W = \frac{1}{2}mv^2 \Rightarrow fs = \frac{1}{2}mu^2$$

$0.5f = 0.5m\,(300)^2 \Rightarrow f = (9000)\,(m)$ and

$$-fs = \frac{1}{2}mv^2 - \frac{1}{2}mv^2 = -(0.25)\,(90000)\,(m) \Rightarrow v^2 = 45000$$

$\Rightarrow v = 212$ m/s

33. A car of mass 1500 kg is travelling at a uniform velocity of 30m/s due east. The driver applies the brakes to slow down the car to a velocity of 10m/s due east. What is the work done in slowing down the car? (UEE: 2007)

A. 15 kJ
B. 300 kJ
C. 600 kJ
D. 750 kJ

Solution: $W = \Delta K_E = \frac{1}{2}m(v^2 - u^2)$

$= \frac{1}{2}(1500)(100 - 900) = 600{,}000\ J$

$W = 600\ KJ$

5.4 Potential Energy

Answer: C

Potential energy is stored by an object when it is in a particular position or condition.

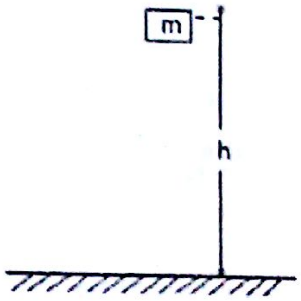
Potential Energy: Is the energy possessed by a body because of its position or configuration

Therefore, potential energy can be gravitational or elastic

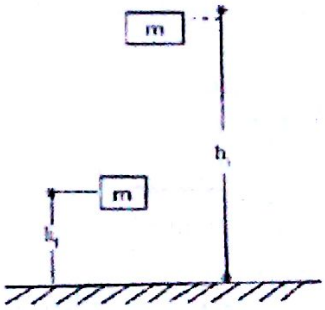
Gravitational Potential Energy: Is the potential energy of a body because of its position from a given surface.

The gravitational potential energy of a body of mass m at a height of h above the given reference surface is

$\boxed{P_E = mgh}$, take $g = 10m/s^2$



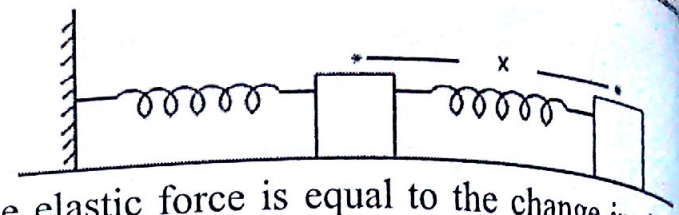
- Work done against gravity is equal to the change in gravitational potential energy.
  $\boxed{W = \Delta PE = mgh_f - mgh_i}$
- So, we can see that
  - The work done to move a body upward is equal to the increase in gravitational potential energy.
  - The work done to move the body downward is equal to the decrease in gravitational potential energy of the body.



Elastic Potential Energy: Is the potential energy of a body such as spring because of its condition or configuration.

The elastic potential energy of a spring of force constant K stretched or compressed by an amount of x from the equilibrium position is given by

$$P_E = \frac{1}{2}Kx^2$$

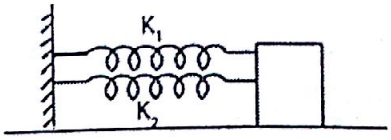


☞ The work done by the elastic force is equal to the change in elastic potential energy.

$$W = \Delta PE = \frac{1}{2}Kx_2^2 - \frac{1}{2}Kx_1^2$$

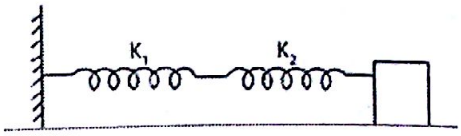
☞ Remember also that, the effective force constants for

i. Springs in parallel



$$K = K_1 + K_2$$

ii. Springs in series



$$\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2}$$

☞ **Important points to remember**

- A stiff spring has a large spring constant. And, the force exerted by the spring is given by, $F = -Kx$, this is often called Hooke's law named after Robert Hooke.
- The increase in elastic potential energy is equal to the work done by the average force needed to stretch the spring

5.5 Conservation of Energy

Recall that, mechanical energy is the sum of kinetic energy and potential energy $ME = PE + KE$

Law of conservation of Energy states that, "In the absence of dissipative force such as friction, mechanical energy remains constant."

$$ME_i = ME_f \Rightarrow KE_i + PE_i = KE_f + PE_f$$

☞ In another way, law of conservation of energy is stated as "energy is neither created nor destroyed, it can simply be transformed from one form to another."

Conceptual Examples

34. Elevators that are lifted by a cable have a counter weight. The elevator and the counter weight are connected by a cable that passes over a drive wheel. The counter weight moves in the

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direction opposite to that of the elevator. What is the purpose of the counter weight?

Explanation: Without the counter weight, very large energy will be transferred to the elevator by the work done from the motor. Thus the presence of the counter weight results in a much smaller change in the elevator's potential energy and less work from the motor.

35. You walk leisurely up a flight of stairs. You then return to the bottom and run up the same set of stairs as fast as you can. Have you performed a different amount of work in these two cases?

Explanation: NO! because, $W = \Delta G_{P_E} = mgh$.

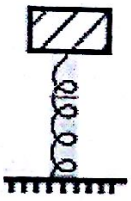
In both cases the change in your potential energy is the same, hence the work you did in lifting your body is the same.

Illustrative Examples

36. a. A 5 kg mass resting on a spring compressed the spring by 5 cm. What is the force constant of the spring?
    b. If the mass is pushed down an additional 15 cm, how much potential energy is stored in the spring.
    c. How high will the mass rise when released from the new position?

Solution: a. $F = kx \Rightarrow mg = kx,\ k = \frac{mg}{x} = \frac{(5)(10)}{(0.05)}$

$$k = 1000\ N/m$$



b. $M_{E_i} = M_{E_f} \Rightarrow M_{E_f} = mgh + \frac{1}{2}kx^2$

$$M_{E_f} = mgx_2 + \frac{1}{2}kx_2^2 = (5)(10)(0.15) + \frac{1}{2}(1000)(0.15)^2 = 18.75J$$

c. $mgh = M_{E_f} = 18.75 \Rightarrow h = \frac{18.75}{(5)(10)} = 0.4m$

36. A moving 2.5 kg block collides with a horizontal spring of force constant 320 N/m and compresses it by 7.5 cm from its rest position. If $\mu_k = 0.25$ between the block and the horizontal surface.

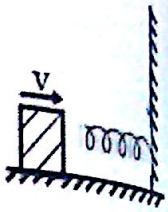
a) How much work is done by the spring in bringing the block to rest?

b) How much energy is dissipated by the friction force?
c) What was the speed of the block when it hit the spring?

Solution: a. $W = \Delta E_{PE} = \frac{1}{2}kx^2 = \frac{1}{2}(320)(0.075)^2 = -0.9\,J$

b. $W_f = \mu mgx = (0.25)(10)(0.075) = 0.47\,J$

c. $M_{E_i} = M_{E_f} \Rightarrow \frac{1}{2}mv^2 = \mu mgx + \frac{1}{2}kx^2,$



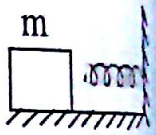
$$v = \sqrt{\frac{2(0.9+0.47)}{2.5}} = 1\ m/s$$

37. A 2 kg block slides to the right with a speed of 5 m/s a long a level surface having $\mu_k = 0.25$, collides with a spring of force constant 150 N/m. It compressed the spring and then pushes back by restoring force. Find:

a) the deformation
b) the speed of the block as it pass the rest position to the left
c) the distance the block moved beyond the rest position as it moved to the left

Solution: a. Transformation of energy

$$K_E = E_{P_E} + W_f \Rightarrow \frac{1}{2}mu^2 = \frac{1}{2}kx^2 + \mu mgx$$



$$\frac{1}{2}(2)(5)^2 = \frac{1}{2}(150)(x^2) + 0.25(2)(10)(x) \Rightarrow x = 0.5m$$

b. $\frac{1}{2}kx^2 = \frac{1}{2}mv^2 + \mu mgx$

$$\frac{1}{2}(150)(0.5)^2 = \frac{1}{2}(2)v^2 + 0.25(2)(10)(0.5),\ v = 4m/s$$

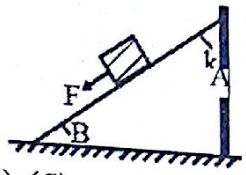
c. $W_f = -\Delta k_E \Rightarrow \mu mgs = \frac{1}{2}mv^2$

$$v^2 = 2\mu gs \Rightarrow s = \frac{v^2}{2\mu g} = \frac{4^2}{(0.25)(2)(10)} = 3.2m$$

38. A block is moved down an incline 5m from point A to point B by a force of 2N. If a 10N friction force is acting on the block and the kinetic energy is increased by 35J between A and B, the

much work is done on the block by its weight as the block moves from A to B?

Solution: From Newton's second law,



$$F + mg \sin\theta - f = ma$$
$$Fs + (mg \sin\theta)\, s - fs = (ma)s$$
$$Fs + mgh - fs = \Delta KE = 35 + (10)(5) - (2)(5)$$
$$\Delta G_{PE} = 75J$$

39. A 2 kg block is placed against a spring of force constant 20 N/cm on a frictionless 30° incline. If the spring is compressed 20 cm and then released.

a) What is the elastic potential energy of the compressed spring?

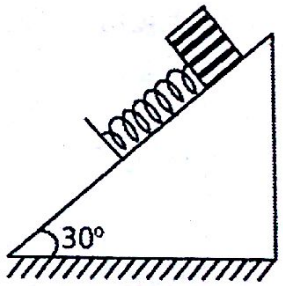
b) What is the change in the gravitational potential energy of the block as it moves from the release point to its highest point on the incline?

c) How far along the incline is the highest point from the release point?

Solution: a. $E_{PE} = \frac{1}{2}kx^2 = \frac{1}{2}(2000)(0.2)^2 = 40J$

b. $E_{PE} = G_{PE} = 40J$

c. $mgh = 40 \Rightarrow h = \frac{40}{2(10)} = 2m$



$$h = S \sin\theta \Rightarrow s = \frac{h}{\sin\theta} = \frac{2}{\sin 30} = 4m$$

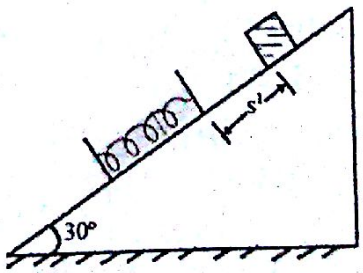
40. A 10 kg block is released from rest on an incline of 30°. Below the block is a spring that can be compressed 2 cm by a force of 200N. The block momentarily stops when it compresses the spring 5 cm.

a) How far has the block moved down the incline to this stopping point?

b) What is the speed of the block just as it touches the spring?

Solution: a. $mg\,(0.05 + s^1) \sin\theta$

$$= \frac{1}{2}k\,(0.05)^2$$
$$(10)(10)(0.05 + s^1) \sin 30$$

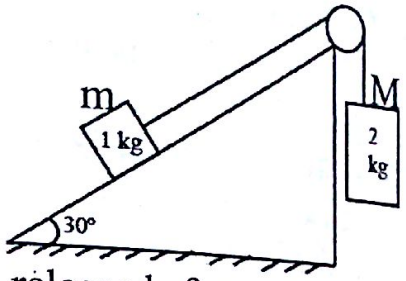


$$= \frac{1}{2}(10000)(0.05)^2$$

$s^1 = 0.2m$ but $s = x + s^1 = 0.05 + 0.2 = 0.25m$

b. $\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow 5v^2 = 5000\ (0.05)^2 = 1.6$ m/s

41. If the pulley is massless and the surfaces are frictionless, what is their total kinetic energy when the 2 kg mass has fallen 25 cm. Assume that the masses are released from rest?



Solution: $\Delta k_E = \Delta G_{PE} \Rightarrow \Delta k_E$

$= Mgh - mgh = (2)\ (10)$

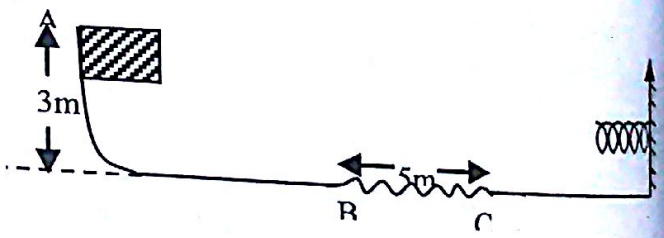
$(0.25) - (1)\ (10)\ (0.25)$

$(0.5) = 3.75J$

42. A block of mass 10 kg is released from A. The track is frictionless except portion BC of length 5m. The block moves down the track and hits a spring of force constant 2200 N/m and compresses it 0.4m from its rest position. What is the coefficient of friction between surface BC and the block?

Solution: From A to B,

$\frac{1}{2}mv_B^2 = mgh$



From B to C

$\frac{1}{2}mv^2{}_B = \mu mgs + \frac{1}{2}mv^2{}_C$

From C to spring in compression

$\frac{1}{2}mv_C^2 = \frac{1}{2}kx^2$

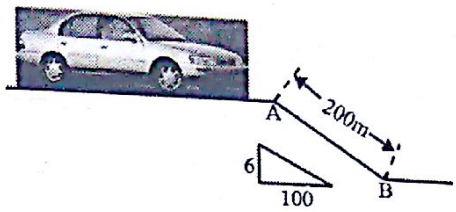
$\Rightarrow mgh = \mu mgs + \frac{1}{2}kx^2$

$\Rightarrow \mu = \frac{2mgh - kx^2}{2mgs}$

$= \frac{(2)(10)(10)(3) - 2200(0.4)^2}{2(10)(10)(5)}$

$\mu_k = 0.25$

43. A 1400 kg car is traveling at a speed of 100 km/h as it passes point A and then the car goes down the 6 percent incline. The driver applies her brakes so as to make the car speed 20 km/h at point B. Calculate the energy dispated from the brakes in the form of heat.



Solution:

$$\tan\theta = \frac{6}{100} \Rightarrow \theta = 3.4°$$

$$(mg\sin\theta)s + \frac{1}{2}mu^2 = \frac{1}{2}mv^2 + W_f$$

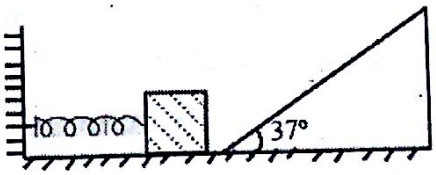
$$(1400)(10)(0.06)(200) + \frac{1}{2}(1400)(27.8)^2 = \frac{1}{2}(1400)(5.6)^2 + W_f$$

$$\Rightarrow W_f = 6.9 * 10^5 J$$

44. A 2 kg block is pushed against a spring of force constant 400 N/m, compressing it 0.21m. When the block is released, it moves along a frictionless horizontal surface and then up a frictionless incline of slope 37°.

a) What is the speed of the block as it slides along the horizontal surface?

b) How far does the block travel up the incline before starting to slide back down?



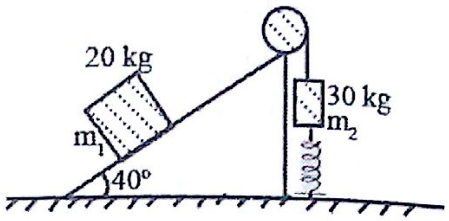
Solution: a. $\frac{1}{2}mv^2 = \frac{1}{2}kx^2$

$$v = x\sqrt{\frac{k}{m}} = 0.21\sqrt{\frac{400}{2}} = 3m/s$$

b. $\frac{1}{2}mv^2 = mgh \Rightarrow h = \frac{v^2}{2g} = \frac{9}{2(10)} = 0.45$

$$s = \frac{h}{\sin\theta} = \frac{0.45}{0.6} = 0.75m$$

45. A 20 kg block is connected to a 30 kg block by a string that passes over a frictionless pulley. The 30 kg block is attached to a spring that has a force constant of 250 N/m. If the 20 kg block is pulled 20 cm down the incline and released from rest, find the speed



of each block as they reach their original position. Assume the incline is frictionless.

Solution: $k_E + p_E$

$$\frac{1}{2}(m_1 + m_2)v^2 = m_1gh_1 + m_2gh_2 + \frac{1}{2}kx^2$$

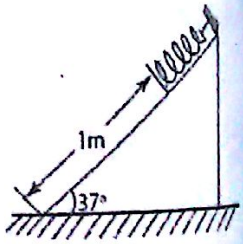
$$v = \sqrt{\frac{2xg(m_2 - m_1 \sin\theta) + kx^2}{(m_1 + m_2)}}, \; h_1 = x \sin\theta, \; h_2 = x$$

$$v = \sqrt{\frac{2(0.02)(10)(30 - 20(0.6) + 250(0.02)^2}{(20+30)}} = 0.4 \text{ m/s}$$

46. A spring with k = 170 N/m is at the top of a frictionless incline of 37°. The lower end of the incline is 1m from the end of the spring. A 2 kg block is pushed against the spring until the spring is compressed 0.2m and released from rest.

a) What is the speed of the block at the instant the spring comes to its rest position?

b) What is the speed of the block when it reaches the lower end of the incline?



Solution: a. $\frac{1}{2}mu^2 = (mg \sin\theta)(x) + \frac{1}{2}kx^2$

$$\frac{1}{2}(2)\,u^2 = (2)(10)(0.6)(0.2) + \frac{1}{2}(170)(0.2)^2 \Rightarrow u = 2.4 \text{ m/s}$$

b. $\frac{1}{2}mv^2 = (mg \sin\theta)(s) + \frac{1}{2}mv^2 = (20)(10)(0.6)(1) + \frac{1}{2}$

$(2)(2.4)^2 \Rightarrow v = 4.2$ m/s

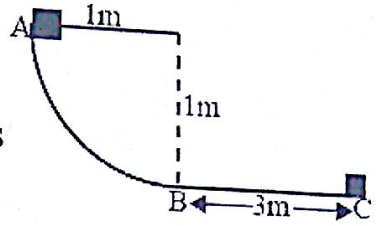
47. In the track shown, section AB is a quadrant of a circle of radius 1m. If a block is released at A and slides without friction until it reaches point B.

a) How fast is it moving at B?

b) If the horizontal surface is rough, and the block comes to rest 3m from B, what is the coefficient of friction is position BC.

Solution: a. $mgh_A = \frac{1}{2}mv_B^2$

$$v_B = \sqrt{2gh_A} = \sqrt{2(10)(1)} = 4.5\text{m/s}$$



b. $W_f = \frac{1}{2}mv_B^2 \Rightarrow \mu mgs = \frac{1}{2}mv_B^2 \Rightarrow$

$$\mu = \frac{v_B^2}{2gs} = \frac{(4.5)^2}{2(10)(3)} = 0.34$$

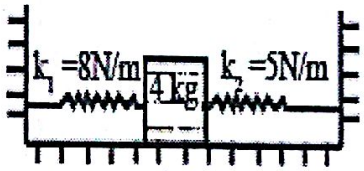
48. If the mass is displaced 20 cm to the right and then released.

a) What is the speed of the block as it passes its rest position in its way to the left?

b) How far does the block go to the left before stopping?

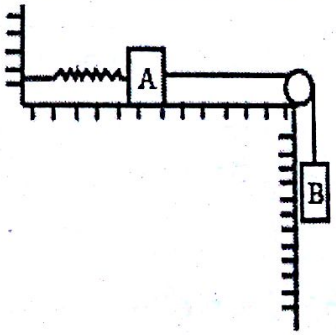
Solution: a. $E_{PEB} = k_{EA} \Rightarrow \frac{8(0.04)+5(0.04)}{2} = \frac{4v^2}{2} \Rightarrow v = 0.36$ m/s

b. $k_1(s) + k_2(s) = \frac{1}{2}mv^2$



$$8(S) + 5(S) + \frac{4}{2}v^2 = 2(0.36)^2 \Rightarrow s = 0.02\text{m}$$

49. Block A of mass 2.3 kg lies on a horizontal surface and is connected to a spring of force constant 300 N/m. If the system is released from rest and Block B of mass 4.9 kg falls 30 cm before coming to rest. What is the coefficient of kinetic friction between block A and the horizontal surface?

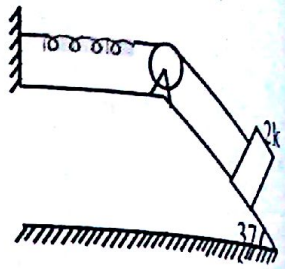


Solution: $P_{Ei} = P_{Ef} + W_f$

$$m_B gh = \frac{1}{2}kh^2 + \mu m_A gh$$

$$\mu = \frac{m_B g - \frac{1}{2}kh}{m_A g} = \frac{(4.9)(10) - \frac{1}{2}(100)(0.3)}{(2.3)(10)} = 0.2$$

50. A 2 kg block placed on a rough incline is connected to a spring of force constant 100 N/m. The block is released from rest and it moves 20 cm down the incline. Find the coefficient of friction between block and incline.



Solution: $P_{Ei} = P_{Ef} + W_f$

$$mgh = \frac{1}{2}kx^2 + \mu mgs$$

$$(mg \sin\theta)s = \frac{1}{2}ks^2 + \mu mgs$$

$$\mu = \frac{mg\sin\theta - \frac{1}{2}ks}{mg} = \frac{(2)(10)(0.6) - \frac{1}{2}(100)(0.2)}{(2)(10)}$$

$$\mu_k = 0.1$$

51. A large storage tank is filled to a height h_o. If the tank is punctured at a height h from the bottom of the tank, how far from the tank will the stream land?

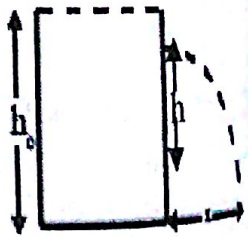
A. $2\sqrt{h(h_o + h)}$ C. $\sqrt{h(h_o + h)}$

B. $\sqrt{h(h_o - h)}$ D. $2\sqrt{h(h_o - h)}$

Solution: $ME_i = M_{Ef}$ $mgh_o = mgh + \frac{1}{2}mv^2$

$$v = \sqrt{2g(h_o - h)}$$

$$h = \frac{1}{2}gt^2, t = \sqrt{\frac{2h}{g}}, \text{ and } x = vt$$



$$x = \sqrt{2g(h_o - h)}\left(\sqrt{\frac{2h}{g}}\right) = 2\sqrt{h(h_o - h)}$$

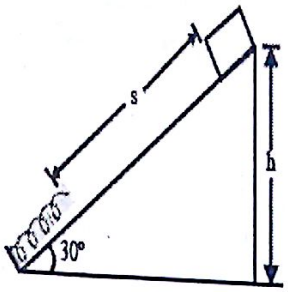
Answer: D

52. As shown below, an ideal massless spring is compressed 1m by a force of 100N. This same spring is placed at the bottom of a frictionless inclined plane of 30° with the horizontal. A 10 kg

mass is released from rest at the top of the incline and come to rest after compressing the spring 2m. Through what distance does the mass slide before coming to rest?

A. 1m B. 2m C. 4m D. 0.6m

Solution: $mgh = \frac{1}{2}kx^2, h = S \sin\theta$



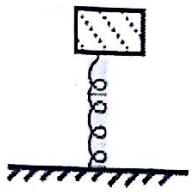
$mg\ s \sin\theta = \frac{1}{2}kx^2,$

$k = \frac{F}{x} = \frac{100}{1} = 100N/m$

$S = 4m$

Answer: C

53. A block of mass 0.25 kg is placed on a light vertical spring of force constant 5000 N/m and pushed down ward so that the spring compressed to 0.1m. To what maximum height above the point of release it rise?



A. 10.2m B. 9.2m C. 8m D. 9m

Solution: $mgh = \frac{1}{2}kx^2 \Rightarrow h = 10.2m$

Answer: A

54. A 1 kg mass is attached to a spring of stiffness 100 N/m. What is the maximum displacement of the mass from its equilibrium position if its speed is 3 m/s at the equilibrium point?

A. 0.1m B. 0.3m C. 3m D. 10m

Solution: $kE = E_{PE} \Rightarrow \frac{1}{2}kx^2 = \frac{1}{2}mv^2 \Rightarrow x = 0.3m$ **Answer: B**

55. A force of 100N is required to stretch a spring that obey hook's law by 50 cm. What is the work done in stretching the spring.

A. 500J B. 50J C. 25J D. 0J

Solution: $W = F_{av}X = \left(\frac{0+100}{2}\right)0.05 = 25J$ **Answer: C**

56. Two men, A and B, are loading identical boxes on to a truck. A lifts his box straight up from the ground to the bed of the truck, whereas B slides his box up a rough ramp to the truck. Which one is correct answer

A. A and B change the potential energy by the same amount
B. A changes more potential energy than B
C. B changes more potential energy than A
D. None. Since the friction and inclination are unknown.

Answer: A

57. Which one of the following statements is true about a spring mass system oscillating on a horizontal frictionless surface? (UEE: 2007)
A. The kinetic and potential energies are equal at all times
B. The kinetic and potential energy are both constant
C. The maximum potential energy is achieved when the mass passes through its equilibrium position.
D. The maximum kinetic energy and maximum potential energy are equal

Answer: D

60. A block of mass 10 kg is pulled up a rough inclined plane with a force of 100N parallel to the inclined plane. The inclined plane makes an angle of 37° with the horizontal. At the instant when the force is applied the block has a speed of 1.5 m/s. If the coefficient of kinetic friction is 0.4 and the block is pulled up 5m, what is the change in the kinetic energy of the block? (UEE: 2007)
A. 20 J B. 30 J C. 40 J D. 45.5 J

Solution: $F - \mu mg\cos\theta - mg\sin\theta = ma \Rightarrow a = \frac{18}{10}\ m/s^2$

But $v^2 = v^2 + 2as \Rightarrow v^2 = 10.25$

$$\Delta K_E = \frac{1}{2}m\left(v^2 - v^2\right) = \frac{1}{2}(10)[10.25 - 2.25] = 40J$$

Answer: C

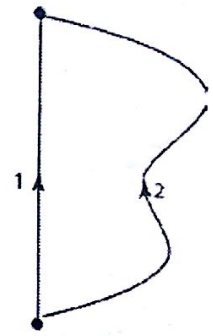
5.6 Conservative and Non Conservative Forces

The key distinction is that, when a conservation force acts, the work it does is stored in the form of energy that can be released at a later time. For example, imagine you are lifting a box of mass m from the floor to a height h. To lift the box with constant speed, the force you must exert against gravity is mg. Since the upward distance is h, the work you do on the box is mgh. If you now let go the box, gravity does the same work mgh, and in the process gives the box an equivalent amount of kinetic energy, this is the nature of conservative force.
However to slide a box of mass m across the floor of friction coefficient μ with constant speed, you must exert a force of μmg. After sliding the

box a distance s, the work you have done is μmgs. Here, if you let go the box, it simply stays put. Friction does no work on it after you let go. Thus, the work done by a non-conservative force cannot be recovered later as kinetic energy, instead, it is converted to other forms of energy such as, sound, heat, … etc

Conservative Force: Is a force that does no work when a body moves on a closed path. Example: gravitational force, spring force.

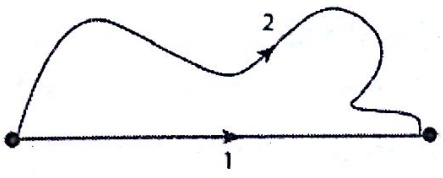
☞**Important points to remember**



- Work done by a conservative force is path independent. Therefore, the work done along path 1 and path 2 is the same and is equal to the change is potential energy
- The work done by a conservative force around a closed path is zero.
- Work done by a conservative force is completely recoverable. That is, the magnitude of the work done, say in the upward and downward direction is the same
- For a conservative force, mechanical energy is conserved ($\Delta ME = 0$). That means, the work done when the body come back to its initial position is zero.

Non conservative Force: Is a force that does work when a body moves on a closed path. Example: friction, drag force, Air resistance,iscosity

☞ **Important points to remember**



- Work done by a non-conservative or dissipative force is path or length dependent. Therefore, work done by friction along path 1 is less than that along path 2
- Work done by non-conservative force is not recoverable. That is, there is lose, mechanical energy is not conserved, the lose in mechanical energy is equal to the work done against friction. ($\Delta ME = Wf$)

Conceptual Examples

61. Our body muscles exert forces when we lift, push, run, jump and so forth. Are these forces conservative?
Explanation: No! There is always lose in the form of heat.

62. If three conservative forces and one non conservative force act on a system, how many potential energy terms appear in the problem?

Explanation: Three! Each conservative force can be related to potential energy

Illustrative Examples

63. Deep in the forest, a 17g leaf falls from a tree and drops straight to the ground. If its initial height was 5.3m and its speed on landing was 3m/s, how much non conservative work was done on the leaf?

Solution: $MEi = mgh = (0.017)\,(10)\,(5.3) = 0.9J$

$$ME_f = \frac{1}{2}mv^2 = \frac{1}{2}(0.017)(3)^2 = 0.08J$$

Work done by non-conservative force $= ME_f - ME_i = -0.82J$

5.7 Power

Power is a measure of how quickly work is done. To be precise, we can say:

Power: Is the rate at which work is done or energy is transformed to other form.

Therefore, we can have

$$\text{Power} = \frac{\text{work done}}{\text{time taken}} = \frac{\text{Energy transformed}}{\text{time taken}}$$

The SI unit of power is Watt (W), after James Watt the scotish engineer and inventor who played a key role in the development of steam engine.

Hence, $1W = 1\,J/S$

☞ **Remember**, the faster the work done, the more powerful the agent that does the work okey!

Notice also that, from the relation $P = \frac{W}{t} = \frac{FS}{t} = FV$ from this, we can see that, power is the dot product of force and velocity vectors. That is, if we have,

$$\vec{F} = F_x i + F_y j + F_z k \text{ and}$$

$$\vec{V} = V_x i + V_y j + V_z k, \text{ then}$$

$$P = \vec{F} \times \vec{V} = F_x V_x + F_y V_y + F_z V_z = Fv \cos\theta$$

Conceptual Examples

64. In everyday language, A strong person is called powerful. What is wrong with this expression?

Explanation: Being muscular or energetic doesn't mean powerful. What matters is how fast he can do a given work relative to others.

65. An advertisement claims that the diesel engine produce 2800 hp to drive a generator that produces 30MW electric power. Is this possible?

Explanation: No! 1 hp = 764W $\Rightarrow$2800 hp = 20MW

Illustrative Examples

66. Water flows over a section of Abay Falls at a rate of 1.2×10^6 kg/s from 50m height. How many 60W bulbs can be lit with this power?

Solution: $P = \frac{W}{t} = \frac{mgh}{t} = \frac{1.2\times10^6 \times 10 \times 50}{1} = 0.6\text{Gw}$

$$n = \frac{P}{60} = 10{,}000{,}000 \text{ bulbs}$$

67. A Grand Dam in Ethiopia is 1270m long and 170m high. The electrical power output from generators of its base is about 2000MW. How many cubic meters of water must flow from the top of the dam per second to produce this power if 92% of the work done on the water by gravity is converted to electrical energy? Take $1m^3 = 1000$ kg.

Solution: $P = \frac{W}{t} = \frac{mgh}{t} \Rightarrow 0.92\left(\frac{m}{t}\right)(10)(170) = 2000\text{MW}$

$$\Rightarrow \frac{m}{t} = 1.3\times10^3 m^3/s$$

68. A pump is required to lift 800 kg of water per minute from a well 14m deep and eject it with a speed of 18 m/s.
    a) How much work is done per minute in lifting the water?
    b) How much work is done in giving the water the kinetic energy it has when ejected?

c) What must be the power output of the pump?

Solution: a. $W = mgh = (800)(10)(14) = 1.1 \times 10^5 J$

b. $W = \Delta k_E = \frac{1}{2}mv^2 = \frac{1}{2}(800)(18)^2 = 1.3 \times 10^5 J$

c. $P = \frac{W}{t} = \frac{mgh + \frac{1}{2}mV^2}{t} = \frac{(1.1+1.3)10^5}{60} = 4KW$

69. At a certain instant, a particle experiences a force $F = (4i - 2j + 9k)$ N while having a velocity $v = (-2i + 4k)$ m/s. What is the instantaneous rate at which the force does work on the particle?

Solution: $P = \vec{F} \cdot \vec{v} = (4i - 2j + 9k).(-2i + 4k) = 28W$

70. A 100 kg block is pulled at a constant speed of 5m/s across a horizontal floor by an applied force of 122N directed 37^0 above the horizontal. What is the… rate at which the force does work on the block?

Solution: $P = \vec{F} \cdot \vec{v} = (F \cos\theta)(v) = (122)(0.8)(5) = 488W$

71. An elevator has a mass of 4500 kg and can carry a maximum load of 1800 kg. If the elevator is moving upward at full load at 4 m/s. What power is required of the force moving the elevator to maintain that speed?

Solution: $P = Fv = (m_1 + m_2)gv = (6300)(10)(4) = 252$ KW

72. What power must a man expend on a 100kg log that he is dragging down a hill side at a speed of 0.5m/s? The slope makes 20° with the horizontal and $\mu_k = 0.5$.

Solution: $W + W_f = \Delta G_{PE} \Rightarrow W = \mu mg\cos\theta - mg\sin\theta$

$$P = \frac{W}{t} = Fv = mgv(\mu\cos\theta - \sin\theta)$$

$$P = (100)(10)(0.5)(0.5 \times 0.9 - 0.3) = 75W$$

73. A 2kg object is accelerated uniformly from rest to a speed of 8 m/s in 4sec. what power is delivered to the object just after 3 sec?

A. 12 w B. 16 w C. 24 w D. 32 w

Solution: $a = \frac{\Delta v}{\Delta t} = \frac{8}{4} = 2\ m/s^2$

$V = u + at = 0 + (2)(3) = 6m/s$

$F = ma = (2)(2) = 4N$

$\therefore P = FV = (4)(6) = 24$ w.

Answer: C

Unit: 5 Additional Examples

74. An automobile travelling with a speed of 60 km/h can brake to stop within 20m. If the car is moving twice as fast, the stopping distance will be?

A. 60 m B. 40 m C. 20 m D. 80 m

Solution: $\frac{1}{2}mv^2 = fs \Rightarrow s \sim v^2$

Answer: D

75. If momentum is increased by 100%, the percentage increase in kinetic energy is

A. 100 B. 200 C. 300 D. 400

Solution: $K_E = \frac{P^2}{2m} \therefore \frac{K_{E1}}{K_{E2}} = \left(\frac{P_1}{P_2}\right)^2 = \frac{1}{4} \Rightarrow K_{E2} = 4K_{E1}$

$$\% = \left(\frac{KE_2 - KE_1}{K_{E1}}\right)100\%$$

Answer: C

76. A 4kg body is thrown upward with a speed of 5m/s. Its kinetic energy just before hitting the ground is?

A. 25 J B. 50 J C. 75 J D. 100 J

Solution: $\frac{1}{2}mv^2 = \frac{1}{2}(4)(5)^2 = 50J$

Answer: B

77. The work done in stretching a spring of force constant K from length ℓ_1 to ℓ_2 is

A. $K(\ell_2^2 - \ell_1^2)$ B. $\frac{1}{2}K(\ell_2^2 - \ell_1^2)$

C. $K(\ell_2 - \ell_1)^2$ D. $\frac{1}{2}K(\ell_2 + \ell_1)$

Solution: $W = PE_2 - PE_1 = \frac{1}{2}K\ell_2^2 - \frac{1}{2}K\ell_1^2 = \frac{1}{2}K(\ell_2^2 - \ell_1^2)$

Answer: B

78. A machine which is 70% efficient raises a 10kg body through a certain distance and spends 100J of energy. The body is then released. On reaching the ground its kinetic energy will be?

A. 50 J B. 35 J C. 70 J D. 100 J

Answer: C

Hint: its efficiency is 70%

79. An engine develops 10kw power. How much time will it take to lift a mass of 200 kg to a height of 40m

A. 4 sec B. 5 sec
C. 8 sec D. 10 sec

Solution: $P = \frac{w}{P} = \frac{mgh}{P} = \frac{(200)(10)(40)}{(10)(1000)} = 8\text{sec}$ **Answer: C**

80. The power developed by a 50 kg boy when moving with a velocity of 2m/s upward in a direction making an angle of 60° with the vertical is

A. 1980 w B. 500 w
C. $490\sqrt{3}$w D. 245 w

Solution: $P = mgh = (50)(10)(2)\cos 60^\circ = 500\text{w}$ **Answer: B**

81. A 4kg mass is moving on a frictionless horizontal table with a velocity of 20m/s. It strikes an ideal spring and comes to rest. If the spring constant is 100 N/m, then the compression of the spring is?

A. 4 mm B. 4 cm C. 4m D. 0.4m

Solution: $\frac{1}{2}kx^2 = \frac{1}{2}mv^2 \Rightarrow x = v\sqrt{\frac{m}{k}} = 20\sqrt{\frac{4}{100}} = 4\text{m}$ **Answer: C**

82. A bullet fired into a target loss half its velocity after penetrating 25cm, how much further will it penetrate before coming to rest?

A. 5 cm B. 25 cm C. 8.3 cm D. 5.3cm

Solution: 1st case, $\frac{1}{2}mv^2 - \frac{1}{2}m\left(\frac{v}{2}\right)^2 = 25f$

$$\frac{3}{8}mv^2 = 25\,f \quad \ldots\ldots(*)$$

2nd case, $\frac{1}{2}m\left(\frac{v}{2}\right)^2 = fs$

$$\frac{1}{8}mv^2 = fs \quad \ldots\ldots(**)$$

take ratio, $s = 8.3$ cm **Answer: C**

83. A body of mass 2kg is thrown up with kinetic energy of 490J. The height at which the kinetic energy of the body becomes half its original value is?

A. 50m B. 25m C. 12.2m D. 10m

Solution: $\frac{490}{2} = 245 = mgh \Rightarrow h = 12.25m$ **Answer: C**

84. A ball is allowed to fall from a height of 10 m. If there is 40% loss of energy due to impact, then after one impact, the ball will go upto

A. 10 m B. 8m C. 4m D. 60m

Solution: 60% energy is retained

$$h=\left(\frac{60}{100}\right)(10) = 6m$$

Answer: D

85. Two spring have force constants K_1 and K_2. If they are stretched till their energies are equal with forces F_1 and F_2 then F_1: F_2 is

A. $K_1 : K_2$ B. $K_2 : K_1$

C. $\sqrt{k_1} : \sqrt{k_2}$ D. $K_1^2 : K_2^2$

Solution: $PE = \frac{F^2}{2k} \Rightarrow F \sim \sqrt{k}$ **Answer: C**

86. Two balls at the same temperature collide. What is conserved?

A. momentum B. kinetic energy

C. velocity D. temperature

Answer: A

87. If momentum is increased by 20%, then kinetic energy is increased by a percent of

A. 40% B. 44% C. 48% D. 56%

Solution: $\frac{KE^1}{KE}=\left(1+\frac{1}{5}\right)^2=\frac{36}{25}$

$$\left(\frac{KE^1}{KE}-1\right)100=\left(\frac{36}{25}-1\right)100 = 44\%$$

Answer: B

88. A body is dropped from a height of 20m and rebounds to a height of 10m. The loss of energy is

A. 10% B. 45% C. 50% D. 75%

Solution: $h^1=\frac{1}{2}h \Rightarrow mgh^1=\frac{1}{2}mgh$, 50% loss **Answer: C**

89. A force $\vec{F} = (5i - 3j)$ N is applied over a particle which displaces it from the origin to the point $\vec{r} = (2i - j)$ m. The work done on the particle is?

A. -7 J B. 7 J C. 13 J D. 11 J

Solution: $W = \vec{F} \cdot \vec{S} = (Ji + 3j) \cdot (2i - j) = 10 - 3 = 7J$ **Answer: B**

90. The acceleration due to gravity on plant A is 9 times the acceleration due to gravity on plant B. A man Jumps to a height of 2m on the surface of A. What is the height of jump by the same person on plant B

A. $\frac{2}{9}$ m B. 18m

C. 6 m D. $\frac{2}{3}$ m

Solution: $\sqrt{2g_A h_A} = \sqrt{2g_B h_B}$ or $h_B = \frac{\rho_A h_A}{\rho B} = 18m$ **Answer: B**

92. When a long spring is stretched by 2 cm, its potential energy is U. If the spring is stretched by 10 cm, the potential energy stored will be?

A. $\frac{U}{5}$ B. 5U C. 10 U D. 25 U

Solution: $PE_1 = \frac{1}{2} KX_1^2$ and $PE_2 = \frac{1}{2} KX_2^2$, $X_2 = 5x_1$

$$PE_2 = \frac{1}{2} K(5x_1)^2 = 25\left(\frac{1}{2} KX_1^2\right) = 25U$$ **Answer: D**

93. Consider the following two statements

a. linear momentum of a system of particles is zero

b. kinetic energy of a system of particles is zero

Then:

A. A implies B and B implies A

B. A does not imply B and B does not imply A

C. A implies B but B does not imply A

D. A does not imply B but B implies A

Explanation: If KE is zero, velocity is zero and hence momentum is zero. **Answer: D**

94. A spring of force constant 500 N/m is stretched initially by 5 cm from the un stretched position. Then, the work required to stretch it further by another 5 cm is

A. 6.25 N.m B. 12.50 Nm
C. 18.75 Nm D. 25. 0Nm

Solution:

$$W = \frac{1}{2}K\left(x_2^{\,2} - x_1^{\,2}\right) = \frac{1}{2}(5000)\left[\left(10\times10^{-2}\right)^2 - \left(5\times10^{-2}\right)^2\right]$$

$= 18.75$ Nm

Answer: C

95. A mass of 0.5 kg moving with a speed of 1.5 m/s on a horizontal smooth surface collides with a spring of force constant 50 N/m. The maximum compression of the spring would be

A. 0.12 m
B. 1.5 m
C. 0.5 m
D. 0.15 m

Solution: $\frac{1}{2}kx^2 = \frac{1}{2}mv^2 \Rightarrow x = 0.15m$ **Answer: D**

96. A uniform chain of length 2m is kept on a table such that a length of 60cm hangs freely from the edge of the table. The total mass of the chain is 4kg. What is the work done in pulling the entire chain on the table?

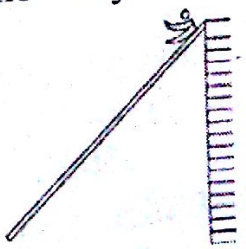
A. 120 J B. 3.6 J C. 7.2 J D. 1200 J

Solution: mass of hanging portion $= \frac{4}{2}(0.6) = 1.2$kg

Distance the center of mass moved = 0.3 m

$W = mgh = (1.2)\,(10)\,(0.3) = 3.6$ J **Answer: B**

97. In a children's park, there is a slide which has a total length of 10m and a height of 8m. A boy of 200N climbs up and slides down. The average friction of the slide is three – tenth of his weight. The work done by the slide on the body as he comes down is?



A. 0 J B. 600J
C. −600J D. 1600 J

Solution: $w = -fs = \frac{3}{10}mgs$

$$W = \frac{3}{10}(200)(10) = -600J$$

Answer: C

98. A pendulum bob has a speed of 3m/s at its lowest position. The pendulum is 0.5m long. The speed of the bob when the length makes 60^o to the vertical will be?

A. 3m/s B. $\frac{1}{3}$ m/s C. $\frac{1}{2}$ m/s D. 2 m/s

Solution: $\frac{1}{2}mv^2 + mg\ell(1-\cos\theta) = \frac{1}{2}mv_{max}^2$

$$\frac{1}{2}V^2 + 5\left(1-\frac{1}{2}\right) = \frac{1}{2}(3)^2 \Rightarrow V = 2m/s$$

Answer: D

99. A ball of mass lkg collides with a wall with speed 8m/s and rebounds with same speed. The work done by the ball on the wall is?

A. 6 J B. 8 J C. 9 J D. zero

Answer: D

100. The energy required to accelerate a car from rest to 10m/s is E. what energy is required to accelerate the car from 10m/s to 20m/s

A. E B. 3E C. 5E D. 7E

Solution: $E = \frac{1}{2}m(10)^2$ and $E^1 = \frac{1}{2}m\left(20^2 - 10^2\right) = \frac{1}{2}m(300)$

$$\frac{E_1}{E} = 3 \text{ or } E^1 = 3E$$

Answer: B

101. Ten liter of water per second is lifted from a well through 10m and delivered with a velocity of 10m/s. Then the power of the motor is?

A. 1kw B. 1.5kw C. 2kw D. 2.5kw

Solution $P = \frac{\frac{1}{2}mv^2 + mgh}{t} = \frac{\frac{1}{2}(10)(10)^2 + (10)(10)(10)}{1} = 1.5kw$

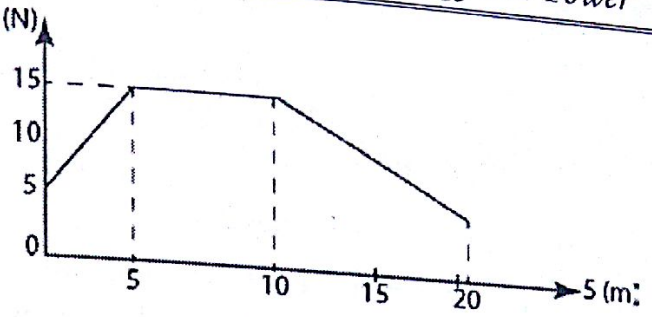
Answer: B

102. From the frictional force versus displacement. The loss of kinetic energy in travelling over s = 0 to s = 20m is

A. 250 J B. 200J C. 150 J D. 10J

Solution: work = Area

$= 250$ J



Answer: A

103. A pump motor is used to deliver water at a certain rate from a given pipe. To obtain thrice as much water from the same pipe in the same time, power of the motor has to be increased

A. 3 times B. 9 times
C. 27 times D. 81 times

Explanation: $P = \dfrac{\frac{1}{2}mv^2}{t}$, increase m and v by a factor of 3

Answer: C

104. A bomb of mass 9 kg at rest explodes into two pieces of masses 3kg and 6kg. The velocity of mass 3 kg is 16m/s. the kinetic energy of mass 6 kg is

A. 96 J B. 192 J C. 384J D. 768 J

Solution: $6v = (3)(16) = 48 \Rightarrow v = 8$m/s

$$K_E = \frac{1}{2}(6)(8)^2 = 192\text{J}$$

Answer: B

105. The momentum of a body of kinetic energy E is doubled. The new kinetic energy is

A. E B. 4E C. 16 E D. 32 E

Answer: B

106. If momentum of a body decrease by 10%, then the percentage decease in kinetic energy is

A. 19 B. 21 C. 37 D. 42

Solution: momentum decreases by 10%, velocity decreases by 10% $V_2 = V - \dfrac{10}{100} = \frac{9}{10}v$

$$\% \text{ decrease in } K_E = \frac{\frac{1}{2}mv^2 - \frac{1}{2}m\left(\frac{9}{10}v\right)^2}{\frac{1}{2}mv^2} = 19$$ Answer: A

107. If distance is plotted against x – axis and kinetic energy against y – axis, then the slop of the graph so obtained is proportional to.
A. distance B. kinetic energy
C. velocity D. acceleration
Answer: D

108. An automobile moving at 30km/h can stop in 8m. If the same automobile is moving at 60km/h, its stopping distance is?
A. 8m B. 16m C. 24m D. 32

Solution: $\frac{1}{2}mv^2 = fs \Rightarrow fs \Rightarrow \frac{m}{2f} = \frac{s}{v^2}$

$$\frac{S_1}{V_1^2} = \frac{S_2}{V_2^2} \Rightarrow S_2 = 32m$$ Answer: D

109. Which of the following statements is incorrect?
A. Kinetic energy may be zero, positive or negative
B. Potential energy may be zero, positive or negative
C. Power, energy and work are all scalars
D. Ballistic pendulum measures speed of bullets
Hint: Kinetic energy can never be negative Answer: A

110. A ball is thrown upward in free space. Its total mechanical energy.
A. remains constant throughout
B. increases during ascent and decreases during descent
C. is zero at maximum height
D. is equal to kinetic energy Answer: A

111. Two spring P and Q having stiffness constants K_1 and K_2 Where $K_2 < K_1$ respectively are stretched equally. Then
A. more work is done on Q
B. more work is done on P
C. their force constants will become equal
D. equal work is done on both the springs

Explanation: Since $K_1 > K_2$, P will develop greater restoring force than Q. So more work will be done on P.

Answer: B

112. A water pump driven by petrol raises water at a rate of 0.5 m^3/min from a depth of 30m. If the pump is 70% efficient, what power is developed by the engine?

A. 1750w B. 2450w
C. 3500w D. 7000w

Solution: $\frac{\Delta m}{\Delta t} = 500\text{kg/min}$, $0.7\ p = \frac{mgh}{t}$

$$P = \frac{mgh}{0.7t} = 3500\ w$$

Answer: C

113. The mechanical energy required to accelerate a body from 10m/s to 20m/s compared with that required to accelerate from 0 to 10m/s in the same time and distance is

A. two times B. three times
C. four times D. five times

Solution: 1st case, $K_E = \frac{1}{2}m\left(V^2 - U^2\right) = \frac{1}{2}m\,(300)$

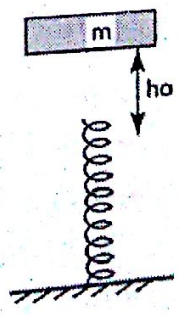
2nd case, $K_E = \frac{1}{2}m\left(V^2 - U^2\right) = \frac{1}{2}m\,(100)$ **Answer: B**

114. Water is falling on the blades of a turbine at a rate of 6000kg/min. The height of the fall is 100m. What is the power gained by the turbine?

A. 10kw B. 6 kw
C. 100 kw D. 600 kw

Solution: $P = \frac{mgh}{t} = \left(\frac{6000}{60}\right)(10)(100) = 100\text{kw}$ **Answer: C**

115. A slab of mass m is released from a height h_o from the top of a spring of force constant K. The maximum compression of the spring is obtained from



A. $mgh_o = \frac{1}{2}Kx^2$

B. $mg\,(h_o - x) = \frac{1}{2}Kx^2$

C. $mgh_o = \frac{1}{2}(h_o + x)^2$

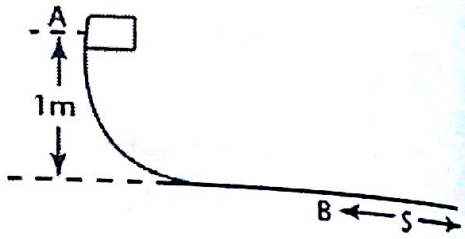
D. $mg(h_o + x) = \frac{1}{2}Kx^2$

Hint: transformation of energy:

Answer: D

116. A block weighing 10N travels down a smooth track AB joined to a rough surface of $\mu = 0.2$ and distance s. What is the length of the rough surface

A. 1m
B. 2m
C. 3m
D. 5m



Solution: $mgh = \mu mgs$

$$S = \frac{h}{\mu} = \frac{1}{0.2} = 5m$$

Answer: D

117. A 30g bullet initially travelling horizontally at 50 m/s penetrates 10 cm into a wooden block. The average force it exerts is?

A. 4000N
B. 375N
C. 2500N
D. 4500N

Solution: $W = \Delta KE \Rightarrow fs = \frac{1}{2}m(V^2 - U^2)$

$$f = \frac{mv^2}{2s} = \frac{(0.03)(2500)}{2(0.1)} = 375N$$

Answer: B

118. The velocity of a 2 kg body is changed from (4i + 3j) m/s to $\hat{K}\,m/s$. The work done on the body is

A. 9J
B. 11J
C. 1J
D. 0J

Solution: $u = \sqrt{4^2 + 3^2} = 5m/s, \quad V = 6m/s$

$$W = \Delta K_E = \frac{1}{2}m(V^2 - U^2) = \frac{1}{2}(2)(6^2 - 5^2) = 11J$$

Answer: B

119. A ball is allowed to fall from a height of 10m. There is 40% loss of energy every time it hits the ground. After second impact with the ground, the ball will rise up to

A. 10m
B. 6m
C. 3.6m
D. 2.4m

Solution: $h_1 = \left(\frac{60}{100}\right)(10) = 6m$ and $h_2 = \left(\frac{60}{100}\right)(6) = 3.6m$

Answer: C

120. A tennis ball dropped from a height of 2m rebounds to 1.5m. What fraction of its energy is lost due to impact?

A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. $\frac{1}{3}$ D. $\frac{1}{8}$

Solution: Fraction loss $= \frac{mgh - mgh^1}{mgh} = \frac{2-1.5}{2} = \frac{1}{4}$

Answer: A

121. A 3kg body is dropped from the top of a tower of height 135m. The kinetic energy of the body after 3 sec is?

A. 950 J B. 10 J
C. 1150 J D. 1350 J

Solution: speed after 3sec, v = gt = (10) (3) = 30m/s

$$K_E = \frac{1}{2}mv^2 = \frac{1}{2}(3)(30)^2 = 1350J$$ **Answer: D**

122. A bomb of mass 12kg at rest explodes into two pieces of masses 4kg and 8kg. The velocity of 8 kg is 6m/s. The kinetic energy of the other mass is

A. 48 J B. 32 J C. 24 J D. 288J

Solution: 12 (0) = (8) (6) + 4 (v) $\Rightarrow$ v = 12 m/s

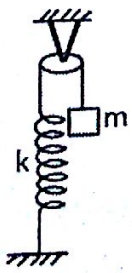
$$K_E = \frac{1}{2}(4)(12)^2 = 288J$$ **Answer: D**

123. Initially the spring is unstretched. When the system is released from rest, the maximum elongation of the spring is?

A. $\frac{mg}{k}$ B. $\frac{2mg}{k}$
C. $\frac{3mg}{k}$ D. $\frac{4mg}{k}$

Solution: $\frac{1}{2}kx^2 = mgx$

$x = \frac{2mg}{k}$



Answer: B

124. A body at rest may have
A. energy B. momentum
C. speed D. velocity

Answer: A

125. A particle of mass 1 g is rotated in a horizontal circle of radius 1m and moves with a velocity of 2m/s in magnitude. The work done in one revelation is
A. zero B. 1 J C. 10 J D. 2J

Explanation: work done by a centripetal force is zero

Answer: A

126. At 1 kg body accelerates uniformly from ret to a velocity of 1m/s in 15 sec. the kinetic energy will be $\frac{2}{6}$J at a time of:
A. 4 sec B. 8 sec
C. 10 sec D. 12 sec

Solution: $a = \frac{1-0}{15} = \frac{1}{15} m/s^2$

$$\frac{1}{2}mu^2 = \frac{2}{9} \Rightarrow \frac{1}{2}(1)(v^2) = \frac{2}{9} \text{ or } v = \frac{2}{3}$$

$$V = u + at \text{ or } \frac{2}{3} = 0 + \frac{1}{15}t \Rightarrow t = 10sec$$

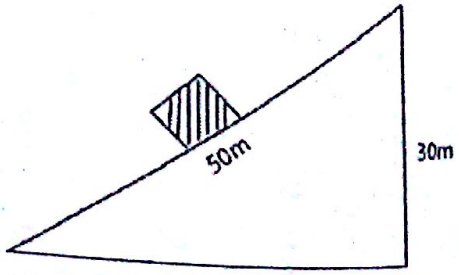
Answer: C

127. If the momentum of a body is numerically equal to its kinetic energy, then the speed of the body is?
A. 1m/s B. 2 m/s C. 4m/s D. 8m/s

Solution: $\frac{1}{2}mv^2 = mv \Rightarrow v = 2m/s$

Answer: B

128. A block of mass 50kg is pulled up along an inclined plane of length 50m. If the coefficient of fiction is 0.4, the work done will be?
A. 10 J
B. 23 kJ
C. 25 kJ
D. 28 kJ



Solution: - work done against gravity
= mgh = (50) (10) (30) = 15kJ
- work done against friction

$= \mu mg \cos\theta\,(s) = (0.4)\,(50)\,(10)\,(50)\left(\frac{40}{50}\right) = 8kJ$

Total work done = 15 kJ + 8 kJ = 23 kJ **Answer: B**

129. Which of the following is a form of energy?
A. pressure B. momentum
C. light D. power

Answer: C

130. If the force acting on a body is inversely proportional to its speed, the kinetic energy of the body is?
A. constant
B. directly proportional to time
C. inversely proportional to time
D. directly proportional to square of time

Hint: $F \sim \frac{1}{v} \Rightarrow FV = \text{constant}$ **Answer: B**

131. A body falling from a height of 10m rebounds losing 20% of energy in the impact, how much will it rebound?
A. 10m B. 8m C. 12m D. 20m

Solution: $\frac{20}{100} = \frac{10 - h^1}{10} \Rightarrow h^1 = 8m$ **Answer: B**

132. The displacement of a body of mass 2kg as a function of time is given by $x = 2t^2 + 5$. The increase in its kinetic energy one second after the start of motion is
A. 8J B. 16J C. 32 J D. 64J

Solution: $V = \frac{dx}{dt} = 4t$, at $t = 1sec$, $V = 4m/s$

$$K_E = \frac{1}{2}mv^2 = \frac{1}{2}(2)(4)^2 = 16J$$ **Answer: B**

133. There will be a decrease in potential energy of a body acted on by conservative force if positive work is done
A. by conservative force
B. by non-conservative force
C. by conservative or non-conservative force
D. P.E never decreases

Explanation: If conservative force acting on a body does positive work, its K_E increases and P_E decreases

Answer: A

134. A particle of mass 100g is thrown upward with a speed of 5m/s. The work done by gravity on the particle is?

A. -0.5 J B. -1.25 J
C. 1.25 J D. 0.5 J

Solution: $w = -mgh = -\frac{1}{2}mv^2 = -\frac{1}{2}(0.1)(5)^2 = -1.25$ J

Answer: B

135. The work done by a force is defined by $W = \vec{F} \cdot \vec{S}$. If $\vec{F}$ and $\vec{S}$ are not zero but work done is zero, we calculate that.

A. $\vec{F} > \vec{S}$
B. $\vec{F}$ and $\vec{S}$ are in same direction
C. $\vec{F}$ and $\vec{S}$ in opposite direct
D. $\vec{F}$ and $\vec{S}$ are at right angle

Hint: $\vec{F} \cdot \vec{S} = FS \cos\theta = 0$ if $\theta = 90°$

Answer: D

136. A force of $\vec{F} = (-2i + 15j + 6k)$ N displaces a body 10m along the Y axis. The work done is

A. 20 J B. 150 J C. 160 J D. 190 J

Solution: $W = \vec{F} \cdot \vec{S} = (15)(10) = 150$J

Answer: B

137. A force $\vec{F} = 3i - 2J + 4k$ displaces a body from a point A (8, −2, −3) to B (−2, 0, 6). The work done is?

A. 1 unit B. 2 units
C. 3 units D. 4 units

Solution: $W = \vec{F} \cdot \vec{S}$ and $\vec{S} = (-2-8, 0-(-2), 6-(-3)) = (-10, 2, 9)$

$W = (-10)(3) + (-2)(2) + (4)(9) = 2$ units

Answer: B

138. A force of $(10i - 3j + 6k)$ N acts on a body of 100g and displaces it from $(6i + 5j - 3k)$m to $(10i - 2j + 7k)$m The work done is

A. 21 J B. 121 J C. 361 J D. 1000 J

Solution:

$W = \vec{F} \cdot \vec{S}$, $\vec{S} = ((10-6)i + (-2-5)j + (7-(-3)K) = 4i - 7J + 10$

$W = (4)(10) + (-3)(-7) + (6)(10) = 121$ J

Answer: B

139. A force $\vec{F} = 2i + 2j$ N displaces a particle through $\vec{S} = 2i + 2\hat{k}$ m in 16 second. The power developed by $\vec{F}$ is

A. 0.25 w B. 25 w
C. 225 w D. 450 w

Solution: $P = \frac{w}{t} = \frac{\vec{F}\cdot\vec{S}}{t} = \frac{4}{16} = 0.25w$ **Answer: A**

140. A spring is stretched by 2cm and its potential energy is u. If the spring is stretched by 10cm, the potential energy will be?
A. u/25 B. u/5 C. 5u D. 25u

Hint: $P_E = \frac{1}{2}kx^2$ and K= constant **Answer: D**

141. It is easier to draw up a wooden block along an inclined plane than to have it up vertically because
A. The friction is reduced
B. The mass becomes smaller
C. only part of it is over come
D. g becomes less **Answer: C**

142. An object of mass 2 kg falls from a height of 20m. The loss of potential energy just before it strikes the ground
A. 20 J B. 40 J C. 200 J D. 400 J **Answer: D**

143. Under the action of a force, an 8kg body moves $x = \frac{t^2}{2}$. The work done by the force in the first two seconds is?
A. 1600 J B. 160 J C. 16 J D. 1.6 J

Solution: $w = FS = mas, \quad a = \frac{d^2x}{dt^2} = 1m/s^2$

at $t = 2, \quad x = 2m$ **Answer: C**

144. A 0.5 kg ball is moving with certain kinetic energy. If it is to lose maximum energy by elastic collision with a second ball at rest, the second ball should have a mass
A. greater than 0.5 kg B. less than 0.5 kg
C. equal to 0.5 kg D. infinite **Answer: C**

Unit 5: Conceptual Questions

1. An Earth Satellite is in a circular orbit at an altitude of 5000km. What is the work done by gravity on the satellite?
2. Spring A is stiffer than spring B. Which spring store more energy if the springs are compressed by the same force?
3. If you cut a spring in half, the spring constant of either half will be changed by _____ factor of the original spring.

4. Can the gravitational potential energy of a system be negative? Explain
5. An automobile strikes a tree and comes to rest. The total energy in the car is now less than before. How and why?
6. An automobile strikes a tree and comes to rest. The total energy in the car is now less than before. How and why?
7. A rope tied to a body causes the body to accelerate. But Newton's 3rd law says the body pulls back. Is the total work done zero? Explain
8. A compressed spring is clamped in its compressed position and then is dissolved in acid. What becomes of its potential energy?

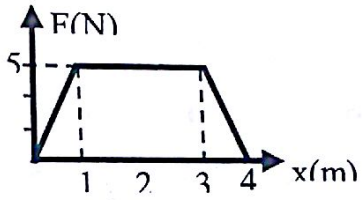
Unit 5: Problems

Work as a Scalar Product

9. A man drags a box $\begin{bmatrix} 20 \\ -1 \end{bmatrix}$ m with a force of $\begin{bmatrix} 35 \\ 25 \end{bmatrix}$ N. How much work does the man do on the box?
10. A force of 20N is applied on a body at an angle of 60° to the direction of the displacement. If the body displaced by 8m, how much work is done by the force.
11. Starting from rest, a 5 kg block slides 2.5m down a rough 30° incline. The coefficient of kinetic friction between the block and the incline is 0.4, determine
    a) Work done by gravity
    b) The work done by the friction force.
    c) The work done by the normal force.
12. A 102 kg object moves in a horizontal straight line path with an initial speed of 53 m/s. If it is stopped along that line with a declaration of 2 m/s^2.
    a) What distance does it travel while decelerating?
    b) What work is done by the decelerating force?
13. A machine weighing 100 kg is to be loaded on a truck 2.5m high by a 10 m long incline. If it requires 300N to overcome friction the work performed is.
14. A 2000 kg car is being pulled up a 37° incline at a constant speed by a cable parallel to the incline. If the friction force is 800N. What is the work done by the cable on the car?

Work Done by Variable Force

15. A variable force $F = (x + 4)N/m$ is applied on a body and displace it by 5m from its original position. How much work is done by the force?
16. The force required to stretch a certain spring a distance x beyond its unscratched length is given by $F = 100x$. How much work is required to stretch the spring by 0.2m?
17. What work is done by a force $F = (2x)i + 3jN$ that moves a particle from a position $r_i = 2i + 3jm$ to $r_f = -4i - 3jm$.
18. A force $F = (4xi + 3yj)$ N acts on a particle as the particle moves in the x direction from the origin to $x = 5m$. What is the work done by the force on the particle?
19. The x directed force that acts on a body is shown as a function of x. What is the work done in the interval?
    a) $0 \leq x \leq 1$
    b) $1 \leq x \leq 3$
    c) $0 \leq x \leq 4$



Work Energy Theorem

20. A football of mass 550 g is at rest on the ground. The ball is kicked with a force of 108N. The footballer's foot is in contact with the ball for 0.3m.
    a) What is the kinetic energy of the ball?
    b) What is the ball's velocity at the moment it loses contact?
21. How much work is required to increase the speed of a 4 kg object from 20 m/s to 40 m/s?
22. A force of 5N acts on a 2 kg body moving in the direction of the force with 4 m/s. Over what distance must the force act to change the bodies speed to 6 m/s.
23. A 2000 kg car moves starting from rest along a level highway under the action of a 1000N forward force and a 950N resistive force. What is the speed of the car after it has moved a distance of 20m?

Potential Energy

24. A boy walks up a hill. His displacement from his starting point is (800, 150)m. How much potential energy he gained?
25. A spring has force constant of 150 N/m and a mass of 100g is attached to its. The spring sits on a horizontal smooth surface and

the other end of the spring is attached to a fixed bar. If the mass is pulled by 10 cm and let go, what is the highest velocity of the mass.

26. A spring is stretched by 50mm by the application of a force. Find the work done if the force required to stretch 1mm of the spring is 10N.

27. A pump lifts water from a lake to a large tank 20m above the lake. How much work does the pump do as it transfers $5m^3$ of water to the tank? take $1m^3 = 1000$ kg

28. When a 0.2 kg load is attached to a spring, the energy stored is 0.5J. If an additional mass of 0.3 kg is added, what will be its stored energy?

29. A spring of force constant 2000 N/m was put vertically in a tall cylinder. If a mass of 4 kg that was 1m above from the top of the spring was released, what will be the maximum compression of the spring?

Conservation of Energy

30. A ball of mass 500g is kicked into the air at an angle of 45°. It reaches a height of 12m. What was its initial velocity?

31. A pendulum bob has a mass of 1 kg. The bob is pulled to one side to an angle of 10° from the vertical.
    a) What is the velocity of the pendulum bob as it swings through its lowest point?
    b) What is the height the bob will rise?

32. A pool ball of mass 100g and velocity $\begin{bmatrix} 5 \\ -4 \end{bmatrix}$ m/s collides with a stationary pool ball of the same mass. After the collision, the pool balls have velocities $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$ m/s. Is the collision elastic? Explain

33. A block of mass 5 kg is released from rest on a an inclined plane as shown. What is the maximum compression of the spring if the spring constant is 1 N/mm and $\mu_k = 0.2$.

34. A 60 kg man dives vertically downwards into a swimming pool from a 20m height. He go down in the water by 2m and started rising. What is the resistance of the water?

Power

35. A weightlifter lifts 200 kg through 1.8m in 2 sec.
    a) What is his power?
    b) Why is this actual power likely to be higher than this?

36. A petrol engine rises 200ℓ of water in a well from a depth of 7m in 6 sec. What is the power of the engine?
37. A net force of 10N acts on a 20 kg body that is initially at rest. What is the instantaneous power exerted by the force just after 2 sec?
38. A worker is given a job to lift 30 kg crates 0.9m from the ground and load on to a bed of truck. How many crates would he have to load onto the truck in 1 minute if his average power output to be 0.5 hp?
39. Water flows from a reservoir at the rate of 3000 kg/min to a turbine 120m below. If the efficiency of the turbine is 80%. What is the power output of the turbine?

Unit 5: Explanation to Questions

1. It is zero. Because, displacement and force are at right angle
2. $E_{PB} > E_{PA}$
3. Twice, by a factor of two
4. Yes, change in gravitational potential energy becomes negative when the body is displaced to a lower elevation negative when the body is displaced to a lower elevation
5. i. If a box is placed on the bed of a truck, the static friction gives the box the same acceleration is the truck
   ii. The friction force on an accelerating car's tires by the road causes an increase in the car is kinetic energy
6. Because, energy of the car is transferred in a number of ways, such as deformation, sound, heat, vibration, matter. etc
7. No. mass difference will causes net work done
8. Change its form, such as into sound, heat, vibration. etc.

Unit 5: Solution to Problems

9. $W = \vec{F}.\vec{S} = (20)(35) + (-1)(25) = 675J$
10. $W = FS\cos\theta = (20)(8)(0.5) = 80\ J$
11. a. $Wg = (mgs)\sin\theta = 62.5\ J$
    b. $Wf = \mu\,(mgs)\cos\theta = 43.30\ J$
    c. zero
12. a. $V^2 = u^2 + 2as \Rightarrow s = 702m$ b. $w = FS = -1.4\times10^5\ J$
13. $w = wf + wg = fs + mgh = 28000\ J$
14. $w = wf + wg = fs + (mgs)\sin\theta = 384\ kJ$

15. $W = (Fav)(S) = \frac{(0+4)+(5+4)}{2}(5) = 32.5J$

16. $W = \frac{1}{2}kx^2,\ K = f/x = \frac{2(2)+2(-4)}{2}i + \frac{(3+3)}{2}j = -2i+3j$

17. $W = (F_{av})(\Delta x),\quad F_{av} = \frac{2(2)+2(-4)}{2}i + \frac{(3+3)}{2}j = -2i+3j$

$\Delta x = -6i - 6j \Rightarrow w = (-2i+3j)\cdot(-6i-6j) = -6J$

18. $W = F\cdot S = F_{av}\ S = \frac{4(0)+4(5)}{2}(5) = 50J$

19. a. $w = \frac{1}{2}FS = \frac{1}{2}(5)(1) = 2.5J$

b. $W = Fs = (5)\ (2) = 10J$

c. 15 J

20. a. $\Delta KE = FS = (108)\ (0.3) = 32.\ 4J$

b. $W = \frac{1}{2}mv^2 = 32.4 \Rightarrow v = 12m/s$

21. $W = \Delta K_E = \frac{1}{2}m\left(v^2 - u^2\right) = 2400J$

22. $W = \Delta K_E = FS \Rightarrow S = 4m$

23. $W = \Delta K_E \Rightarrow F_S = \frac{1}{2}mu^2 \Rightarrow V = 1m/s\ ,\ F = 50N$

24. $PE = mgh = (10)\ (150)\ (m)\ J$

25. $\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow v = 3.9\ m/s$

26. $E_{PE} = \frac{1}{2}kx^2 = 12.5\ J$

27. $W = mgh = 1000,\ 000J$

28. $mg = kx, 0.5 = \frac{mg}{2k} \Rightarrow k = 2\ N/\ m,\quad E_{PE} = \frac{5}{4} = 1.25J$

29. $mg(h+x) = \frac{1}{2}kx^2 \Rightarrow x = 0.22m$

30. $\frac{1}{2}m{u_y}^2 = mgh \Rightarrow u_y = 15.5\ m/s \Rightarrow u = 21.9\ m/s$

31. a. 0.8 m/s . b. 0.03 m

32. No. kinetic energy is NOT conserved here

33. $mg\,(0.8 + x)\sin\theta = \frac{1}{2}kx^2 + \mu mg\cos\theta\,(0.8 + x) \Rightarrow x = 160$ mm

34. $mgh_1 = Fh_2 \Rightarrow F = (60)\,(10)\,(20)/\,2 = 6000N$

35. a. $P = mgh/t = 1800$ W

b. Because, he might momentarly stop

36. $P = mg\,h/t = (200)\,(10)\,(7)\,/\,6 = 2333$ w

37. $P = FV = F(at) = \frac{F^2}{m}t = 10W$

38. $P = mg\,h/t \Rightarrow (0.5)(746) = \frac{n(30\times10\times0.9)}{60} \Rightarrow n = 83$ crates

39. $P = mgh/t \Rightarrow \frac{(3000)(10)(120)}{60} = 60000$ w and 80%

$P = P_{out\ put} = 0.8\ (60000) = 4800w = 64.3$ hp

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Rotational Motion

It is certainly no exaggeration to say that rotation is part of
life. After all, we live on a planet that rotates about its axis
and that revolves about the sun once a year. Rotation occurs at
from the motions of electron in atoms to the motions of entire
In addition, engines that power vehicles and machines do have
parts, as do CDs, DVDs, not to mention the tumbling rotating
in the air we breathe. Thus, a study of rotation yields results
to a great variety of natural phenomena.

6.1 Rotational Kinematics

Real world bodies can be very complicated, the forces that act
can deform them – stretching, twisting, and squeezing them.
neglect these deformations for now and assume that the body
perfectly definite and unchanging shape and size. We call this
model a *rigid body*.

To describe rotational motion, we need to define the most basic
quantities – angular displacement, angular velocity and
acceleration.

i) Angular Displacement

Angular Displacement (θ, theta): Is the angle through which
body rotates about a fixed axis.

The relationship between the angular displacement (θ) and
displacement (s) is given by

$$s = r\theta$$

Where, r = radius of the circle
S = arc length or tangential displacement
θ = central angle or angular displacement

The angular displacement of a body can be expressed in radian,
revolution or cycle. But the SI unit of angular displacement is
(rad).

Radian: Is the angle for which the arc length on a circle of radius r is equal to the radius of the circle.

☞ From this definition it follows that, for an angle of one radian, the arc length is equal to the radius, s = r. Similarly, an angle of two radians corresponds to an arc length of two radii, s = 2r and so on. Thus, the angle θ in radians for an arbitrary arc length s on a circle of radius r is given by

$$\theta(\text{in radians}) = \frac{\text{Arc length}}{\text{Radius}} = \frac{s}{r}$$

Accordingly, the angle in radians is the ratio of two lengths. In calculations, therefore, the radian is treated as a number without units and has no effect on other units that it multiplies or divides.

Moreover, the conversion among radian, revolution and degree is possible as follows.

- 1 rev = 360°
- 1 rev = 2π rad
- 2π rad = 360°

Therefore, the number of degrees in one radian is,

$$1 \text{ rad} = \frac{360°}{2\pi} = 57.3°$$

☞ **Note**, Radians, as well as degrees and revolutions are dimensionless.

Remember also that, angular displacement is a vector quantity its direction may be clockwise or counterclockwise. And, standard convention calls a clockwise displacement negative and a counterclockwise displacement positive.

ii) Angular Velocity

Angular velocity (ω, omega): Is the time rate of change of angular displacement.

That is, Angluar velocity $= \frac{\text{Angular displacement}}{\text{time taken}}$

$$\omega = \frac{\Delta\theta}{\Delta t}$$

The SI unit of angular velocity is radian per second, rad/s. Other units such as revolution per minute (rev/min or rpm) are often used. Since 1 rev = 2π rad, we can have;

$$1 \text{ rev/s} = 2\pi \text{ rad/s and } 1 \text{ rev/min} = 1\text{rpm} = \frac{2\pi}{60} \text{ rad/s}$$

That is, 1 rad/s is about 10 rpm.

☞ In agreement with sign convention adopted for angular displacement, angular velocity is positive when the rotation is counterclockwise and negative when it is clockwise.

Angular velocity is a vector quantity whose direction is determined using the right hand rule.

Right hand rule: "If you curl your right hand fingers in the direction of rotation, then your thumb points in the direction of angular velocity."

Therefore, the angular velocity points in the positive z direction for a counter – clockwise rotation and it points in the negative z direction for a clockwise rotation.

Hence, $\omega > 0$ counterclockwise rotation

$\omega < 0$ clockwise rotation

Right hand rule

☞ **Remember,** all points on a rotating rigid body rotates through the same angle in the same time. Hence, at any instant, every part of a rotating rigid body has the same angular velocity.

iii) Angular Acceleration

Angular Acceleration (α, alpha): Is the rate at which angular velocity of a body changes with time.

That is, Angular Acceleration = $\dfrac{\text{change in Angular velocity}}{\text{time taken}}$

$$\boxed{\alpha = \frac{\Delta\omega}{\Delta t}} \text{ or, } \boxed{\alpha = \frac{\omega - \omega_o}{t}} \qquad \omega = \frac{\Delta\theta}{t}$$

The SI unit of angular acceleration is rad/s^2. An angular acceleration of 5 rad/s^2, for example, means that the angular velocity of the body is increasing by 5 rad/s every second.

☞ Angular acceleration is a vector quantity with the property that:

- The direction of the angular acceleration is in the same direction as the angular velocity when the rotation speeds up and,
- The direction of the angular acceleration is in opposite direction to the angular velocity when the rotation slows down.

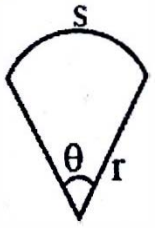
The Equations of Rotational Kinematics

Equations of motion for uniformly accelerated angular motion are analogous to those equations of motion for uniformly accelerated linear motion except the replacement of linear variables with angular variables. Therefore, we replace the initial velocity u with ω_o, the finial velocity v with ω, the displacement s with θ and the acceleration a with α.

As a result, the equations of motion of constant linear acceleration and the corresponding equations of motion for constant angular acceleration are given below.

Linear Equation (a = constant)	**Angular Equation** (α = constant)
$v = u + at$	$\omega = \omega_o + \alpha t$
$s = \left(\frac{v+u}{2}\right)t$	$\theta = \left(\frac{\omega_o + \omega}{2}\right)t$
$s = ut + \frac{1}{2}at^2$	$\theta = \omega_o t + \frac{1}{2}\alpha t^2$
$v^2 = u^2 + 2as$	$\omega^2 = \omega_o^2 + 2\alpha\theta$

Angular Variables and Tangential Variables



When a body moves an arc length s on a circle of radius r in a time t

- The tangential displacement, s and the angular displacement, θ are related by

$$\boxed{s = r\theta}$$

- The tangential speed, v and the angular speed, ω are related as follows if we divide both sides of the above equation by t

$$\frac{s}{t} = r\left(\frac{\theta}{t}\right), \frac{s}{t} = v \text{ and } \frac{\theta}{t} = \omega$$

$$\boxed{v = r\omega}$$